Surviving the Future

Crisis, Climate, and Human Resilience

A Practical Guide to Living Through the Crises Caused by Global Warming

Author: Setyo Wisodo | Year: 2026

Surviving the Future: Crisis, Climate, and Human Resilience

A Detailed Reference for Living Through the Crises Caused by Global Warming

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Preface: Why Future Global Warming Is Certain

Global warming is not a hypothesis; it is an observed, measured, and physically explained reality. The certainty of future warming rests on three pillars: (1) the physics of the greenhouse effect, (2) the observed accumulation of greenhouse gases in the atmosphere, and (3) the inertia of the climate system. Even if all emissions stopped today, committed warming from past emissions would continue for decades. This book is written to be debated in academic spaces: every major claim is tied to peer-reviewed literature, data, and expert statements.

The Three Pillars of Certainty

- **Physics:** Greenhouse gases absorb outgoing infrared radiation. This is established physics, tested for over a century (Arrhenius, 1896; modern radiative transfer models).
- **Observation:** Atmospheric CO2 has risen from ~280 ppm (pre-industrial) to ~420 ppm (2024). Global mean temperature has risen ~1.1-1.2 C. These are direct measurements.
- **Inertia:** The ocean stores heat; ice sheets respond slowly. Even with zero emissions, warming continues for decades (committed warming).

What the Data Say

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Consensus

> Dr. James Hansen, climate scientist: 'The science is clear: continued emissions will drive dangerous warming.'

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

Current Conditions and News

- 2023 was the hottest year on record, with global temperatures 1.48 C above pre-industrial levels (WMO, 2024).

- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.
- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.
- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.
- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.

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Chapter 1: Introduction: The Warming World

Theme: the warming world and the coming crises

This section examines this chapter in the context of the warming world and the coming crises. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of the warming world and the coming crises. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the

chapter.

1.1 Why This Book

Evidence from peer-reviewed research shows that Why This Book is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Why This Book

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Why This Book

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Why This Book

- Assess your current situation regarding why this book and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

1.2 What 'Survive' Really Means

For individuals and communities, the practical implications of What 'Survive' Really Means include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for What 'Survive' Really Means

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: What 'Survive' Really Means

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for What 'Survive' Really Means

- Assess your current situation regarding what 'survive' really means and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

1.3 How to Use This Book

Historical experience and recent disasters demonstrate that preparedness related to How to Use This Book reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for How to Use This Book

- Assess local conditions and identify the most relevant hazards.

- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: How to Use This Book

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for How to Use This Book

- Assess your current situation regarding how to use this book and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

1.4 The Scope of the Crisis

Policy and governance play a decisive role in The Scope of the Crisis. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for The Scope of the Crisis

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: The Scope of the Crisis

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for The Scope of the Crisis

- Assess your current situation regarding the scope of the crisis and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

1.5 Personal, Community, and Societal Survival

Technology can support Personal, Community, and Societal Survival, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Personal, Community, and Societal Survival

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Personal, Community, and Societal Survival

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Personal, Community, and Societal Survival

- Assess your current situation regarding personal, community, and societal survival and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

1.6 Knowledge as a Survival Tool

Education and training strengthen Knowledge as a Survival Tool. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Knowledge as a Survival Tool

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Knowledge as a Survival Tool

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Knowledge as a Survival Tool

- Assess your current situation regarding knowledge as a survival tool and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

1.7 The Role of Education

Monitoring and evaluation ensure that efforts related to The Role of Education remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for The Role of Education

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: The Role of Education

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for The Role of Education

- Assess your current situation regarding the role of education and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

1.8 Preparing for Uncertainty

This section examines Preparing for Uncertainty in the context of the warming world and the coming crises. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Preparing for Uncertainty

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Preparing for Uncertainty

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Preparing for Uncertainty

- Assess your current situation regarding preparing for uncertainty and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

1.9 Building Resilience

Evidence from peer-reviewed research shows that Building Resilience is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Building Resilience

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Building Resilience

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Building Resilience

- Assess your current situation regarding building resilience and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

1.10 A Roadmap for Readers

For individuals and communities, the practical implications of A Roadmap for Readers include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for A Roadmap for Readers

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: A Roadmap for Readers

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for A Roadmap for Readers

- Assess your current situation regarding a roadmap for readers and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Introduction: The Warming World

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Introduction: The Warming World

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Introduction: The Warming World

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

Current Conditions and News: Introduction: The Warming World

- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.
- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.

Future Scenarios and Predictions: Introduction: The Warming World

Under a high-emissions scenario (SSP5-8.5), the warming world and the coming crises is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to the warming world and the coming crises.

- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to the warming world and the coming crises.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Introduction: The Warming World

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 1

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 1

- IPCC. (2023). Climate Change 2023: Synthesis Report. IPCC, Geneva.
- IPCC. (2021). Climate Change 2021: The Physical Science Basis (AR6 WG1). Cambridge University Press.

Exercises — Chapter 1

- Reflect on how the warming world and the coming crises affects your own community and write a one-page plan.

Chapter 2: The Science of Global Warming

Theme: the physical science of global warming

This section examines this chapter in the context of the physical science of global warming. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of the physical science of global warming. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

2.1 The Greenhouse Effect

Evidence from peer-reviewed research shows that The Greenhouse Effect is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for The Greenhouse Effect

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: The Greenhouse Effect

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for The Greenhouse Effect

- Assess your current situation regarding the greenhouse effect and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

2.2 Greenhouse Gases and Their Sources

For individuals and communities, the practical implications of Greenhouse Gases and Their Sources include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Greenhouse Gases and Their Sources

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Greenhouse Gases and Their Sources

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Greenhouse Gases and Their Sources

- Assess your current situation regarding greenhouse gases and their sources and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

2.3 Evidence of Warming

Historical experience and recent disasters demonstrate that preparedness related to Evidence of Warming reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Evidence of Warming

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Evidence of Warming

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Evidence of Warming

- Assess your current situation regarding evidence of warming and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

2.4 Temperature Records and Trends

Policy and governance play a decisive role in Temperature Records and Trends. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Temperature Records and Trends

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Temperature Records and Trends

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Temperature Records and Trends

- Assess your current situation regarding temperature records and trends and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

2.5 Ocean Heat and Sea Level

Technology can support Ocean Heat and Sea Level, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Ocean Heat and Sea Level

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Ocean Heat and Sea Level

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Ocean Heat and Sea Level

- Assess your current situation regarding ocean heat and sea level and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

2.6 Ice Loss and Cryosphere

Education and training strengthen Ice Loss and Cryosphere. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Ice Loss and Cryosphere

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Ice Loss and Cryosphere

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Ice Loss and Cryosphere

- Assess your current situation regarding ice loss and cryosphere and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

2.7 Carbon Cycle and Feedbacks

Monitoring and evaluation ensure that efforts related to Carbon Cycle and Feedbacks remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Carbon Cycle and Feedbacks

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Carbon Cycle and Feedbacks

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Carbon Cycle and Feedbacks

- Assess your current situation regarding carbon cycle and feedbacks and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

2.8 Climate Sensitivity

This section examines Climate Sensitivity in the context of the physical science of global warming. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Climate Sensitivity

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate Sensitivity

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Climate Sensitivity

- Assess your current situation regarding climate sensitivity and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

2.9 Attribution of Warming

Evidence from peer-reviewed research shows that Attribution of Warming is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Attribution of Warming

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Attribution of Warming

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Attribution of Warming

- Assess your current situation regarding attribution of warming and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

2.10 Key Scientific Conclusions

For individuals and communities, the practical implications of Key Scientific Conclusions include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Key Scientific Conclusions

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Key Scientific Conclusions

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Key Scientific Conclusions

- Assess your current situation regarding key scientific conclusions and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: The Science of Global Warming

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: The Science of Global Warming

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: The Science of Global Warming

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

Current Conditions and News: The Science of Global Warming

- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.
- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.

Future Scenarios and Predictions: The Science of Global Warming

Under a high-emissions scenario (SSP5-8.5), the physical science of global warming is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to the physical science of global warming.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to the physical science of global warming.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: The Science of Global Warming

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 2

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 2

- IPCC. (2022). Climate Change 2022: Impacts, Adaptation and Vulnerability (AR6 WG2). Cambridge University Press.
- IPCC. (2022). Climate Change 2022: Mitigation of Climate Change (AR6 WG3). Cambridge University Press.

Exercises — Chapter 2

- Reflect on how the physical science of global warming affects your own community and write a one-page plan.

Chapter 3: History of Climate Change

Theme: the history of climate change and human response

This section examines this chapter in the context of the history of climate change and human response. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of the history of climate change and human response. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

3.1 Early Climate Science

Evidence from peer-reviewed research shows that Early Climate Science is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Early Climate Science

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Early Climate Science

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Early Climate Science

- Assess your current situation regarding early climate science and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

3.2 Industrial Revolution and Emissions

For individuals and communities, the practical implications of Industrial Revolution and Emissions include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Industrial Revolution and Emissions

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.

- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Industrial Revolution and Emissions

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Industrial Revolution and Emissions

- Assess your current situation regarding industrial revolution and emissions and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

3.3 20th Century Warming

Historical experience and recent disasters demonstrate that preparedness related to 20th Century Warming reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for 20th Century Warming

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: 20th Century Warming

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for 20th Century Warming

- Assess your current situation regarding 20th century warming and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

3.4 International Awareness

Policy and governance play a decisive role in International Awareness. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for International Awareness

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: International Awareness

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for International Awareness

- Assess your current situation regarding international awareness and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

3.5 IPCC Assessments

Technology can support IPCC Assessments, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for IPCC Assessments

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: IPCC Assessments

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for IPCC Assessments

- Assess your current situation regarding ipcc assessments and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

3.6 Paris Agreement Era

Education and training strengthen Paris Agreement Era. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Paris Agreement Era

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Paris Agreement Era

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Paris Agreement Era

- Assess your current situation regarding paris agreement era and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

3.7 Current Emission Trends

Monitoring and evaluation ensure that efforts related to Current Emission Trends remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Current Emission Trends

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Current Emission Trends

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Current Emission Trends

- Assess your current situation regarding current emission trends and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

3.8 Lessons from History

This section examines Lessons from History in the context of the history of climate change and human response. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Lessons from History

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Lessons from History

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Lessons from History

- Assess your current situation regarding lessons from history and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

3.9 Failed and Successful Responses

Evidence from peer-reviewed research shows that Failed and Successful Responses is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Failed and Successful Responses

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.

- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Failed and Successful Responses

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Failed and Successful Responses

- Assess your current situation regarding failed and successful responses and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

3.10 Looking Forward

For individuals and communities, the practical implications of Looking Forward include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Looking Forward

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Looking Forward

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Looking Forward

- Assess your current situation regarding looking forward and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: History of Climate Change

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: History of Climate Change

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: History of Climate Change

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

Current Conditions and News: History of Climate Change

- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.
- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.

Future Scenarios and Predictions: History of Climate Change

Under a high-emissions scenario (SSP5-8.5), the history of climate change and human response is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to the history of climate change and human response.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to the history of climate change and human response.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: History of Climate Change

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 3

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 3

- IPCC. (2018). Global Warming of 1.5 C (SR15). IPCC, Geneva.
- IPCC. (2019). Special Report on the Ocean and Cryosphere in a Changing Climate (SROCC). IPCC, Geneva.

Exercises — Chapter 3

- Reflect on how the history of climate change and human response affects your own community and write a one-page plan.

Chapter 4: Climate Systems and Feedback Loops

Theme: climate systems and feedback mechanisms

This section examines this chapter in the context of climate systems and feedback mechanisms. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of climate systems and feedback mechanisms. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

4.1 The Climate System

Evidence from peer-reviewed research shows that The Climate System is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for The Climate System

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: The Climate System

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for The Climate System

- Assess your current situation regarding the climate system and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

4.2 Atmosphere and Circulation

For individuals and communities, the practical implications of Atmosphere and Circulation include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Atmosphere and Circulation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Atmosphere and Circulation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Atmosphere and Circulation

- Assess your current situation regarding atmosphere and circulation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

4.3 Ocean Circulation

Historical experience and recent disasters demonstrate that preparedness related to Ocean Circulation reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Ocean Circulation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Ocean Circulation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Ocean Circulation

- Assess your current situation regarding ocean circulation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

4.4 Cryosphere Dynamics

Policy and governance play a decisive role in Cryosphere Dynamics. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Cryosphere Dynamics

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cryosphere Dynamics

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cryosphere Dynamics

- Assess your current situation regarding cryosphere dynamics and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

4.5 Positive Feedback Loops

Technology can support Positive Feedback Loops, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Positive Feedback Loops

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Positive Feedback Loops

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Positive Feedback Loops

- Assess your current situation regarding positive feedback loops and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

4.6 Negative Feedback Loops

Education and training strengthen Negative Feedback Loops. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Negative Feedback Loops

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Negative Feedback Loops

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Negative Feedback Loops

- Assess your current situation regarding negative feedback loops and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

4.7 Tipping Points

Monitoring and evaluation ensure that efforts related to Tipping Points remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Tipping Points

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Tipping Points

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Tipping Points

- Assess your current situation regarding tipping points and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

4.8 Abrupt Change

This section examines Abrupt Change in the context of climate systems and feedback mechanisms. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Abrupt Change

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Abrupt Change

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Abrupt Change

- Assess your current situation regarding abrupt change and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

4.9 Regional Variations

Evidence from peer-reviewed research shows that Regional Variations is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Regional Variations

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Regional Variations

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Regional Variations

- Assess your current situation regarding regional variations and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

4.10 Implications for Survival

For individuals and communities, the practical implications of Implications for Survival include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Implications for Survival

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Implications for Survival

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Implications for Survival

- Assess your current situation regarding implications for survival and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Climate Systems and Feedback Loops

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate Systems and Feedback Loops

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Climate Systems and Feedback Loops

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

Current Conditions and News: Climate Systems and Feedback Loops

- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.
- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.

Future Scenarios and Predictions: Climate Systems and Feedback Loops

Under a high-emissions scenario (SSP5-8.5), climate systems and feedback mechanisms is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to climate systems and feedback mechanisms.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to climate systems and feedback mechanisms.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Climate Systems and Feedback Loops

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 4

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 4

- IPCC. (2019). Climate Change and Land (SRCCL). IPCC, Geneva.
- NASA. (2024). Global Climate Change: Vital Signs. climate.nasa.gov.

Exercises — Chapter 4

- Reflect on how climate systems and feedback mechanisms affects your own community and write a one-page plan.

Chapter 5: Measurement and Monitoring

Theme: climate measurement and monitoring

This section examines this chapter in the context of climate measurement and monitoring. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of climate measurement and monitoring. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

5.1 Temperature Monitoring

Evidence from peer-reviewed research shows that Temperature Monitoring is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Temperature Monitoring

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Temperature Monitoring

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Temperature Monitoring

- Assess your current situation regarding temperature monitoring and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

5.2 Satellite Observations

For individuals and communities, the practical implications of Satellite Observations include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Satellite Observations

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Satellite Observations

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Satellite Observations

- Assess your current situation regarding satellite observations and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

5.3 Ocean Monitoring

Historical experience and recent disasters demonstrate that preparedness related to Ocean Monitoring reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Ocean Monitoring

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Ocean Monitoring

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Ocean Monitoring

- Assess your current situation regarding ocean monitoring and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

5.4 Ice and Snow Monitoring

Policy and governance play a decisive role in Ice and Snow Monitoring. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Ice and Snow Monitoring

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Ice and Snow Monitoring

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Ice and Snow Monitoring

- Assess your current situation regarding ice and snow monitoring and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

5.5 Greenhouse Gas Inventories

Technology can support Greenhouse Gas Inventories, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Greenhouse Gas Inventories

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Greenhouse Gas Inventories

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Greenhouse Gas Inventories

- Assess your current situation regarding greenhouse gas inventories and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

5.6 Extreme Event Attribution

Education and training strengthen Extreme Event Attribution. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Extreme Event Attribution

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Extreme Event Attribution

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Extreme Event Attribution

- Assess your current situation regarding extreme event attribution and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

5.7 Early Warning Systems

Monitoring and evaluation ensure that efforts related to Early Warning Systems remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Early Warning Systems

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Early Warning Systems

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Early Warning Systems

- Assess your current situation regarding early warning systems and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

5.8 Data Quality and Uncertainty

This section examines Data Quality and Uncertainty in the context of climate measurement and monitoring. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Data Quality and Uncertainty

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Data Quality and Uncertainty

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Data Quality and Uncertainty

- Assess your current situation regarding data quality and uncertainty and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

5.9 Open Data and Citizen Science

Evidence from peer-reviewed research shows that Open Data and Citizen Science is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Open Data and Citizen Science

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Open Data and Citizen Science

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Open Data and Citizen Science

- Assess your current situation regarding open data and citizen science and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

5.10 Using Data for Decisions

For individuals and communities, the practical implications of Using Data for Decisions include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Using Data for Decisions

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Using Data for Decisions

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Using Data for Decisions

- Assess your current situation regarding using data for decisions and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Measurement and Monitoring

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Measurement and Monitoring

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Measurement and Monitoring

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

Current Conditions and News: Measurement and Monitoring

- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.
- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).

Future Scenarios and Predictions: Measurement and Monitoring

Under a high-emissions scenario (SSP5-8.5), climate measurement and monitoring is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to climate measurement and monitoring.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to climate measurement and monitoring.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Measurement

and Monitoring

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 5

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 5

- NOAA. (2024). State of the Climate. ncei.noaa.gov.
- WMO. (2023). State of the Global Climate 2023. World Meteorological Organization, Geneva.

Exercises — Chapter 5

- Reflect on how climate measurement and monitoring affects your own community and write a one-page plan.

Chapter 6: Extreme Heat

Theme: extreme heat and heat waves

This section examines this chapter in the context of extreme heat and heat waves. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of extreme heat and heat waves. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

6.1 Heat Wave Definition and Trends

Evidence from peer-reviewed research shows that Heat Wave Definition and Trends is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Heat Wave Definition and Trends

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Heat Wave Definition and Trends

Indicator	Current Status	Future Outlook
Risk level	Moderate to high	Increasing
Preparedness	Uneven	Needs improvement
Resource availability	Declining in stressed regions	Variable
Community capacity	Moderate	Can be strengthened

Practical Steps for Heat Wave Definition and Trends

- Assess your current situation regarding heat wave definition and trends and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

6.2 Physiology of Heat Stress

For individuals and communities, the practical implications of Physiology of Heat Stress include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Physiology of Heat Stress

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Physiology of Heat Stress

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Physiology of Heat Stress

- Assess your current situation regarding physiology of heat stress and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

6.3 Heat Illness Spectrum

Historical experience and recent disasters demonstrate that preparedness related to Heat Illness Spectrum reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Heat Illness Spectrum

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Heat Illness Spectrum

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Heat Illness Spectrum

- Assess your current situation regarding heat illness spectrum and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

6.4 Vulnerable Populations

Policy and governance play a decisive role in Vulnerable Populations. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Vulnerable Populations

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Vulnerable Populations

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Vulnerable Populations

- Assess your current situation regarding vulnerable populations and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

6.5 Urban Heat Islands

Technology can support Urban Heat Islands, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Urban Heat Islands

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Urban Heat Islands

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Urban Heat Islands

- Assess your current situation regarding urban heat islands and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

6.6 Infrastructure Impacts

Education and training strengthen Infrastructure Impacts. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Infrastructure Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Infrastructure Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Infrastructure Impacts

- Assess your current situation regarding infrastructure impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

6.7 Labor and Productivity

Monitoring and evaluation ensure that efforts related to Labor and Productivity remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Labor and Productivity

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Labor and Productivity

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Labor and Productivity

- Assess your current situation regarding labor and productivity and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

6.8 Heat Action Plans

This section examines Heat Action Plans in the context of extreme heat and heat waves. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Heat Action Plans

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Heat Action Plans

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Heat Action Plans

- Assess your current situation regarding heat action plans and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

6.9 Cooling Strategies

Evidence from peer-reviewed research shows that Cooling Strategies is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Cooling Strategies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cooling Strategies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cooling Strategies

- Assess your current situation regarding cooling strategies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

6.10 Long-Term Adaptation to Heat

For individuals and communities, the practical implications of Long-Term Adaptation to Heat include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Long-Term Adaptation to Heat

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Long-Term Adaptation to Heat

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Long-Term Adaptation to Heat

- Assess your current situation regarding long-term adaptation to heat and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Extreme Heat

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Extreme Heat

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Extreme Heat

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

> Dr. Maria Neira, WHO: 'Climate change is the greatest health challenge of the 21st century.'

Current Conditions and News: Extreme Heat

- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).
- Coral bleaching events are becoming more frequent, threatening marine ecosystems.

Future Scenarios and Predictions: Extreme Heat

Under a high-emissions scenario (SSP5-8.5), extreme heat and heat waves is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to extreme heat and heat waves.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to extreme heat and heat waves.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Extreme Heat

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 6

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 6

- WHO. (2023). Climate Change and Health Fact Sheet. who.int.
- WHO. (2021). COP26 Special Report on Climate Change and Health. WHO, Geneva.

Exercises — Chapter 6

- Reflect on how extreme heat and heat waves affects your own community and write a one-page plan.

Chapter 7: Water Scarcity and Drought

Theme: water scarcity and drought

This section examines this chapter in the context of water scarcity and drought. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of water scarcity and drought. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

7.1 Global Water Cycle

Evidence from peer-reviewed research shows that Global Water Cycle is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Global Water Cycle

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Global Water Cycle

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Global Water Cycle

- Assess your current situation regarding global water cycle and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

7.2 Drought Types and Indices

For individuals and communities, the practical implications of Drought Types and Indices include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Drought Types and Indices

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Drought Types and Indices

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Drought Types and Indices

- Assess your current situation regarding drought types and indices and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

7.3 Causes of Water Scarcity

Historical experience and recent disasters demonstrate that preparedness related to Causes of Water Scarcity reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Causes of Water Scarcity

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Causes of Water Scarcity

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Causes of Water Scarcity

- Assess your current situation regarding causes of water scarcity and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

7.4 Agricultural Water Use

Policy and governance play a decisive role in Agricultural Water Use. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Agricultural Water Use

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Agricultural Water Use

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Agricultural Water Use

- Assess your current situation regarding agricultural water use and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

7.5 Urban Water Stress

Technology can support Urban Water Stress, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Urban Water Stress

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Urban Water Stress

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Urban Water Stress

- Assess your current situation regarding urban water stress and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

7.6 Groundwater Depletion

Education and training strengthen Groundwater Depletion. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Groundwater Depletion

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Groundwater Depletion

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Groundwater Depletion

- Assess your current situation regarding groundwater depletion and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

7.7 Water Quality

Monitoring and evaluation ensure that efforts related to Water Quality remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Water Quality

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water Quality

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water Quality

- Assess your current situation regarding water quality and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

7.8 Water Governance

This section examines Water Governance in the context of water scarcity and drought. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Water Governance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water Governance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water Governance

- Assess your current situation regarding water governance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

7.9 Conservation and Efficiency

Evidence from peer-reviewed research shows that Conservation and Efficiency is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Conservation and Efficiency

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Conservation and Efficiency

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Conservation and Efficiency

- Assess your current situation regarding conservation and efficiency and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

7.10 Alternative Water Sources

For individuals and communities, the practical implications of Alternative Water Sources include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Alternative Water Sources

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Alternative Water Sources

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Alternative Water Sources

- Assess your current situation regarding alternative water sources and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Water Scarcity and Drought

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water Scarcity and Drought

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Water Scarcity and Drought

> Dr. Maria Neira, WHO: 'Climate change is the greatest health challenge of the 21st century.'

> Prof. Hans-Otto Portner, IPCC: 'The window for action is closing rapidly.'

Current Conditions and News: Water Scarcity and Drought

- Coral bleaching events are becoming more frequent, threatening marine ecosystems.
- Arctic sea ice is declining at a rate of about 13% per decade.

Future Scenarios and Predictions: Water Scarcity and Drought

Under a high-emissions scenario (SSP5-8.5), water scarcity and drought is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to water scarcity and drought.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to water scarcity and drought.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Water Scarcity and Drought

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 7

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 7

- FAO. (2023). The State of Food Security and Nutrition in the World. FAO, Rome.
- FAO. (2022). The State of the World's Land and Water Resources (SOLAW 2021). FAO, Rome.

Exercises — Chapter 7

- Reflect on how water scarcity and drought affects your own community and write a one-page plan.

Chapter 8: Floods and Sea Level Rise

Theme: floods and sea level rise

This section examines this chapter in the context of floods and sea level rise. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of floods and sea level rise. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

8.1 Flood Types and Causes

Evidence from peer-reviewed research shows that Flood Types and Causes is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Flood Types and Causes

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Flood Types and Causes

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Flood Types and Causes

- Assess your current situation regarding flood types and causes and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

8.2 Sea Level Rise Mechanisms

For individuals and communities, the practical implications of Sea Level Rise Mechanisms include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Sea Level Rise Mechanisms

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sea Level Rise Mechanisms

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sea Level Rise Mechanisms

- Assess your current situation regarding sea level rise mechanisms and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

8.3 Coastal Flood Risk

Historical experience and recent disasters demonstrate that preparedness related to Coastal Flood Risk reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Coastal Flood Risk

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Coastal Flood Risk

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Coastal Flood Risk

- Assess your current situation regarding coastal flood risk and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

8.4 Riverine and Flash Floods

Policy and governance play a decisive role in Riverine and Flash Floods. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Riverine and Flash Floods

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Riverine and Flash Floods

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Riverine and Flash Floods

- Assess your current situation regarding riverine and flash floods and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

8.5 Urban Flooding

Technology can support Urban Flooding, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Urban Flooding

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Urban Flooding

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Urban Flooding

- Assess your current situation regarding urban flooding and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

8.6 Flood Impacts on Health

Education and training strengthen Flood Impacts on Health. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Flood Impacts on Health

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Flood Impacts on Health

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Flood Impacts on Health

- Assess your current situation regarding flood impacts on health and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

8.7 Flood Defenses

Monitoring and evaluation ensure that efforts related to Flood Defenses remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Flood Defenses

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Flood Defenses

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Flood Defenses

- Assess your current situation regarding flood defenses and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

8.8 Managed Retreat

This section examines Managed Retreat in the context of floods and sea level rise. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Managed Retreat

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Managed Retreat

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Managed Retreat

- Assess your current situation regarding managed retreat and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

8.9 Flood Insurance and Finance

Evidence from peer-reviewed research shows that Flood Insurance and Finance is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Flood Insurance and Finance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Flood Insurance and Finance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Flood Insurance and Finance

- Assess your current situation regarding flood insurance and finance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

8.10 Long-Term Coastal Adaptation

For individuals and communities, the practical implications of Long-Term Coastal Adaptation include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Long-Term Coastal Adaptation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Long-Term Coastal Adaptation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Long-Term Coastal Adaptation

- Assess your current situation regarding long-term coastal adaptation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Floods and Sea Level Rise

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Floods and Sea Level Rise

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Floods and Sea Level Rise

> Prof. Hans-Otto Portner, IPCC: 'The window for action is closing rapidly.'

> Dr. Debra Roberts, IPCC: 'Adaptation and mitigation must go hand in hand.'

Current Conditions and News: Floods and Sea Level Rise

- Arctic sea ice is declining at a rate of about 13% per decade.
- Glaciers worldwide are losing mass, contributing to sea level rise.

Future Scenarios and Predictions: Floods and Sea Level Rise

Under a high-emissions scenario (SSP5-8.5), floods and sea level rise is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to floods and sea level rise.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to floods and sea level rise.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Floods and Sea Level Rise

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 8

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 8

- World Bank. (2021). Groundswell: Acting on Internal Climate Migration. World Bank, Washington DC.
- UNHCR. (2023). Climate Change and Displacement. unhcr.org.

Exercises — Chapter 8

- Reflect on how floods and sea level rise affects your own community and write a one-page plan.

Chapter 9: Food and Agriculture

Theme: food security and agriculture under climate change

This section examines this chapter in the context of food security and agriculture under climate change. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of food security and agriculture under climate change. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

9.1 Food Systems Overview

Evidence from peer-reviewed research shows that Food Systems Overview is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Food Systems Overview

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Food Systems Overview

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Food Systems Overview

- Assess your current situation regarding food systems overview and identify gaps.

- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

9.2 Crop Yield Impacts

For individuals and communities, the practical implications of Crop Yield Impacts include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Crop Yield Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Crop Yield Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Crop Yield Impacts

- Assess your current situation regarding crop yield impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

9.3 Heat and Water Stress on Crops

Historical experience and recent disasters demonstrate that preparedness related to Heat and Water Stress on Crops reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Heat and Water Stress on Crops

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Heat and Water Stress on Crops

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Heat and Water Stress on Crops

- Assess your current situation regarding heat and water stress on crops and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

9.4 Pests and Diseases

Policy and governance play a decisive role in Pests and Diseases. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Pests and Diseases

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Pests and Diseases

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Pests and Diseases

- Assess your current situation regarding pests and diseases and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

9.5 Livestock and Fisheries

Technology can support Livestock and Fisheries, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Livestock and Fisheries

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Livestock and Fisheries

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Livestock and Fisheries

- Assess your current situation regarding livestock and fisheries and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

9.6 Supply Chain Vulnerability

Education and training strengthen Supply Chain Vulnerability. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Supply Chain Vulnerability

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Supply Chain Vulnerability

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Supply Chain Vulnerability

- Assess your current situation regarding supply chain vulnerability and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

9.7 Price Volatility and Hunger

Monitoring and evaluation ensure that efforts related to Price Volatility and Hunger remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Price Volatility and Hunger

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Price Volatility and Hunger

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Price Volatility and Hunger

- Assess your current situation regarding price volatility and hunger and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

9.8 Adaptive Agriculture

This section examines Adaptive Agriculture in the context of food security and agriculture under climate change. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Adaptive Agriculture

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Adaptive Agriculture

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Adaptive Agriculture

- Assess your current situation regarding adaptive agriculture and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

9.9 Food Storage and Preservation

Evidence from peer-reviewed research shows that Food Storage and Preservation is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Food Storage and Preservation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Food Storage and Preservation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Food Storage and Preservation

- Assess your current situation regarding food storage and preservation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

9.10 Nutrition and Dietary Shifts

For individuals and communities, the practical implications of Nutrition and Dietary Shifts include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Nutrition and Dietary Shifts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Nutrition and Dietary Shifts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Nutrition and Dietary Shifts

- Assess your current situation regarding nutrition and dietary shifts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Food and Agriculture

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Food and Agriculture

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Food and Agriculture

> Dr. Debra Roberts, IPCC: 'Adaptation and mitigation must go hand in hand.'

Current Conditions and News: Food and Agriculture

- Glaciers worldwide are losing mass, contributing to sea level rise.

Future Scenarios and Predictions: Food and Agriculture

Under a high-emissions scenario (SSP5-8.5), food security and agriculture under climate change is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to food security and agriculture under climate change.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to food security and agriculture under climate change.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Food and Agriculture

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 9

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 9

- UNEP. (2023). Adaptation Gap Report 2023. UNEP, Nairobi.
- UNDRR. (2015). Sendai Framework for Disaster Risk Reduction 2015-2030. UNDRR, Geneva.

Exercises — Chapter 9

- Reflect on how food security and agriculture under climate change affects your own community and write a one-page plan.

Chapter 10: Ecosystems and Biodiversity

Theme: ecosystems and biodiversity loss

This section examines this chapter in the context of ecosystems and biodiversity loss. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of ecosystems and biodiversity loss. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

10.1 Biodiversity and Ecosystem Services

Evidence from peer-reviewed research shows that Biodiversity and Ecosystem Services is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Biodiversity and Ecosystem Services

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Biodiversity and Ecosystem Services

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Biodiversity and Ecosystem Services

- Assess your current situation regarding biodiversity and ecosystem services and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

10.2 Habitat Loss

For individuals and communities, the practical implications of Habitat Loss include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Habitat Loss

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Habitat Loss

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Habitat Loss

- Assess your current situation regarding habitat loss and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

10.3 Species Extinction Risk

Historical experience and recent disasters demonstrate that preparedness related to Species Extinction Risk reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Species Extinction Risk

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Species Extinction Risk

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Species Extinction Risk

- Assess your current situation regarding species extinction risk and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

10.4 Forest Ecosystems

Policy and governance play a decisive role in Forest Ecosystems. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Forest Ecosystems

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Forest Ecosystems

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Forest Ecosystems

- Assess your current situation regarding forest ecosystems and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

10.5 Marine Ecosystems

Technology can support Marine Ecosystems, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Marine Ecosystems

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Marine Ecosystems

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Marine Ecosystems

- Assess your current situation regarding marine ecosystems and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

10.6 Wetlands and Freshwater

Education and training strengthen Wetlands and Freshwater. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Wetlands and Freshwater

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Wetlands and Freshwater

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Wetlands and Freshwater

- Assess your current situation regarding wetlands and freshwater and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

10.7 Invasive Species

Monitoring and evaluation ensure that efforts related to Invasive Species remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Invasive Species

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Invasive Species

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Invasive Species

- Assess your current situation regarding invasive species and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

10.8 Ecosystem-Based Adaptation

This section examines Ecosystem-Based Adaptation in the context of ecosystems and biodiversity loss. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Ecosystem-Based Adaptation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Ecosystem-Based Adaptation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Ecosystem-Based Adaptation

- Assess your current situation regarding ecosystem-based adaptation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

10.9 Conservation Strategies

Evidence from peer-reviewed research shows that Conservation Strategies is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Conservation Strategies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Conservation Strategies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Conservation Strategies

- Assess your current situation regarding conservation strategies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

10.10 Human Dependence on Nature

For individuals and communities, the practical implications of Human Dependence on Nature include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Human Dependence on Nature

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Human Dependence on Nature

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Human Dependence on Nature

- Assess your current situation regarding human dependence on nature and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Ecosystems and Biodiversity

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Ecosystems and Biodiversity

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Ecosystems and Biodiversity

> Dr. James Hansen, climate scientist: 'The science is clear: continued emissions will drive dangerous warming.'

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

Current Conditions and News: Ecosystems and

Biodiversity

- 2023 was the hottest year on record, with global temperatures 1.48 C above pre-industrial levels (WMO, 2024).
- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.

Future Scenarios and Predictions: Ecosystems and Biodiversity

Under a high-emissions scenario (SSP5-8.5), ecosystems and biodiversity loss is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to ecosystems and biodiversity loss.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to ecosystems and biodiversity loss.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Ecosystems and Biodiversity

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 10

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 10

- United Nations. (2015). Paris Agreement. unfccc.int.
- United Nations. (2015). 2030 Agenda for Sustainable Development. UN, New York.

Exercises — Chapter 10

- Reflect on how ecosystems and biodiversity loss affects your own community and write a one-page plan.

Chapter 11: Health and Disease

Theme: health impacts of climate change

This section examines this chapter in the context of health impacts of climate change. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of health impacts of climate change. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

11.1 Direct and Indirect Health Impacts

Evidence from peer-reviewed research shows that Direct and Indirect Health Impacts is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Direct and Indirect Health Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Direct and Indirect Health Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Direct and Indirect Health Impacts

- Assess your current situation regarding direct and indirect health impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

11.2 Heat-Related Illness

For individuals and communities, the practical implications of Heat-Related Illness include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Heat-Related Illness

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Heat-Related Illness

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Heat-Related Illness

- Assess your current situation regarding heat-related illness and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

11.3 Vector-Borne Diseases

Historical experience and recent disasters demonstrate that preparedness related to Vector-Borne Diseases reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Vector-Borne Diseases

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Vector-Borne Diseases

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Vector-Borne Diseases

- Assess your current situation regarding vector-borne diseases and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

11.4 Water-Borne Diseases

Policy and governance play a decisive role in Water-Borne Diseases. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Water-Borne Diseases

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water-Borne Diseases

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water-Borne Diseases

- Assess your current situation regarding water-borne diseases and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

11.5 Air Quality and Respiratory Health

Technology can support Air Quality and Respiratory Health, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Air Quality and Respiratory Health

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Air Quality and Respiratory Health

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Air Quality and Respiratory Health

- Assess your current situation regarding air quality and respiratory health and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

11.6 Mental Health Impacts

Education and training strengthen Mental Health Impacts. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Mental Health Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Mental Health Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Mental Health Impacts

- Assess your current situation regarding mental health impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

11.7 Health System Resilience

Monitoring and evaluation ensure that efforts related to Health System Resilience remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Health System Resilience

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Health System Resilience

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Health System Resilience

- Assess your current situation regarding health system resilience and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

11.8 Vulnerable Groups

This section examines Vulnerable Groups in the context of health impacts of climate change. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Vulnerable Groups

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Vulnerable Groups

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Vulnerable Groups

- Assess your current situation regarding vulnerable groups and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

11.9 Prevention and Preparedness

Evidence from peer-reviewed research shows that Prevention and Preparedness is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Prevention and Preparedness

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Prevention and Preparedness

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Prevention and Preparedness

- Assess your current situation regarding prevention and preparedness and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

11.10 Health Equity

For individuals and communities, the practical implications of Health Equity include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Health Equity

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Health Equity

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Health Equity

- Assess your current situation regarding health equity and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Health and Disease

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Health and Disease

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Health and Disease

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

Current Conditions and News: Health and Disease

- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.
- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.

Future Scenarios and Predictions: Health and Disease

Under a high-emissions scenario (SSP5-8.5), health impacts of climate change is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to health impacts of climate change.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to health impacts of climate change.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Health and Disease

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 11

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 11

- Stern, N. (2007). The Economics of Climate Change: The Stern Review. Cambridge University Press.
- Watts, N. et al. (2021). The 2021 report of the Lancet Countdown on health and climate change. The Lancet, 398(10311), 1619-1662.

Exercises — Chapter 11

- Reflect on how health impacts of climate change affects your own community and write a one-page plan.

Chapter 12: Economic Impacts

Theme: economic impacts of climate change

This section examines this chapter in the context of economic impacts of climate change. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of economic impacts of climate change. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

12.1 Damage Cost Estimates

Evidence from peer-reviewed research shows that Damage Cost Estimates is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Damage Cost Estimates

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Damage Cost Estimates

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Damage Cost Estimates

- Assess your current situation regarding damage cost estimates and identify gaps.

- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

12.2 Sectoral Impacts

For individuals and communities, the practical implications of Sectoral Impacts include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Sectoral Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sectoral Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sectoral Impacts

- Assess your current situation regarding sectoral impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

12.3 Labor Productivity Loss

Historical experience and recent disasters demonstrate that preparedness related to Labor Productivity Loss reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Labor Productivity Loss

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Labor Productivity Loss

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Labor Productivity Loss

- Assess your current situation regarding labor productivity loss and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

12.4 Infrastructure Damage

Policy and governance play a decisive role in Infrastructure Damage. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Infrastructure Damage

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Infrastructure Damage

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Infrastructure Damage

- Assess your current situation regarding infrastructure damage and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

12.5 Insurance and Financial Risk

Technology can support Insurance and Financial Risk, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Insurance and Financial Risk

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Insurance and Financial Risk

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Insurance and Financial Risk

- Assess your current situation regarding insurance and financial risk and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

12.6 Macroeconomic Effects

Education and training strengthen Macroeconomic Effects. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Macroeconomic Effects

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Macroeconomic Effects

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Macroeconomic Effects

- Assess your current situation regarding macroeconomic effects and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

12.7 Poverty and Inequality

Monitoring and evaluation ensure that efforts related to Poverty and Inequality remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Poverty and Inequality

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Poverty and Inequality

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Poverty and Inequality

- Assess your current situation regarding poverty and inequality and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

12.8 Cost of Adaptation

This section examines Cost of Adaptation in the context of economic impacts of climate change. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Cost of Adaptation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cost of Adaptation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cost of Adaptation

- Assess your current situation regarding cost of adaptation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

12.9 Cost of Inaction

Evidence from peer-reviewed research shows that Cost of Inaction is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Cost of Inaction

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cost of Inaction

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cost of Inaction

- Assess your current situation regarding cost of inaction and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

12.10 Economic Resilience Strategies

For individuals and communities, the practical implications of Economic Resilience Strategies include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Economic Resilience Strategies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Economic Resilience Strategies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Economic Resilience Strategies

- Assess your current situation regarding economic resilience strategies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Economic Impacts

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Economic Impacts

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Economic Impacts

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

Current Conditions and News: Economic Impacts

- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.
- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.

Future Scenarios and Predictions: Economic Impacts

Under a high-emissions scenario (SSP5-8.5), economic impacts of climate change is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to economic impacts of climate change.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to economic impacts of climate change.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Economic Impacts

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 12

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 12

- Romanello, M. et al. (2023). The 2023 report of the Lancet Countdown. The Lancet, 402(10419), 2346-2394.
- Ebi, K.L. et al. (2021). Hot weather and heat extremes: health risks. The Lancet, 398(10301), 698-708.

Exercises — Chapter 12

- Reflect on how economic impacts of climate change affects your own community and write a one-page plan.

Chapter 13: Migration and Displacement

Theme: climate migration and displacement

This section examines this chapter in the context of climate migration and displacement. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of climate migration and displacement. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

13.1 Drivers of Climate Migration

Evidence from peer-reviewed research shows that Drivers of Climate Migration is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Drivers of Climate Migration

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Drivers of Climate Migration

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Drivers of Climate Migration

- Assess your current situation regarding drivers of climate migration and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

13.2 Internal Displacement

For individuals and communities, the practical implications of Internal Displacement include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Internal Displacement

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Internal Displacement

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Internal Displacement

- Assess your current situation regarding internal displacement and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

13.3 Cross-Border Movement

Historical experience and recent disasters demonstrate that preparedness related to Cross-Border Movement reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Cross-Border Movement

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cross-Border Movement

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cross-Border Movement

- Assess your current situation regarding cross-border movement and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

13.4 Legal Status and Rights

Policy and governance play a decisive role in Legal Status and Rights. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Legal Status and Rights

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Legal Status and Rights

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Legal Status and Rights

- Assess your current situation regarding legal status and rights and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

13.5 Urbanization Pressures

Technology can support Urbanization Pressures, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Urbanization Pressures

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Urbanization Pressures

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Urbanization Pressures

- Assess your current situation regarding urbanization pressures and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

13.6 Receiving Communities

Education and training strengthen Receiving Communities. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Receiving Communities

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Receiving Communities

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Receiving Communities

- Assess your current situation regarding receiving communities and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

13.7 Planned Relocation

Monitoring and evaluation ensure that efforts related to Planned Relocation remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Planned Relocation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Planned Relocation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Planned Relocation

- Assess your current situation regarding planned relocation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

13.8 Loss and Damage

This section examines Loss and Damage in the context of climate migration and displacement. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Loss and Damage

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Loss and Damage

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Loss and Damage

- Assess your current situation regarding loss and damage and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

13.9 Social Cohesion

Evidence from peer-reviewed research shows that Social Cohesion is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Social Cohesion

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Social Cohesion

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Social Cohesion

- Assess your current situation regarding social cohesion and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

13.10 Policy Responses

For individuals and communities, the practical implications of Policy Responses include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Policy Responses

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Policy Responses

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Policy Responses

- Assess your current situation regarding policy responses and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Migration and Displacement

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Migration and Displacement

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Migration and Displacement

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

Current Conditions and News: Migration and Displacement

- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.
- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.

Future Scenarios and Predictions: Migration and Displacement

Under a high-emissions scenario (SSP5-8.5), climate migration and displacement is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to climate migration and displacement.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to climate migration and displacement.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Migration and Displacement

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 13

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 13

- Vicedo-Cabrera, A.M. et al. (2021). Burden of heat-related mortality attributable to recent human-induced climate change. Nature Climate Change, 11, 492-500.
- Hsiang, S.M. & Burke, M. (2014). Climate, conflict, and social stability. Climatic Change, 123, 39-55.

Exercises — Chapter 13

- Reflect on how climate migration and displacement affects your own community and write a one-page plan.

Chapter 14: Conflict and Security

Theme: climate change and conflict

This section examines this chapter in the context of climate change and conflict. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of climate change and conflict. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

14.1 Resource Competition

Evidence from peer-reviewed research shows that Resource Competition is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Resource Competition

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Resource Competition

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Resource Competition

- Assess your current situation regarding resource competition and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

14.2 Climate and Violence Evidence

For individuals and communities, the practical implications of Climate and Violence Evidence include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Climate and Violence Evidence

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate and Violence Evidence

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Climate and Violence Evidence

- Assess your current situation regarding climate and violence evidence and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

14.3 Fragile States

Historical experience and recent disasters demonstrate that preparedness related to Fragile States reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Fragile States

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Fragile States

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Fragile States

- Assess your current situation regarding fragile states and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

14.4 Food and Water Conflict

Policy and governance play a decisive role in Food and Water Conflict. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Food and Water Conflict

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Food and Water Conflict

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Food and Water Conflict

- Assess your current situation regarding food and water conflict and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

14.5 Migration-Conflict Links

Technology can support Migration-Conflict Links, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Migration-Conflict Links

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Migration-Conflict Links

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Migration-Conflict Links

- Assess your current situation regarding migration-conflict links and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

14.6 Security Sector Responses

Education and training strengthen Security Sector Responses. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Security Sector Responses

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Security Sector Responses

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Security Sector Responses

- Assess your current situation regarding security sector responses and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

14.7 Peacebuilding and Adaptation

Monitoring and evaluation ensure that efforts related to Peacebuilding and Adaptation remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Peacebuilding and Adaptation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Peacebuilding and Adaptation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Peacebuilding and Adaptation

- Assess your current situation regarding peacebuilding and adaptation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

14.8 Early Warning for Conflict

This section examines Early Warning for Conflict in the context of climate change and conflict. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Early Warning for Conflict

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Early Warning for Conflict

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Early Warning for Conflict

- Assess your current situation regarding early warning for conflict and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

14.9 Governance and Stability

Evidence from peer-reviewed research shows that Governance and Stability is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Governance and Stability

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Governance and Stability

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Governance and Stability

- Assess your current situation regarding governance and stability and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

14.10 Reducing Risk

For individuals and communities, the practical implications of Reducing Risk include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Reducing Risk

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Reducing Risk

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Reducing Risk

- Assess your current situation regarding reducing risk and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Conflict and Security

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Conflict and Security

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Conflict and Security

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

Current Conditions and News: Conflict and Security

- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.
- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.

Future Scenarios and Predictions: Conflict and Security

Under a high-emissions scenario (SSP5-8.5), climate change and conflict is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to climate change and conflict.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to climate change and conflict.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Conflict and

Security

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 14

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 14

- Burke, M., Hsiang, S.M. & Miguel, E. (2015). Global non-linear effect of temperature on economic production. Nature, 527, 235-239.
- Lobell, D.B., Schlenker, W. & Costa-Roberts, J. (2011). Climate trends and global crop production since 1980. Science, 333(6042), 616-620.

Exercises — Chapter 14

- Reflect on how climate change and conflict affects your own community and write a one-page plan.

Chapter 15: Infrastructure Failure

Theme: infrastructure resilience

This section examines this chapter in the context of infrastructure resilience. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of infrastructure resilience. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

15.1 Critical Infrastructure

Evidence from peer-reviewed research shows that Critical Infrastructure is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Critical Infrastructure

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Critical Infrastructure

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Critical Infrastructure

- Assess your current situation regarding critical infrastructure and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

15.2 Power Grids

For individuals and communities, the practical implications of Power Grids include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Power Grids

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Power Grids

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Power Grids

- Assess your current situation regarding power grids and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

15.3 Water and Sanitation Systems

Historical experience and recent disasters demonstrate that preparedness related to Water and Sanitation Systems reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Water and Sanitation Systems

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water and Sanitation Systems

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water and Sanitation Systems

- Assess your current situation regarding water and sanitation systems and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

15.4 Transport Networks

Policy and governance play a decisive role in Transport Networks. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Transport Networks

- Assess local conditions and identify the most relevant hazards.

- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Transport Networks

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Transport Networks

- Assess your current situation regarding transport networks and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

15.5 Communication Systems

Technology can support Communication Systems, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Communication Systems

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Communication Systems

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Communication Systems

- Assess your current situation regarding communication systems and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

15.6 Building Standards

Education and training strengthen Building Standards. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Building Standards

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Building Standards

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Building Standards

- Assess your current situation regarding building standards and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

15.7 Cascading Failures

Monitoring and evaluation ensure that efforts related to Cascading Failures remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Cascading Failures

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cascading Failures

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cascading Failures

- Assess your current situation regarding cascading failures and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

15.8 Maintenance and Investment

This section examines Maintenance and Investment in the context of infrastructure resilience. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Maintenance and Investment

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Maintenance and Investment

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Maintenance and Investment

- Assess your current situation regarding maintenance and investment and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

15.9 Redundancy and Backup

Evidence from peer-reviewed research shows that Redundancy and Backup is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Redundancy and Backup

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Redundancy and Backup

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Redundancy and Backup

- Assess your current situation regarding redundancy and backup and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

15.10 Future-Proofing Infrastructure

For individuals and communities, the practical implications of Future-Proofing Infrastructure include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Future-Proofing Infrastructure

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Future-Proofing Infrastructure

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Future-Proofing Infrastructure

- Assess your current situation regarding future-proofing infrastructure and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Infrastructure Failure

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Infrastructure Failure

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Infrastructure Failure

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

Current Conditions and News: Infrastructure Failure

- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.
- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).

Future Scenarios and Predictions: Infrastructure Failure

Under a high-emissions scenario (SSP5-8.5), infrastructure resilience is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to infrastructure resilience.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to infrastructure resilience.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Infrastructure Failure

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 15

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 15

- Zhao, C. et al. (2017). Temperature increase reduces global yields of major crops. PNAS, 114(35), 9326-9331.
- Wheeler, T. & von Braun, J. (2013). Climate change impacts on global food security. Science, 341(6145), 508-513.

Exercises — Chapter 15

- Reflect on how infrastructure resilience affects your own community and write a one-page plan.

Chapter 16: Climate Literacy

Theme: climate literacy and education

This section examines this chapter in the context of climate literacy and education. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of climate literacy and education. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

16.1 What Is Climate Literacy

Evidence from peer-reviewed research shows that What Is Climate Literacy is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for What Is Climate Literacy

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: What Is Climate Literacy

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for What Is Climate Literacy

- Assess your current situation regarding what is climate literacy and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

16.2 Core Concepts

For individuals and communities, the practical implications of Core Concepts include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Core Concepts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Core Concepts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Core Concepts

- Assess your current situation regarding core concepts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

16.3 Age-Appropriate Education

Historical experience and recent disasters demonstrate that preparedness related to Age-Appropriate Education reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Age-Appropriate Education

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Age-Appropriate Education

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Age-Appropriate Education

- Assess your current situation regarding age-appropriate education and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

16.4 Formal Education

Policy and governance play a decisive role in Formal Education. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Formal Education

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Formal Education

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Formal Education

- Assess your current situation regarding formal education and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

16.5 Informal and Community Education

Technology can support Informal and Community Education, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Informal and Community Education

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Informal and Community Education

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Informal and Community Education

- Assess your current situation regarding informal and community education and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

16.6 Teacher Training

Education and training strengthen Teacher Training. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Teacher Training

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Teacher Training

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Teacher Training

- Assess your current situation regarding teacher training and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

16.7 Curriculum Design

Monitoring and evaluation ensure that efforts related to Curriculum Design remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Curriculum Design

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Curriculum Design

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Curriculum Design

- Assess your current situation regarding curriculum design and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

16.8 Assessment

This section examines Assessment in the context of climate literacy and education. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Assessment

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Assessment

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Assessment

- Assess your current situation regarding assessment and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

16.9 Lifelong Learning

Evidence from peer-reviewed research shows that Lifelong Learning is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Lifelong Learning

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Lifelong Learning

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Lifelong Learning

- Assess your current situation regarding lifelong learning and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

16.10 Measuring Literacy Outcomes

For individuals and communities, the practical implications of Measuring Literacy Outcomes include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Measuring Literacy Outcomes

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Measuring Literacy Outcomes

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Measuring Literacy Outcomes

- Assess your current situation regarding measuring literacy outcomes and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Climate Literacy

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate Literacy

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Climate Literacy

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

> Dr. Maria Neira, WHO: 'Climate change is the greatest health challenge of the 21st century.'

Current Conditions and News: Climate Literacy

- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).
- Coral bleaching events are becoming more frequent, threatening marine ecosystems.

Future Scenarios and Predictions: Climate Literacy

Under a high-emissions scenario (SSP5-8.5), climate literacy and education is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to climate literacy and education.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to climate literacy and education.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Climate Literacy

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 16

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 16

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Exercises — Chapter 16

- Reflect on how climate literacy and education affects your own community and write a one-page plan.

Chapter 17: Reference Paper Education

Theme: reference paper education and scientific literacy

This section examines this chapter in the context of reference paper education and scientific literacy. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of reference paper education and scientific literacy. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

17.1 What Is a Reference Paper

Evidence from peer-reviewed research shows that What Is a Reference Paper is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for What Is a Reference Paper

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: What Is a Reference Paper

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for What Is a Reference Paper

- Assess your current situation regarding what is a reference paper and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

17.2 Peer Review Process

For individuals and communities, the practical implications of Peer Review Process include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Peer Review Process

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Peer Review Process

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Peer Review Process

- Assess your current situation regarding peer review process and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

17.3 Reading Scientific Literature

Historical experience and recent disasters demonstrate that preparedness related to Reading Scientific Literature reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Reading Scientific Literature

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Reading Scientific Literature

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Reading Scientific Literature

- Assess your current situation regarding reading scientific literature and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

17.4 Evaluating Evidence

Policy and governance play a decisive role in Evaluating Evidence. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Evaluating Evidence

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Evaluating Evidence

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Evaluating Evidence

- Assess your current situation regarding evaluating evidence and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

17.5 Understanding Methods

Technology can support Understanding Methods, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Understanding Methods

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Understanding Methods

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Understanding Methods

- Assess your current situation regarding understanding methods and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

17.6 Statistics for Citizens

Education and training strengthen Statistics for Citizens. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Statistics for Citizens

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Statistics for Citizens

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Statistics for Citizens

- Assess your current situation regarding statistics for citizens and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

17.7 Identifying Bias

Monitoring and evaluation ensure that efforts related to Identifying Bias remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Identifying Bias

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Identifying Bias

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Identifying Bias

- Assess your current situation regarding identifying bias and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

17.8 Building a Knowledge Base

This section examines Building a Knowledge Base in the context of reference paper education and scientific literacy. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Building a Knowledge Base

- Assess local conditions and identify the most relevant hazards.

- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Building a Knowledge Base

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Building a Knowledge Base

- Assess your current situation regarding building a knowledge base and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

17.9 Sharing Knowledge

Evidence from peer-reviewed research shows that Sharing Knowledge is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Sharing Knowledge

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sharing Knowledge

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sharing Knowledge

- Assess your current situation regarding sharing knowledge and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

17.10 Teaching Others

For individuals and communities, the practical implications of Teaching Others include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Teaching Others

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Teaching Others

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Teaching Others

- Assess your current situation regarding teaching others and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Reference Paper Education

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Reference Paper Education

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Reference Paper Education

> Dr. Maria Neira, WHO: 'Climate change is the greatest health challenge of the 21st century.'

> Prof. Hans-Otto Portner, IPCC: 'The window for action is closing rapidly.'

Current Conditions and News: Reference Paper Education

- Coral bleaching events are becoming more frequent, threatening marine ecosystems.
- Arctic sea ice is declining at a rate of about 13% per decade.

Future Scenarios and Predictions: Reference Paper Education

Under a high-emissions scenario (SSP5-8.5), reference paper education and scientific literacy is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to reference paper education and scientific literacy.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to reference paper education and scientific literacy.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Reference Paper

Education

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 17

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 17

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- Obradovich, N. et al. (2018). Empirical evidence of mental health risks posed by climate change. PNAS, 115(43), 10953-10958.

Exercises — Chapter 17

- Reflect on how reference paper education and scientific literacy affects your own community and write a one-page plan.

Chapter 18: How to Read Scientific Literature

Theme: reading and understanding scientific papers

This section examines this chapter in the context of reading and understanding scientific papers. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of reading and understanding scientific papers. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

18.1 Paper Structure

Evidence from peer-reviewed research shows that Paper Structure is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Paper Structure

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Paper Structure

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Paper Structure

- Assess your current situation regarding paper structure and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

18.2 Abstract and Introduction

For individuals and communities, the practical implications of Abstract and Introduction include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Abstract and Introduction

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Abstract and Introduction

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Abstract and Introduction

- Assess your current situation regarding abstract and introduction and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

18.3 Methods Section

Historical experience and recent disasters demonstrate that preparedness related to Methods Section reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Methods Section

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Methods Section

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Methods Section

- Assess your current situation regarding methods section and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

18.4 Results and Figures

Policy and governance play a decisive role in Results and Figures. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Results and Figures

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Results and Figures

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Results and Figures

- Assess your current situation regarding results and figures and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

18.5 Discussion and Conclusions

Technology can support Discussion and Conclusions, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Discussion and Conclusions

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Discussion and Conclusions

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Discussion and Conclusions

- Assess your current situation regarding discussion and conclusions and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

18.6 References and Citations

Education and training strengthen References and Citations. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for References and Citations

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: References and Citations

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for References and Citations

- Assess your current situation regarding references and citations and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

18.7 Systematic Reviews and Meta-Analyses

Monitoring and evaluation ensure that efforts related to Systematic Reviews and Meta-Analyses remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Systematic Reviews and Meta-Analyses

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Systematic Reviews and Meta-Analyses

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Systematic Reviews and Meta-Analyses

- Assess your current situation regarding systematic reviews and meta-analyses and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

18.8 Preprints and Grey Literature

This section examines Preprints and Grey Literature in the context of reading and understanding scientific papers. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Preprints and Grey Literature

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Preprints and Grey Literature

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Preprints and Grey Literature

- Assess your current situation regarding preprints and grey literature and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

18.9 Critical Appraisal Tools

Evidence from peer-reviewed research shows that Critical Appraisal Tools is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Critical Appraisal Tools

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Critical Appraisal Tools

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Critical Appraisal Tools

- Assess your current situation regarding critical appraisal tools and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

18.10 Practical Reading Plan

For individuals and communities, the practical implications of Practical Reading Plan include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Practical Reading Plan

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Practical Reading Plan

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Practical Reading Plan

- Assess your current situation regarding practical reading plan and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: How to Read Scientific Literature

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.

- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: How to Read Scientific Literature

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: How to Read Scientific Literature

> Prof. Hans-Otto Portner, IPCC: 'The window for action is closing rapidly.'

> Dr. Debra Roberts, IPCC: 'Adaptation and mitigation must go hand in hand.'

Current Conditions and News: How to Read Scientific Literature

- Arctic sea ice is declining at a rate of about 13% per decade.
- Glaciers worldwide are losing mass, contributing to sea level rise.

Future Scenarios and Predictions: How to Read Scientific Literature

Under a high-emissions scenario (SSP5-8.5), reading and understanding scientific papers is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to reading and understanding scientific papers.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to reading and understanding scientific papers.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: How to Read Scientific Literature

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 18

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 18

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Exercises — Chapter 18

- Reflect on how reading and understanding scientific papers affects your own community and write a one-page plan.

Chapter 19: Data and Statistics for Citizens

Theme: data and statistics for citizens

This section examines this chapter in the context of data and statistics for citizens. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of data and statistics for citizens. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

19.1 Why Data Matters

Evidence from peer-reviewed research shows that Why Data Matters is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Why Data Matters

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Why Data Matters

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Why Data Matters

- Assess your current situation regarding why data matters and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

19.2 Types of Data

For individuals and communities, the practical implications of Types of Data include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Types of Data

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Types of Data

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Types of Data

- Assess your current situation regarding types of data and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

19.3 Descriptive Statistics

Historical experience and recent disasters demonstrate that preparedness related to Descriptive Statistics reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Descriptive Statistics

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Descriptive Statistics

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Descriptive Statistics

- Assess your current situation regarding descriptive statistics and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

19.4 Uncertainty and Error

Policy and governance play a decisive role in Uncertainty and Error. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Uncertainty and Error

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Uncertainty and Error

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Uncertainty and Error

- Assess your current situation regarding uncertainty and error and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

19.5 Correlation vs Causation

Technology can support Correlation vs Causation, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Correlation vs Causation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Correlation vs Causation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Correlation vs Causation

- Assess your current situation regarding correlation vs causation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

19.6 Graphs and Charts

Education and training strengthen Graphs and Charts. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Graphs and Charts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Graphs and Charts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Graphs and Charts

- Assess your current situation regarding graphs and charts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

19.7 Misleading Statistics

Monitoring and evaluation ensure that efforts related to Misleading Statistics remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Misleading Statistics

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Misleading Statistics

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Misleading Statistics

- Assess your current situation regarding misleading statistics and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

19.8 Open Data Sources

This section examines Open Data Sources in the context of data and statistics for citizens. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Open Data Sources

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Open Data Sources

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Open Data Sources

- Assess your current situation regarding open data sources and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

19.9 Personal Data Collection

Evidence from peer-reviewed research shows that Personal Data Collection is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Personal Data Collection

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Personal Data Collection

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Personal Data Collection

- Assess your current situation regarding personal data collection and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

19.10 Data-Driven Decisions

For individuals and communities, the practical implications of Data-Driven Decisions include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Data-Driven Decisions

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Data-Driven Decisions

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Data-Driven Decisions

- Assess your current situation regarding data-driven decisions and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Data and Statistics for Citizens

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Data and Statistics for Citizens

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Data and Statistics for Citizens

> Dr. Debra Roberts, IPCC: 'Adaptation and mitigation must go hand in hand.'

Current Conditions and News: Data and Statistics for Citizens

- Glaciers worldwide are losing mass, contributing to sea level rise.

Future Scenarios and Predictions: Data and Statistics for Citizens

Under a high-emissions scenario (SSP5-8.5), data and statistics for citizens is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to data and statistics for citizens.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to data and statistics for citizens.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Data and Statistics for Citizens

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 19

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 19

- Aldrich, D.P. & Meyer, M.A. (2015). Social capital and community resilience. American Behavioral Scientist, 59(2), 254-269.
- Berkes, F. & Ross, H. (2013). Community resilience: toward an integrated approach. Society & Natural Resources, 26(1), 5-20.

Exercises — Chapter 19

- Reflect on how data and statistics for citizens affects your own community and write a one-page plan.

Chapter 20: Misinformation and Media Literacy

Theme: misinformation and media literacy

This section examines this chapter in the context of misinformation and media literacy. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of misinformation and media literacy. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

20.1 Climate Misinformation

Evidence from peer-reviewed research shows that Climate Misinformation is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Climate Misinformation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate Misinformation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Climate Misinformation

- Assess your current situation regarding climate misinformation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

20.2 Sources of Misinformation

For individuals and communities, the practical implications of Sources of Misinformation include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Sources of Misinformation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sources of Misinformation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sources of Misinformation

- Assess your current situation regarding sources of misinformation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

20.3 Psychology of Belief

Historical experience and recent disasters demonstrate that preparedness related to Psychology of Belief reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Psychology of Belief

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Psychology of Belief

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Psychology of Belief

- Assess your current situation regarding psychology of belief and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

20.4 Fact-Checking

Policy and governance play a decisive role in Fact-Checking. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Fact-Checking

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Fact-Checking

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Fact-Checking

- Assess your current situation regarding fact-checking and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

20.5 Media Literacy Skills

Technology can support Media Literacy Skills, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Media Literacy Skills

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Media Literacy Skills

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Media Literacy Skills

- Assess your current situation regarding media literacy skills and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

20.6 Social Media Dynamics

Education and training strengthen Social Media Dynamics. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Social Media Dynamics

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Social Media Dynamics

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Social Media Dynamics

- Assess your current situation regarding social media dynamics and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

20.7 Trusted Sources

Monitoring and evaluation ensure that efforts related to Trusted Sources remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Trusted Sources

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Trusted Sources

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Trusted Sources

- Assess your current situation regarding trusted sources and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

20.8 Communicating Science

This section examines Communicating Science in the context of misinformation and media literacy. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Communicating Science

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Communicating Science

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Communicating Science

- Assess your current situation regarding communicating science and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

20.9 Countering Denial

Evidence from peer-reviewed research shows that Countering Denial is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Countering Denial

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Countering Denial

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Countering Denial

- Assess your current situation regarding countering denial and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

20.10 Building Trust

For individuals and communities, the practical implications of Building Trust include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Building Trust

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Building Trust

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Building Trust

- Assess your current situation regarding building trust and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Misinformation and Media Literacy

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Misinformation and Media Literacy

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Misinformation and Media Literacy

> Dr. James Hansen, climate scientist: 'The science is clear: continued emissions will drive dangerous warming.'

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

Current Conditions and News: Misinformation and Media Literacy

- 2023 was the hottest year on record, with global temperatures 1.48 C above pre-industrial levels (WMO, 2024).
- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.

Future Scenarios and Predictions: Misinformation and Media Literacy

Under a high-emissions scenario (SSP5-8.5), misinformation and media literacy is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to misinformation and media literacy.

- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to misinformation and media literacy.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Misinformation and Media Literacy

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 20

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 20

- Folke, C. et al. (2010). Resilience thinking. Ecology and Society, 15(4), 20.
- Oppenheimer, M. et al. (2019). Sea Level Rise and Implications for Low-Lying Islands, Coasts and Communities. SROCC Chapter 4.

Exercises — Chapter 20

- Reflect on how misinformation and media literacy affects your own community and write a one-page plan.

Chapter 21: Risk Assessment for Your Location

Theme: local risk assessment

This section examines this chapter in the context of local risk assessment. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of local risk assessment. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

21.1 Understanding Local Hazards

Evidence from peer-reviewed research shows that Understanding Local Hazards is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Understanding Local Hazards

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Understanding Local Hazards

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Understanding Local Hazards

- Assess your current situation regarding understanding local hazards and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

21.2 Hazard Mapping

For individuals and communities, the practical implications of Hazard Mapping include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Hazard Mapping

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Hazard Mapping

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Hazard Mapping

- Assess your current situation regarding hazard mapping and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

21.3 Exposure and Vulnerability

Historical experience and recent disasters demonstrate that preparedness related to Exposure and Vulnerability reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Exposure and Vulnerability

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Exposure and Vulnerability

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Exposure and Vulnerability

- Assess your current situation regarding exposure and vulnerability and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

21.4 Climate Projections for Your Region

Policy and governance play a decisive role in Climate Projections for Your Region. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Climate Projections for Your Region

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate Projections for Your Region

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Climate Projections for Your Region

- Assess your current situation regarding climate projections for your region and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

21.5 Personal Risk Matrix

Technology can support Personal Risk Matrix, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Personal Risk Matrix

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Personal Risk Matrix

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Personal Risk Matrix

- Assess your current situation regarding personal risk matrix and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

21.6 Household Risk Profile

Education and training strengthen Household Risk Profile. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Household Risk Profile

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Household Risk Profile

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Household Risk Profile

- Assess your current situation regarding household risk profile and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

21.7 Community Risk Assessment

Monitoring and evaluation ensure that efforts related to Community Risk Assessment remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Community Risk Assessment

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community Risk Assessment

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community Risk Assessment

- Assess your current situation regarding community risk assessment and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

21.8 Monitoring and Updating

This section examines Monitoring and Updating in the context of local risk assessment. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Monitoring and Updating

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Monitoring and Updating

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Monitoring and Updating

- Assess your current situation regarding monitoring and updating and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

21.9 Prioritizing Actions

Evidence from peer-reviewed research shows that Prioritizing Actions is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Prioritizing Actions

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Prioritizing Actions

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Prioritizing Actions

- Assess your current situation regarding prioritizing actions and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

21.10 Documenting Your Plan

For individuals and communities, the practical implications of Documenting Your Plan include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Documenting Your Plan

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Documenting Your Plan

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Documenting Your Plan

- Assess your current situation regarding documenting your plan and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

Case Study: Risk Assessment for Your Location

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Risk Assessment for Your Location

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Risk Assessment for Your Location

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

Current Conditions and News: Risk Assessment for Your Location

- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.
- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.

Future Scenarios and Predictions: Risk Assessment for Your Location

Under a high-emissions scenario (SSP5-8.5), local risk assessment is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to local risk assessment.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to local risk assessment.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Risk Assessment for Your Location

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?

- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 21

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 21

- Cisse, G. et al. (2022). Health, Wellbeing, and the Changing Structure of Communities. IPCC AR6 WG2 Chapter 7.
- WHO & UNICEF. (2023). Progress on Household Drinking Water, Sanitation and Hygiene 2000-2022. JMP.

Exercises — Chapter 21

- Reflect on how local risk assessment affects your own community and write a one-page plan.

Chapter 22: Water Security at Home

Theme: household water security

This section examines this chapter in the context of household water security. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of household water security. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

22.1 Daily Water Needs

Evidence from peer-reviewed research shows that Daily Water Needs is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Daily Water Needs

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Daily Water Needs

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Daily Water Needs

- Assess your current situation regarding daily water needs and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

22.2 Water Storage

For individuals and communities, the practical implications of Water Storage include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Water Storage

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water Storage

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water Storage

- Assess your current situation regarding water storage and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

22.3 Purification Methods

Historical experience and recent disasters demonstrate that preparedness related to Purification Methods reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Purification Methods

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Purification Methods

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Purification Methods

- Assess your current situation regarding purification methods and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

22.4 Rainwater Harvesting

Policy and governance play a decisive role in Rainwater Harvesting. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Rainwater Harvesting

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Rainwater Harvesting

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Rainwater Harvesting

- Assess your current situation regarding rainwater harvesting and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

22.5 Greywater Reuse

Technology can support Greywater Reuse, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Greywater Reuse

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Greywater Reuse

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Greywater Reuse

- Assess your current situation regarding greywater reuse and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

22.6 Well and Spring Protection

Education and training strengthen Well and Spring Protection. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Well and Spring Protection

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Well and Spring Protection

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Well and Spring Protection

- Assess your current situation regarding well and spring protection and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

22.7 Contamination Prevention

Monitoring and evaluation ensure that efforts related to Contamination Prevention remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Contamination Prevention

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Contamination Prevention

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Contamination Prevention

- Assess your current situation regarding contamination prevention and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

22.8 Water in Disasters

This section examines Water in Disasters in the context of household water security. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Water in Disasters

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water in Disasters

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water in Disasters

- Assess your current situation regarding water in disasters and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

22.9 Long-Term Water Planning

Evidence from peer-reviewed research shows that Long-Term Water Planning is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Long-Term Water Planning

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Long-Term Water Planning

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Long-Term Water Planning

- Assess your current situation regarding long-term water planning and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

22.10 Water Conservation Habits

For individuals and communities, the practical implications of Water Conservation Habits include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Water Conservation Habits

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water Conservation Habits

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water Conservation Habits

- Assess your current situation regarding water conservation habits and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Water Security at Home

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water Security at Home

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Water Security at Home

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

Current Conditions and News: Water Security at Home

- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.
- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.

Future Scenarios and Predictions: Water Security at Home

Under a high-emissions scenario (SSP5-8.5), household water security is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to household water security.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to household water security.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Water Security at Home

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 22

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 22

- UNEP. (2022). Emissions Gap Report 2022. UNEP, Nairobi.
- IEA. (2023). World Energy Outlook 2023. International Energy Agency, Paris.

Exercises — Chapter 22

- Reflect on how household water security affects your own community and write a one-page plan.

Chapter 23: Food Security at Home

Theme: household food security

This section examines this chapter in the context of household food security. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of household food security. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

23.1 Nutritional Needs

Evidence from peer-reviewed research shows that Nutritional Needs is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Nutritional Needs

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Nutritional Needs

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Nutritional Needs

- Assess your current situation regarding nutritional needs and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

23.2 Food Storage Principles

For individuals and communities, the practical implications of Food Storage Principles include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Food Storage Principles

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Food Storage Principles

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Food Storage Principles

- Assess your current situation regarding food storage principles and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

23.3 Non-Perishable Supplies

Historical experience and recent disasters demonstrate that preparedness related to Non-Perishable Supplies reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Non-Perishable Supplies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Non-Perishable Supplies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Non-Perishable Supplies

- Assess your current situation regarding non-perishable supplies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

23.4 Food Preservation

Policy and governance play a decisive role in Food Preservation. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Food Preservation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Food Preservation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Food Preservation

- Assess your current situation regarding food preservation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

23.5 Home Gardening

Technology can support Home Gardening, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Home Gardening

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Home Gardening

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Home Gardening

- Assess your current situation regarding home gardening and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

23.6 Community Gardens

Education and training strengthen Community Gardens. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Community Gardens

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community Gardens

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community Gardens

- Assess your current situation regarding community gardens and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

23.7 Reducing Food Waste

Monitoring and evaluation ensure that efforts related to Reducing Food Waste remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Reducing Food Waste

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Reducing Food Waste

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Reducing Food Waste

- Assess your current situation regarding reducing food waste and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

23.8 Dietary Adaptation

This section examines Dietary Adaptation in the context of household food security. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Dietary Adaptation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Dietary Adaptation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Dietary Adaptation

- Assess your current situation regarding dietary adaptation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

23.9 Cooking Without Power

Evidence from peer-reviewed research shows that Cooking Without Power is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Cooking Without Power

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cooking Without Power

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cooking Without Power

- Assess your current situation regarding cooking without power and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

23.10 Food Safety in Crises

For individuals and communities, the practical implications of Food Safety in Crises include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Food Safety in Crises

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Food Safety in Crises

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Food Safety in Crises

- Assess your current situation regarding food safety in crises and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Food Security at Home

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Food Security at Home

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Food Security at Home

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

Current Conditions and News: Food Security at Home

- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.
- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.

Future Scenarios and Predictions: Food Security at Home

Under a high-emissions scenario (SSP5-8.5), household food security is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to household food security.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to household food security.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Food Security at Home

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 23

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 23

- IRENA. (2023). World Energy Transitions Outlook 2023. IRENA, Abu Dhabi.
- FAO. (2021). Climate Change: Unpacking the Burden on Food Safety. FAO, Rome.

Exercises — Chapter 23

- Reflect on how household food security affects your own community and write a one-page plan.

Chapter 24: Shelter and Energy

Theme: shelter and energy resilience

This section examines this chapter in the context of shelter and energy resilience. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of shelter and energy resilience. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

24.1 Home Heat Resilience

Evidence from peer-reviewed research shows that Home Heat Resilience is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Home Heat Resilience

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Home Heat Resilience

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Home Heat Resilience

- Assess your current situation regarding home heat resilience and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

24.2 Insulation and Shading

For individuals and communities, the practical implications of Insulation and Shading include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Insulation and Shading

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Insulation and Shading

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Insulation and Shading

- Assess your current situation regarding insulation and shading and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

24.3 Ventilation

Historical experience and recent disasters demonstrate that preparedness related to Ventilation reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Ventilation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Ventilation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Ventilation

- Assess your current situation regarding ventilation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

24.4 Cooling Strategies

Policy and governance play a decisive role in Cooling Strategies. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Cooling Strategies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cooling Strategies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cooling Strategies

- Assess your current situation regarding cooling strategies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

24.5 Heating in Cold Snaps

Technology can support Heating in Cold Snaps, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Heating in Cold Snaps

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Heating in Cold Snaps

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Heating in Cold Snaps

- Assess your current situation regarding heating in cold snaps and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

24.6 Power Outage Planning

Education and training strengthen Power Outage Planning. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Power Outage Planning

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Power Outage Planning

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Power Outage Planning

- Assess your current situation regarding power outage planning and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

24.7 Solar and Battery

Monitoring and evaluation ensure that efforts related to Solar and Battery remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Solar and Battery

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Solar and Battery

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Solar and Battery

- Assess your current situation regarding solar and battery and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

24.8 Generator Safety

This section examines Generator Safety in the context of shelter and energy resilience. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Generator Safety

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Generator Safety

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Generator Safety

- Assess your current situation regarding generator safety and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

24.9 Evacuation Sheltering

Evidence from peer-reviewed research shows that Evacuation Sheltering is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Evacuation Sheltering

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Evacuation Sheltering

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Evacuation Sheltering

- Assess your current situation regarding evacuation sheltering and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

24.10 Long-Term Home Upgrades

For individuals and communities, the practical implications of Long-Term Home Upgrades include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Long-Term Home Upgrades

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Long-Term Home Upgrades

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Long-Term Home Upgrades

- Assess your current situation regarding long-term home upgrades and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Shelter and Energy

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Shelter and Energy

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Shelter and Energy

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

Current Conditions and News: Shelter and Energy

- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.
- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.

Future Scenarios and Predictions: Shelter and Energy

Under a high-emissions scenario (SSP5-8.5), shelter and energy resilience is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to shelter and energy resilience.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to shelter and energy resilience.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Shelter and Energy

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 24

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 24

- World Bank. (2022). What You Need to Know About Food Security and Climate Change. World Bank, Washington DC.
- UNDP. (2022). Human Development Report 2021/22. UNDP, New York.

Exercises — Chapter 24

- Reflect on how shelter and energy resilience affects your own community and write a one-page plan.

Chapter 25: Health and First Aid

Theme: health and first aid preparedness

This section examines this chapter in the context of health and first aid preparedness. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of health and first aid preparedness. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

25.1 First Aid Basics

Evidence from peer-reviewed research shows that First Aid Basics is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for First Aid Basics

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: First Aid Basics

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for First Aid Basics

- Assess your current situation regarding first aid basics and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

25.2 Heat Illness Recognition

For individuals and communities, the practical implications of Heat Illness Recognition include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Heat Illness Recognition

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Heat Illness Recognition

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Heat Illness Recognition

- Assess your current situation regarding heat illness recognition and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

25.3 Dehydration and Rehydration

Historical experience and recent disasters demonstrate that preparedness related to Dehydration and Rehydration reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Dehydration and Rehydration

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Dehydration and Rehydration

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Dehydration and Rehydration

- Assess your current situation regarding dehydration and rehydration and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

25.4 Wound Care

Policy and governance play a decisive role in Wound Care. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Wound Care

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Wound Care

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Wound Care

- Assess your current situation regarding wound care and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

25.5 Infection Control

Technology can support Infection Control, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Infection Control

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Infection Control

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Infection Control

- Assess your current situation regarding infection control and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

25.6 Medication Management

Education and training strengthen Medication Management. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Medication Management

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Medication Management

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Medication Management

- Assess your current situation regarding medication management and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

25.7 Chronic Disease Planning

Monitoring and evaluation ensure that efforts related to Chronic Disease Planning remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Chronic Disease Planning

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Chronic Disease Planning

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Chronic Disease Planning

- Assess your current situation regarding chronic disease planning and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

25.8 Mental Health First Aid

This section examines Mental Health First Aid in the context of health and first aid preparedness. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Mental Health First Aid

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Mental Health First Aid

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Mental Health First Aid

- Assess your current situation regarding mental health first aid and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

25.9 Emergency Medical Services

Evidence from peer-reviewed research shows that Emergency Medical Services is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Emergency Medical Services

- Assess local conditions and identify the most relevant hazards.

- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Emergency Medical Services

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Emergency Medical Services

- Assess your current situation regarding emergency medical services and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

25.10 Health Kits and Supplies

For individuals and communities, the practical implications of Health Kits and Supplies include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Health Kits and Supplies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Health Kits and Supplies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Health Kits and Supplies

- Assess your current situation regarding health kits and supplies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Health and First Aid

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Health and First Aid

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Health and First Aid

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

Current Conditions and News: Health and First Aid

- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.
- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).

Future Scenarios and Predictions: Health and First Aid

Under a high-emissions scenario (SSP5-8.5), health and first aid preparedness is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to health and first aid preparedness.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to health and first aid preparedness.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Health and First Aid

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 25

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 25

- IFRC. (2022). World Disasters Report 2022. IFRC, Geneva.
- Red Cross. (2020). Heat Waves: A Silent Killer. IFRC, Geneva.

Exercises — Chapter 25

- Reflect on how health and first aid preparedness affects your own community and write a one-page plan.

Chapter 26: Financial Preparedness

Theme: financial preparedness

This section examines this chapter in the context of financial preparedness. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of financial preparedness. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

26.1 Emergency Funds

Evidence from peer-reviewed research shows that Emergency Funds is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Emergency Funds

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Emergency Funds

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Emergency Funds

- Assess your current situation regarding emergency funds and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

26.2 Income Diversification

For individuals and communities, the practical implications of Income Diversification include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Income Diversification

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Income Diversification

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Income Diversification

- Assess your current situation regarding income diversification and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

26.3 Insurance Review

Historical experience and recent disasters demonstrate that preparedness related to Insurance Review reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Insurance Review

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Insurance Review

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Insurance Review

- Assess your current situation regarding insurance review and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

26.4 Document Protection

Policy and governance play a decisive role in Document Protection. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Document Protection

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Document Protection

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Document Protection

- Assess your current situation regarding document protection and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

26.5 Debt and Cash Flow

Technology can support Debt and Cash Flow, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Debt and Cash Flow

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Debt and Cash Flow

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Debt and Cash Flow

- Assess your current situation regarding debt and cash flow and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

26.6 Barter and Local Exchange

Education and training strengthen Barter and Local Exchange. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Barter and Local Exchange

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Barter and Local Exchange

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Barter and Local Exchange

- Assess your current situation regarding barter and local exchange and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

26.7 Retirement and Long-Term

Monitoring and evaluation ensure that efforts related to Retirement and Long-Term remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Retirement and Long-Term

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Retirement and Long-Term

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Retirement and Long-Term

- Assess your current situation regarding retirement and long-term and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

26.8 Fraud Prevention

This section examines Fraud Prevention in the context of financial preparedness. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Fraud Prevention

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Fraud Prevention

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Fraud Prevention

- Assess your current situation regarding fraud prevention and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

26.9 Financial Aid and Relief

Evidence from peer-reviewed research shows that Financial Aid and Relief is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Financial Aid and Relief

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Financial Aid and Relief

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Financial Aid and Relief

- Assess your current situation regarding financial aid and relief and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

26.10 Rebuilding Finances After Disaster

For individuals and communities, the practical implications of Rebuilding Finances After Disaster include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Rebuilding Finances After Disaster

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Rebuilding Finances After Disaster

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Rebuilding Finances After Disaster

- Assess your current situation regarding rebuilding finances after disaster and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Financial Preparedness

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Financial Preparedness

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Financial Preparedness

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

> Dr. Maria Neira, WHO: 'Climate change is the greatest health challenge of the 21st century.'

Current Conditions and News: Financial Preparedness

- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).
- Coral bleaching events are becoming more frequent, threatening marine ecosystems.

Future Scenarios and Predictions: Financial Preparedness

Under a high-emissions scenario (SSP5-8.5), financial preparedness is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to financial preparedness.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to financial preparedness.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Financial Preparedness

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 26

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 26

- WMO. (2022). Early Warnings for All. WMO, Geneva.
- UNESCO. (2021). Getting Every School Climate-Ready. UNESCO, Paris.

Exercises — Chapter 26

- Reflect on how financial preparedness affects your own community and write a one-page plan.

Chapter 27: Evacuation and Emergency Response

Theme: evacuation and emergency response

This section examines this chapter in the context of evacuation and emergency response. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of evacuation and emergency response. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

27.1 When to Evacuate

Evidence from peer-reviewed research shows that When to Evacuate is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for When to Evacuate

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.

- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: When to Evacuate

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for When to Evacuate

- Assess your current situation regarding when to evacuate and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

27.2 Evacuation Routes and Plans

For individuals and communities, the practical implications of Evacuation Routes and Plans include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Evacuation Routes and Plans

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Evacuation Routes and Plans

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Evacuation Routes and Plans

- Assess your current situation regarding evacuation routes and plans and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

27.3 Go-Bags and Kits

Historical experience and recent disasters demonstrate that preparedness related to Go-Bags and Kits reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Go-Bags and Kits

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Go-Bags and Kits

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Go-Bags and Kits

- Assess your current situation regarding go-bags and kits and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

27.4 Family Communication

Policy and governance play a decisive role in Family Communication. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Family Communication

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Family Communication

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Family Communication

- Assess your current situation regarding family communication and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

27.5 Pets and Livestock

Technology can support Pets and Livestock, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Pets and Livestock

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Pets and Livestock

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Pets and Livestock

- Assess your current situation regarding pets and livestock and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

27.6 Sheltering in Place

Education and training strengthen Sheltering in Place. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Sheltering in Place

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sheltering in Place

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sheltering in Place

- Assess your current situation regarding sheltering in place and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

27.7 Returning Home Safely

Monitoring and evaluation ensure that efforts related to Returning Home Safely remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Returning Home Safely

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Returning Home Safely

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Returning Home Safely

- Assess your current situation regarding returning home safely and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

27.8 Post-Disaster Safety

This section examines Post-Disaster Safety in the context of evacuation and emergency response. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Post-Disaster Safety

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Post-Disaster Safety

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Post-Disaster Safety

- Assess your current situation regarding post-disaster safety and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

27.9 Psychological First Aid

Evidence from peer-reviewed research shows that Psychological First Aid is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Psychological First Aid

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Psychological First Aid

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Psychological First Aid

- Assess your current situation regarding psychological first aid and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

27.10 Learning from Drills

For individuals and communities, the practical implications of Learning from Drills include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Learning from Drills

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Learning from Drills

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Learning from Drills

- Assess your current situation regarding learning from drills and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Evacuation and Emergency Response

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Evacuation and Emergency Response

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Evacuation and Emergency Response

> Dr. Maria Neira, WHO: 'Climate change is the greatest health challenge of the 21st century.'

> Prof. Hans-Otto Portner, IPCC: 'The window for action is closing rapidly.'

Current Conditions and News: Evacuation and Emergency Response

- Coral bleaching events are becoming more frequent, threatening marine ecosystems.
- Arctic sea ice is declining at a rate of about 13% per decade.

Future Scenarios and Predictions: Evacuation and Emergency Response

Under a high-emissions scenario (SSP5-8.5), evacuation and emergency response is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to evacuation and emergency response.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to evacuation and emergency response.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Evacuation and Emergency Response

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 27

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 27

- UNESCO. (2020). Education for Sustainable Development: A Roadmap. UNESCO, Paris.
- OECD. (2021). Managing Climate Risks, Facing Essential Challenges. OECD, Paris.

Exercises — Chapter 27

- Reflect on how evacuation and emergency response affects your own community and write a one-page plan.

Chapter 28: Special Populations

Theme: protecting vulnerable and special populations

This section examines this chapter in the context of protecting vulnerable and special populations. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of protecting vulnerable and special populations. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

28.1 Children and Infants

Evidence from peer-reviewed research shows that Children and Infants is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Children and Infants

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Children and Infants

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Children and Infants

- Assess your current situation regarding children and infants and identify gaps.

- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

28.2 Elderly

For individuals and communities, the practical implications of Elderly include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Elderly

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Elderly

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Elderly

- Assess your current situation regarding elderly and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

28.3 People with Disabilities

Historical experience and recent disasters demonstrate that preparedness related to People with Disabilities reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for People with Disabilities

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: People with Disabilities

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for People with Disabilities

- Assess your current situation regarding people with disabilities and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

28.4 Chronic Illness

Policy and governance play a decisive role in Chronic Illness. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Chronic Illness

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Chronic Illness

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Chronic Illness

- Assess your current situation regarding chronic illness and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

28.5 Pregnant and Nursing

Technology can support Pregnant and Nursing, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Pregnant and Nursing

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Pregnant and Nursing

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Pregnant and Nursing

- Assess your current situation regarding pregnant and nursing and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

28.6 Low-Income Households

Education and training strengthen Low-Income Households. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Low-Income Households

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Low-Income Households

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Low-Income Households

- Assess your current situation regarding low-income households and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

28.7 Migrants and Refugees

Monitoring and evaluation ensure that efforts related to Migrants and Refugees remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Migrants and Refugees

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Migrants and Refugees

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Migrants and Refugees

- Assess your current situation regarding migrants and refugees and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

28.8 Indigenous Communities

This section examines Indigenous Communities in the context of protecting vulnerable and special populations. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Indigenous Communities

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Indigenous Communities

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Indigenous Communities

- Assess your current situation regarding indigenous communities and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

28.9 Rural Isolation

Evidence from peer-reviewed research shows that Rural Isolation is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Rural Isolation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Rural Isolation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Rural Isolation

- Assess your current situation regarding rural isolation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

28.10 Tailored Preparedness Plans

For individuals and communities, the practical implications of Tailored Preparedness Plans include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Tailored Preparedness Plans

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Tailored Preparedness Plans

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Tailored Preparedness Plans

- Assess your current situation regarding tailored preparedness plans and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Special Populations

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Special Populations

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Special Populations

> Prof. Hans-Otto Portner, IPCC: 'The window for action is closing rapidly.'

> Dr. Debra Roberts, IPCC: 'Adaptation and mitigation must go hand in hand.'

Current Conditions and News: Special Populations

- Arctic sea ice is declining at a rate of about 13% per decade.
- Glaciers worldwide are losing mass, contributing to sea level rise.

Future Scenarios and Predictions: Special Populations

Under a high-emissions scenario (SSP5-8.5), protecting vulnerable and special populations is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to protecting vulnerable and special populations.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to protecting vulnerable and special populations.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Special Populations

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 28

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 28

- OECD. (2023). Climate Finance Provided and Mobilised by Developed Countries. OECD, Paris.
- IMF. (2022). World Economic Outlook: War Sets Back Global Recovery. IMF, Washington DC.

Exercises — Chapter 28

- Reflect on how protecting vulnerable and special populations affects your own community and write a one-page plan.

Chapter 29: Social Capital and Networks

Theme: social capital and community networks

This section examines this chapter in the context of social capital and community networks. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of social capital and community networks. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

29.1 What Is Social Capital

Evidence from peer-reviewed research shows that What Is Social Capital is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for What Is Social Capital

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: What Is Social Capital

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for What Is Social Capital

- Assess your current situation regarding what is social capital and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

29.2 Trust and Reciprocity

For individuals and communities, the practical implications of Trust and Reciprocity include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Trust and Reciprocity

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Trust and Reciprocity

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Trust and Reciprocity

- Assess your current situation regarding trust and reciprocity and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

29.3 Network Mapping

Historical experience and recent disasters demonstrate that preparedness related to Network Mapping reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Network Mapping

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Network Mapping

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Network Mapping

- Assess your current situation regarding network mapping and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

29.4 Bridging and Bonding

Policy and governance play a decisive role in Bridging and Bonding. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Bridging and Bonding

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Bridging and Bonding

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Bridging and Bonding

- Assess your current situation regarding bridging and bonding and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

29.5 Community Leaders

Technology can support Community Leaders, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Community Leaders

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community Leaders

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community Leaders

- Assess your current situation regarding community leaders and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

29.6 Inclusive Participation

Education and training strengthen Inclusive Participation. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Inclusive Participation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Inclusive Participation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Inclusive Participation

- Assess your current situation regarding inclusive participation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

29.7 Digital Networks

Monitoring and evaluation ensure that efforts related to Digital Networks remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Digital Networks

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Digital Networks

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Digital Networks

- Assess your current situation regarding digital networks and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

29.8 Maintaining Networks

This section examines Maintaining Networks in the context of social capital and community networks. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Maintaining Networks

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Maintaining Networks

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Maintaining Networks

- Assess your current situation regarding maintaining networks and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

29.9 Measuring Social Capital

Evidence from peer-reviewed research shows that Measuring Social Capital is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Measuring Social Capital

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Measuring Social Capital

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Measuring Social Capital

- Assess your current situation regarding measuring social capital and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

29.10 Strengthening Ties

For individuals and communities, the practical implications of Strengthening Ties include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Strengthening Ties

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Strengthening Ties

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Strengthening Ties

- Assess your current situation regarding strengthening ties and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Social Capital and Networks

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Social Capital and Networks

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Social Capital and Networks

> Dr. Debra Roberts, IPCC: 'Adaptation and mitigation must go hand in hand.'

Current Conditions and News: Social Capital and Networks

- Glaciers worldwide are losing mass, contributing to sea level rise.

Future Scenarios and Predictions: Social Capital and Networks

Under a high-emissions scenario (SSP5-8.5), social capital and community networks is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to social capital and community networks.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to social capital and community networks.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Social Capital and Networks

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 29

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 29

- IPCC. (2023). Climate Change 2023: Synthesis Report Summary for Policymakers. IPCC, Geneva.
- IPCC. (2023). Climate Change 2023: Synthesis Report. IPCC, Geneva.

Exercises — Chapter 29

- Reflect on how social capital and community networks affects your own community and write a one-page plan.

Chapter 30: Mutual Aid

Theme: mutual aid and community support

This section examines this chapter in the context of mutual aid and community support. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of mutual aid and community support. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

30.1 Principles of Mutual Aid

Evidence from peer-reviewed research shows that Principles of Mutual Aid is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Principles of Mutual Aid

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Principles of Mutual Aid

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Principles of Mutual Aid

- Assess your current situation regarding principles of mutual aid and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

30.2 Starting a Group

For individuals and communities, the practical implications of Starting a Group include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Starting a Group

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Starting a Group

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Starting a Group

- Assess your current situation regarding starting a group and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

30.3 Resource Sharing

Historical experience and recent disasters demonstrate that preparedness related to Resource Sharing reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Resource Sharing

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Resource Sharing

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Resource Sharing

- Assess your current situation regarding resource sharing and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

30.4 Care Networks

Policy and governance play a decisive role in Care Networks. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Care Networks

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Care Networks

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Care Networks

- Assess your current situation regarding care networks and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

30.5 Skill Sharing

Technology can support Skill Sharing, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Skill Sharing

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Skill Sharing

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Skill Sharing

- Assess your current situation regarding skill sharing and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

30.6 Food and Supply Distribution

Education and training strengthen Food and Supply Distribution. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Food and Supply Distribution

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Food and Supply Distribution

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Food and Supply Distribution

- Assess your current situation regarding food and supply distribution and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

30.7 Communication Systems

Monitoring and evaluation ensure that efforts related to Communication Systems remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Communication Systems

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Communication Systems

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Communication Systems

- Assess your current situation regarding communication systems and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

30.8 Conflict Resolution

This section examines Conflict Resolution in the context of mutual aid and community support. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Conflict Resolution

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Conflict Resolution

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Conflict Resolution

- Assess your current situation regarding conflict resolution and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

30.9 Sustainability of Groups

Evidence from peer-reviewed research shows that Sustainability of Groups is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Sustainability of Groups

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sustainability of Groups

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sustainability of Groups

- Assess your current situation regarding sustainability of groups and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

30.10 Mutual Aid and Official Response

For individuals and communities, the practical implications of Mutual Aid and Official Response include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Mutual Aid and Official Response

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Mutual Aid and Official Response

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Mutual Aid and Official Response

- Assess your current situation regarding mutual aid and official response and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Mutual Aid

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Mutual Aid

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Mutual Aid

> Dr. James Hansen, climate scientist: 'The science is clear: continued emissions will drive dangerous warming.'

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

Current Conditions and News: Mutual Aid

- 2023 was the hottest year on record, with global temperatures 1.48 C above pre-industrial levels (WMO, 2024).
- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.

Future Scenarios and Predictions: Mutual Aid

Under a high-emissions scenario (SSP5-8.5), mutual aid and community support is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to mutual aid and community support.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to mutual aid and community support.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Mutual Aid

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 30

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 30

- IPCC. (2021). Climate Change 2021: The Physical Science Basis (AR6 WG1). Cambridge University Press.
- IPCC. (2022). Climate Change 2022: Impacts, Adaptation and Vulnerability (AR6 WG2). Cambridge University Press.

Exercises — Chapter 30

- Reflect on how mutual aid and community support affects your own community and write a one-page plan.

Chapter 31: Local Governance

Theme: local governance and climate action

This section examines this chapter in the context of local governance and climate action. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of local governance and climate action. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

31.1 Role of Local Government

Evidence from peer-reviewed research shows that Role of Local Government is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Role of Local Government

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Role of Local Government

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Role of Local Government

- Assess your current situation regarding role of local government and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

31.2 Emergency Management

For individuals and communities, the practical implications of Emergency Management include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Emergency Management

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Emergency Management

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Emergency Management

- Assess your current situation regarding emergency management and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

31.3 Climate Adaptation Plans

Historical experience and recent disasters demonstrate that preparedness related to Climate Adaptation Plans reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Climate Adaptation Plans

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate Adaptation Plans

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Climate Adaptation Plans

- Assess your current situation regarding climate adaptation plans and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

31.4 Zoning and Land Use

Policy and governance play a decisive role in Zoning and Land Use. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Zoning and Land Use

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Zoning and Land Use

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Zoning and Land Use

- Assess your current situation regarding zoning and land use and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

31.5 Public Services

Technology can support Public Services, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Public Services

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Public Services

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Public Services

- Assess your current situation regarding public services and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

31.6 Budget and Finance

Education and training strengthen Budget and Finance. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Budget and Finance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Budget and Finance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Budget and Finance

- Assess your current situation regarding budget and finance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

31.7 Public Engagement

Monitoring and evaluation ensure that efforts related to Public Engagement remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Public Engagement

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Public Engagement

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Public Engagement

- Assess your current situation regarding public engagement and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

31.8 Accountability

This section examines Accountability in the context of local governance and climate action. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Accountability

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Accountability

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Accountability

- Assess your current situation regarding accountability and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

31.9 Partnerships

Evidence from peer-reviewed research shows that Partnerships is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Partnerships

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Partnerships

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Partnerships

- Assess your current situation regarding partnerships and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

31.10 Innovation at Local Level

For individuals and communities, the practical implications of Innovation at Local Level include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Innovation at Local Level

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Innovation at Local Level

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Innovation at Local Level

- Assess your current situation regarding innovation at local level and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Local Governance

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Local Governance

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Local Governance

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

Current Conditions and News: Local Governance

- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.
- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.

Future Scenarios and Predictions: Local Governance

Under a high-emissions scenario (SSP5-8.5), local governance and climate action is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to local governance and climate action.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to local governance and climate action.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Local Governance

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 31

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 31

- IPCC. (2022). Climate Change 2022: Mitigation of Climate Change (AR6 WG3). Cambridge University Press.
- IPCC. (2018). Global Warming of 1.5 C (SR15). IPCC, Geneva.

Exercises — Chapter 31

- Reflect on how local governance and climate action affects your own community and write a one-page plan.

Chapter 32: Urban Resilience

Theme: urban resilience to climate change

This section examines this chapter in the context of urban resilience to climate change. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of urban resilience to climate change. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

32.1 Urban Heat Islands

Evidence from peer-reviewed research shows that Urban Heat Islands is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Urban Heat Islands

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Urban Heat Islands

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Urban Heat Islands

- Assess your current situation regarding urban heat islands and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

32.2 Green Infrastructure

For individuals and communities, the practical implications of Green Infrastructure include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Green Infrastructure

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Green Infrastructure

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Green Infrastructure

- Assess your current situation regarding green infrastructure and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

32.3 Stormwater Management

Historical experience and recent disasters demonstrate that preparedness related to Stormwater Management reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Stormwater Management

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Stormwater Management

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Stormwater Management

- Assess your current situation regarding stormwater management and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

32.4 Urban Agriculture

Policy and governance play a decisive role in Urban Agriculture. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Urban Agriculture

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Urban Agriculture

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Urban Agriculture

- Assess your current situation regarding urban agriculture and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

32.5 Transport and Mobility

Technology can support Transport and Mobility, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Transport and Mobility

- Assess local conditions and identify the most relevant hazards.

- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Transport and Mobility

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Transport and Mobility

- Assess your current situation regarding transport and mobility and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

32.6 Housing and Buildings

Education and training strengthen Housing and Buildings. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Housing and Buildings

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Housing and Buildings

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Housing and Buildings

- Assess your current situation regarding housing and buildings and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

32.7 Energy Systems

Monitoring and evaluation ensure that efforts related to Energy Systems remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Energy Systems

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Energy Systems

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Energy Systems

- Assess your current situation regarding energy systems and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

32.8 Waste and Sanitation

This section examines Waste and Sanitation in the context of urban resilience to climate change. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Waste and Sanitation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Waste and Sanitation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Waste and Sanitation

- Assess your current situation regarding waste and sanitation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

32.9 Equity in Cities

Evidence from peer-reviewed research shows that Equity in Cities is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Equity in Cities

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Equity in Cities

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Equity in Cities

- Assess your current situation regarding equity in cities and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

32.10 Smart Cities and Data

For individuals and communities, the practical implications of Smart Cities and Data include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Smart Cities and Data

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Smart Cities and Data

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Smart Cities and Data

- Assess your current situation regarding smart cities and data and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Urban Resilience

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Urban Resilience

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Urban Resilience

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

Current Conditions and News: Urban Resilience

- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.
- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.

Future Scenarios and Predictions: Urban Resilience

Under a high-emissions scenario (SSP5-8.5), urban resilience to climate change is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to urban resilience to climate change.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to urban resilience to climate change.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Urban Resilience

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 32

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 32

- IPCC. (2019). Special Report on the Ocean and Cryosphere in a Changing Climate (SROCC). IPCC, Geneva.
- IPCC. (2019). Climate Change and Land (SRCCL). IPCC, Geneva.

Exercises — Chapter 32

- Reflect on how urban resilience to climate change affects your own community and write a one-page plan.

Chapter 33: Rural Resilience

Theme: rural resilience

This section examines this chapter in the context of rural resilience. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of rural resilience. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

33.1 Rural Vulnerabilities

Evidence from peer-reviewed research shows that Rural Vulnerabilities is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Rural Vulnerabilities

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Rural Vulnerabilities

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Rural Vulnerabilities

- Assess your current situation regarding rural vulnerabilities and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

33.2 Agricultural Livelihoods

For individuals and communities, the practical implications of Agricultural Livelihoods include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Agricultural Livelihoods

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Agricultural Livelihoods

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Agricultural Livelihoods

- Assess your current situation regarding agricultural livelihoods and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

33.3 Water and Sanitation

Historical experience and recent disasters demonstrate that preparedness related to Water and Sanitation reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Water and Sanitation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water and Sanitation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water and Sanitation

- Assess your current situation regarding water and sanitation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

33.4 Healthcare Access

Policy and governance play a decisive role in Healthcare Access. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Healthcare Access

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Healthcare Access

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Healthcare Access

- Assess your current situation regarding healthcare access and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

33.5 Connectivity and Transport

Technology can support Connectivity and Transport, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Connectivity and Transport

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Connectivity and Transport

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Connectivity and Transport

- Assess your current situation regarding connectivity and transport and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

33.6 Economic Diversification

Education and training strengthen Economic Diversification. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Economic Diversification

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Economic Diversification

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Economic Diversification

- Assess your current situation regarding economic diversification and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

33.7 Land Management

Monitoring and evaluation ensure that efforts related to Land Management remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Land Management

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Land Management

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Land Management

- Assess your current situation regarding land management and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

33.8 Community Institutions

This section examines Community Institutions in the context of rural resilience. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Community Institutions

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community Institutions

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community Institutions

- Assess your current situation regarding community institutions and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

33.9 Indigenous and Traditional Knowledge

Evidence from peer-reviewed research shows that Indigenous and Traditional Knowledge is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Indigenous and Traditional Knowledge

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Indigenous and Traditional Knowledge

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Indigenous and Traditional Knowledge

- Assess your current situation regarding indigenous and traditional knowledge and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

33.10 Bridging Urban-Rural Gaps

For individuals and communities, the practical implications of Bridging Urban-Rural Gaps include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Bridging Urban-Rural Gaps

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Bridging Urban-Rural Gaps

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Bridging Urban-Rural Gaps

- Assess your current situation regarding bridging urban-rural gaps and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Rural Resilience

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Rural Resilience

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Rural Resilience

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

Current Conditions and News: Rural Resilience

- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.
- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.

Future Scenarios and Predictions: Rural Resilience

Under a high-emissions scenario (SSP5-8.5), rural resilience is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to rural resilience.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to rural resilience.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Rural Resilience

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 33

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 33

- NASA. (2024). Global Climate Change: Vital Signs. climate.nasa.gov.
- NOAA. (2024). State of the Climate. ncei.noaa.gov.

Exercises — Chapter 33

- Reflect on how rural resilience affects your own community and write a one-page plan.

Chapter 34: Indigenous Knowledge

Theme: indigenous and traditional knowledge

This section examines this chapter in the context of indigenous and traditional knowledge. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of indigenous and traditional knowledge. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

34.1 Value of Indigenous Knowledge

Evidence from peer-reviewed research shows that Value of Indigenous Knowledge is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Value of Indigenous Knowledge

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Value of Indigenous Knowledge

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Value of Indigenous Knowledge

- Assess your current situation regarding value of indigenous knowledge and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

34.2 Traditional Ecological Knowledge

For individuals and communities, the practical implications of Traditional Ecological Knowledge include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Traditional Ecological Knowledge

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Traditional Ecological Knowledge

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Traditional Ecological Knowledge

- Assess your current situation regarding traditional ecological knowledge and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

34.3 Weather and Climate Indicators

Historical experience and recent disasters demonstrate that preparedness related to Weather and Climate Indicators reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Weather and Climate Indicators

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Weather and Climate Indicators

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Weather and Climate Indicators

- Assess your current situation regarding weather and climate indicators and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

34.4 Sustainable Resource Management

Policy and governance play a decisive role in Sustainable Resource Management. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Sustainable Resource Management

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sustainable Resource Management

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sustainable Resource Management

- Assess your current situation regarding sustainable resource management and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

34.5 Cultural Resilience

Technology can support Cultural Resilience, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Cultural Resilience

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cultural Resilience

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cultural Resilience

- Assess your current situation regarding cultural resilience and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

34.6 Knowledge Transmission

Education and training strengthen Knowledge Transmission. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Knowledge Transmission

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Knowledge Transmission

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Knowledge Transmission

- Assess your current situation regarding knowledge transmission and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

34.7 Partnerships with Indigenous Peoples

Monitoring and evaluation ensure that efforts related to Partnerships with Indigenous Peoples remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Partnerships with Indigenous Peoples

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Partnerships with Indigenous Peoples

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Partnerships with Indigenous Peoples

- Assess your current situation regarding partnerships with indigenous peoples and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

34.8 Protecting Knowledge

This section examines Protecting Knowledge in the context of indigenous and traditional knowledge. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Protecting Knowledge

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Protecting Knowledge

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Protecting Knowledge

- Assess your current situation regarding protecting knowledge and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

34.9 Case Studies

Evidence from peer-reviewed research shows that Case Studies is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Case Studies

- Assess local conditions and identify the most relevant hazards.

- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Case Studies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Case Studies

- Assess your current situation regarding case studies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

34.10 Integration with Science

For individuals and communities, the practical implications of Integration with Science include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Integration with Science

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Integration with Science

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Integration with Science

- Assess your current situation regarding integration with science and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Indigenous Knowledge

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Indigenous Knowledge

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Indigenous Knowledge

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

Current Conditions and News: Indigenous Knowledge

- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.
- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.

Future Scenarios and Predictions: Indigenous Knowledge

Under a high-emissions scenario (SSP5-8.5), indigenous and traditional knowledge is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to indigenous and traditional knowledge.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to indigenous and traditional knowledge.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Indigenous Knowledge

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 34

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 34

- WMO. (2023). State of the Global Climate 2023. World Meteorological Organization, Geneva.
- WHO. (2023). Climate Change and Health Fact Sheet. who.int.

Exercises — Chapter 34

- Reflect on how indigenous and traditional knowledge affects your own community and write a one-page plan.

Chapter 35: Renewable Energy Systems

Theme: renewable energy for resilience

This section examines this chapter in the context of renewable energy for resilience. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of renewable energy for resilience. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

35.1 Solar Power

Evidence from peer-reviewed research shows that Solar Power is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Solar Power

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Solar Power

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Solar Power

- Assess your current situation regarding solar power and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

35.2 Wind Power

For individuals and communities, the practical implications of Wind Power include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Wind Power

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Wind Power

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Wind Power

- Assess your current situation regarding wind power and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

35.3 Hydropower

Historical experience and recent disasters demonstrate that preparedness related to Hydropower reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Hydropower

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Hydropower

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Hydropower

- Assess your current situation regarding hydropower and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

35.4 Biomass

Policy and governance play a decisive role in Biomass. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Biomass

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Biomass

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Biomass

- Assess your current situation regarding biomass and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

35.5 Energy Storage

Technology can support Energy Storage, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Energy Storage

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Energy Storage

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Energy Storage

- Assess your current situation regarding energy storage and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

35.6 Microgrids

Education and training strengthen Microgrids. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Microgrids

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Microgrids

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Microgrids

- Assess your current situation regarding microgrids and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

35.7 Distributed Generation

Monitoring and evaluation ensure that efforts related to Distributed Generation remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Distributed Generation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Distributed Generation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Distributed Generation

- Assess your current situation regarding distributed generation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

35.8 Energy Efficiency

This section examines Energy Efficiency in the context of renewable energy for resilience. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Energy Efficiency

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Energy Efficiency

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Energy Efficiency

- Assess your current situation regarding energy efficiency and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

35.9 Transition Planning

Evidence from peer-reviewed research shows that Transition Planning is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Transition Planning

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Transition Planning

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Transition Planning

- Assess your current situation regarding transition planning and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

35.10 Equity and Access

For individuals and communities, the practical implications of Equity and Access include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Equity and Access

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Equity and Access

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Equity and Access

- Assess your current situation regarding equity and access and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Renewable Energy Systems

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Renewable Energy Systems

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Renewable Energy Systems

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

Current Conditions and News: Renewable Energy Systems

- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.
- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).

Future Scenarios and Predictions: Renewable Energy

Systems

Under a high-emissions scenario (SSP5-8.5), renewable energy for resilience is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to renewable energy for resilience.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to renewable energy for resilience.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Renewable Energy Systems

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 35

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 35

- WHO. (2021). COP26 Special Report on Climate Change and Health. WHO, Geneva.
- FAO. (2023). The State of Food Security and Nutrition in the World. FAO, Rome.

Exercises — Chapter 35

- Reflect on how renewable energy for resilience affects your own community and write a one-page plan.

Chapter 36: Water Technologies

Theme: water technologies for adaptation

This section examines this chapter in the context of water technologies for adaptation. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of water technologies for adaptation. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

36.1 Rainwater Harvesting

Evidence from peer-reviewed research shows that Rainwater Harvesting is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Rainwater Harvesting

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.

- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Rainwater Harvesting

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Rainwater Harvesting

- Assess your current situation regarding rainwater harvesting and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

36.2 Desalination

For individuals and communities, the practical implications of Desalination include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Desalination

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Desalination

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Desalination

- Assess your current situation regarding desalination and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

36.3 Wastewater Recycling

Historical experience and recent disasters demonstrate that preparedness related to Wastewater Recycling reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Wastewater Recycling

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Wastewater Recycling

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Wastewater Recycling

- Assess your current situation regarding wastewater recycling and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

36.4 Smart Irrigation

Policy and governance play a decisive role in Smart Irrigation. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Smart Irrigation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Smart Irrigation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Smart Irrigation

- Assess your current situation regarding smart irrigation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

36.5 Leak Detection

Technology can support Leak Detection, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Leak Detection

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Leak Detection

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Leak Detection

- Assess your current situation regarding leak detection and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

36.6 Water Quality Monitoring

Education and training strengthen Water Quality Monitoring. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Water Quality Monitoring

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water Quality Monitoring

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water Quality Monitoring

- Assess your current situation regarding water quality monitoring and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

36.7 Low-Tech Purification

Monitoring and evaluation ensure that efforts related to Low-Tech Purification remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Low-Tech Purification

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Low-Tech Purification

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Low-Tech Purification

- Assess your current situation regarding low-tech purification and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

36.8 Storage Innovations

This section examines Storage Innovations in the context of water technologies for adaptation. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Storage Innovations

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Storage Innovations

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Storage Innovations

- Assess your current situation regarding storage innovations and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

36.9 Cost and Scalability

Evidence from peer-reviewed research shows that Cost and Scalability is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Cost and Scalability

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cost and Scalability

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cost and Scalability

- Assess your current situation regarding cost and scalability and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

36.10 Maintenance and Training

For individuals and communities, the practical implications of Maintenance and Training include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Maintenance and Training

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Maintenance and Training

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Maintenance and Training

- Assess your current situation regarding maintenance and training and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Water Technologies

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water Technologies

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Water Technologies

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

> Dr. Maria Neira, WHO: 'Climate change is the greatest health challenge of the 21st century.'

Current Conditions and News: Water Technologies

- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).
- Coral bleaching events are becoming more frequent, threatening marine ecosystems.

Future Scenarios and Predictions: Water Technologies

Under a high-emissions scenario (SSP5-8.5), water technologies for adaptation is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to water technologies for adaptation.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to water technologies for adaptation.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Water Technologies

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 36

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 36

- FAO. (2022). The State of the World's Land and Water Resources (SOLAW 2021). FAO, Rome.
- World Bank. (2021). Groundswell: Acting on Internal Climate Migration. World Bank, Washington DC.

Exercises — Chapter 36

- Reflect on how water technologies for adaptation affects your own community and write a one-page plan.

Chapter 37: Agricultural Technologies

Theme: agricultural technologies

This section examines this chapter in the context of agricultural technologies. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of agricultural technologies. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

37.1 Precision Agriculture

Evidence from peer-reviewed research shows that Precision Agriculture is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Precision Agriculture

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Precision Agriculture

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Precision Agriculture

- Assess your current situation regarding precision agriculture and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

37.2 Drought-Resistant Varieties

For individuals and communities, the practical implications of Drought-Resistant Varieties include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Drought-Resistant Varieties

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Drought-Resistant Varieties

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Drought-Resistant Varieties

- Assess your current situation regarding drought-resistant varieties and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

37.3 Soil Health

Historical experience and recent disasters demonstrate that preparedness related to Soil Health reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Soil Health

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Soil Health

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Soil Health

- Assess your current situation regarding soil health and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

37.4 Agroforestry

Policy and governance play a decisive role in Agroforestry. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Agroforestry

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Agroforestry

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Agroforestry

- Assess your current situation regarding agroforestry and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

37.5 Vertical and Urban Farming

Technology can support Vertical and Urban Farming, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Vertical and Urban Farming

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Vertical and Urban Farming

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Vertical and Urban Farming

- Assess your current situation regarding vertical and urban farming and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

37.6 Hydroponics and Aquaponics

Education and training strengthen Hydroponics and Aquaponics. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Hydroponics and Aquaponics

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Hydroponics and Aquaponics

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Hydroponics and Aquaponics

- Assess your current situation regarding hydroponics and aquaponics and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

37.7 Pest Management

Monitoring and evaluation ensure that efforts related to Pest Management remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Pest Management

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Pest Management

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Pest Management

- Assess your current situation regarding pest management and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

37.8 Post-Harvest Tech

This section examines Post-Harvest Tech in the context of agricultural technologies. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Post-Harvest Tech

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Post-Harvest Tech

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Post-Harvest Tech

- Assess your current situation regarding post-harvest tech and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

37.9 Digital Tools for Farmers

Evidence from peer-reviewed research shows that Digital Tools for Farmers is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Digital Tools for Farmers

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Digital Tools for Farmers

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Digital Tools for Farmers

- Assess your current situation regarding digital tools for farmers and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

37.10 Access and Equity

For individuals and communities, the practical implications of Access and Equity include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Access and Equity

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Access and Equity

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Access and Equity

- Assess your current situation regarding access and equity and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Agricultural Technologies

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Agricultural Technologies

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Agricultural Technologies

> Dr. Maria Neira, WHO: 'Climate change is the greatest health challenge of the 21st century.'

> Prof. Hans-Otto Portner, IPCC: 'The window for action is closing rapidly.'

Current Conditions and News: Agricultural Technologies

- Coral bleaching events are becoming more frequent, threatening marine ecosystems.
- Arctic sea ice is declining at a rate of about 13% per decade.

Future Scenarios and Predictions: Agricultural Technologies

Under a high-emissions scenario (SSP5-8.5), agricultural technologies is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to agricultural technologies.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to agricultural technologies.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Agricultural Technologies

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 37

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 37

- UNHCR. (2023). Climate Change and Displacement. unhcr.org.
- UNEP. (2023). Adaptation Gap Report 2023. UNEP, Nairobi.

Exercises — Chapter 37

- Reflect on how agricultural technologies affects your own community and write a one-page plan.

Chapter 38: Early Warning Systems

Theme: early warning systems

This section examines this chapter in the context of early warning systems. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of early warning systems. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

38.1 What Is an Early Warning System

Evidence from peer-reviewed research shows that What Is an Early Warning System is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for What Is an Early Warning System

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: What Is an Early Warning System

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for What Is an Early Warning System

- Assess your current situation regarding what is an early warning system and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

38.2 Hazard Monitoring

For individuals and communities, the practical implications of Hazard Monitoring include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Hazard Monitoring

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Hazard Monitoring

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Hazard Monitoring

- Assess your current situation regarding hazard monitoring and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

38.3 Forecasting and Modeling

Historical experience and recent disasters demonstrate that preparedness related to Forecasting and Modeling reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Forecasting and Modeling

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Forecasting and Modeling

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Forecasting and Modeling

- Assess your current situation regarding forecasting and modeling and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

38.4 Alert Dissemination

Policy and governance play a decisive role in Alert Dissemination. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Alert Dissemination

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Alert Dissemination

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Alert Dissemination

- Assess your current situation regarding alert dissemination and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

38.5 Community Response

Technology can support Community Response, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Community Response

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community Response

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community Response

- Assess your current situation regarding community response and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

38.6 Heat Health Warnings

Education and training strengthen Heat Health Warnings. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Heat Health Warnings

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Heat Health Warnings

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Heat Health Warnings

- Assess your current situation regarding heat health warnings and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

38.7 Flood Warnings

Monitoring and evaluation ensure that efforts related to Flood Warnings remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Flood Warnings

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Flood Warnings

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Flood Warnings

- Assess your current situation regarding flood warnings and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

38.8 Wildfire Warnings

This section examines Wildfire Warnings in the context of early warning systems. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Wildfire Warnings

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Wildfire Warnings

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Wildfire Warnings

- Assess your current situation regarding wildfire warnings and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

38.9 Inclusive Warnings

Evidence from peer-reviewed research shows that Inclusive Warnings is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Inclusive Warnings

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Inclusive Warnings

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Inclusive Warnings

- Assess your current situation regarding inclusive warnings and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

38.10 Evaluating Effectiveness

For individuals and communities, the practical implications of Evaluating Effectiveness include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Evaluating Effectiveness

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Evaluating Effectiveness

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Evaluating Effectiveness

- Assess your current situation regarding evaluating effectiveness and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Early Warning Systems

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Early Warning Systems

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Early Warning Systems

> Prof. Hans-Otto Portner, IPCC: 'The window for action is closing rapidly.'

> Dr. Debra Roberts, IPCC: 'Adaptation and mitigation must go hand in hand.'

Current Conditions and News: Early Warning Systems

- Arctic sea ice is declining at a rate of about 13% per decade.
- Glaciers worldwide are losing mass, contributing to sea level rise.

Future Scenarios and Predictions: Early Warning Systems

Under a high-emissions scenario (SSP5-8.5), early warning systems is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to early warning systems.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to early warning systems.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Early Warning Systems

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 38

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 38

- UNDRR. (2015). Sendai Framework for Disaster Risk Reduction 2015-2030. UNDRR, Geneva.
- United Nations. (2015). Paris Agreement. unfccc.int.

Exercises — Chapter 38

- Reflect on how early warning systems affects your own community and write a one-page plan.

Chapter 39: AI and Data for Climate

Theme: AI and data for climate action

This section examines this chapter in the context of AI and data for climate action. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of AI and data for climate action. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

39.1 AI Applications in Climate

Evidence from peer-reviewed research shows that AI Applications in Climate is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for AI Applications in Climate

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.

- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: AI Applications in Climate

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for AI Applications in Climate

- Assess your current situation regarding ai applications in climate and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

39.2 Forecasting and Prediction

For individuals and communities, the practical implications of Forecasting and Prediction include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Forecasting and Prediction

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Forecasting and Prediction

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Forecasting and Prediction

- Assess your current situation regarding forecasting and prediction and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

39.3 Resource Optimization

Historical experience and recent disasters demonstrate that preparedness related to Resource Optimization reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Resource Optimization

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Resource Optimization

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Resource Optimization

- Assess your current situation regarding resource optimization and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

39.4 Remote Sensing

Policy and governance play a decisive role in Remote Sensing. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Remote Sensing

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Remote Sensing

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Remote Sensing

- Assess your current situation regarding remote sensing and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

39.5 Citizen Science Data

Technology can support Citizen Science Data, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Citizen Science Data

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Citizen Science Data

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Citizen Science Data

- Assess your current situation regarding citizen science data and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

39.6 Ethics and Bias

Education and training strengthen Ethics and Bias. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Ethics and Bias

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Ethics and Bias

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Ethics and Bias

- Assess your current situation regarding ethics and bias and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

39.7 Data Governance

Monitoring and evaluation ensure that efforts related to Data Governance remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Data Governance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Data Governance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Data Governance

- Assess your current situation regarding data governance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

39.8 Accessibility and Equity

This section examines Accessibility and Equity in the context of AI and data for climate action. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Accessibility and Equity

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Accessibility and Equity

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Accessibility and Equity

- Assess your current situation regarding accessibility and equity and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

39.9 Risks and Limitations

Evidence from peer-reviewed research shows that Risks and Limitations is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Risks and Limitations

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Risks and Limitations

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Risks and Limitations

- Assess your current situation regarding risks and limitations and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

39.10 Future Directions

For individuals and communities, the practical implications of Future Directions include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Future Directions

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Future Directions

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Future Directions

- Assess your current situation regarding future directions and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: AI and Data for Climate

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: AI and Data for Climate

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: AI and Data for Climate

> Dr. Debra Roberts, IPCC: 'Adaptation and mitigation must go hand in hand.'

Current Conditions and News: AI and Data for Climate

- Glaciers worldwide are losing mass, contributing to sea level rise.

Future Scenarios and Predictions: AI and Data for Climate

Under a high-emissions scenario (SSP5-8.5), AI and data for climate action is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to AI and data for climate action.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to AI and data for climate action.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: AI and Data for Climate

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?

- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 39

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 39

- United Nations. (2015). 2030 Agenda for Sustainable Development. UN, New York.
- Stern, N. (2007). The Economics of Climate Change: The Stern Review. Cambridge University Press.

Exercises — Chapter 39

- Reflect on how AI and data for climate action affects your own community and write a one-page plan.

Chapter 40: Low-Tech and Appropriate Technology

Theme: low-tech and appropriate technology

This section examines this chapter in the context of low-tech and appropriate technology. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of low-tech and appropriate technology. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

40.1 Principles of Appropriate Tech

Evidence from peer-reviewed research shows that Principles of Appropriate Tech is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Principles of Appropriate Tech

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Principles of Appropriate Tech

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Principles of Appropriate Tech

- Assess your current situation regarding principles of appropriate tech and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

40.2 Passive Cooling

For individuals and communities, the practical implications of Passive Cooling include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Passive Cooling

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Passive Cooling

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Passive Cooling

- Assess your current situation regarding passive cooling and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

40.3 Simple Water Filters

Historical experience and recent disasters demonstrate that preparedness related to Simple Water Filters reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Simple Water Filters

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Simple Water Filters

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Simple Water Filters

- Assess your current situation regarding simple water filters and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

40.4 Solar Cookers

Policy and governance play a decisive role in Solar Cookers. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Solar Cookers

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Solar Cookers

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Solar Cookers

- Assess your current situation regarding solar cookers and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

40.5 Manual Tools

Technology can support Manual Tools, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Manual Tools

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Manual Tools

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Manual Tools

- Assess your current situation regarding manual tools and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

40.6 Maintenance and Repair

Education and training strengthen Maintenance and Repair. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Maintenance and Repair

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Maintenance and Repair

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Maintenance and Repair

- Assess your current situation regarding maintenance and repair and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

40.7 Local Materials

Monitoring and evaluation ensure that efforts related to Local Materials remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Local Materials

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Local Materials

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Local Materials

- Assess your current situation regarding local materials and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

40.8 Skills and Training

This section examines Skills and Training in the context of low-tech and appropriate technology. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Skills and Training

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Skills and Training

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Skills and Training

- Assess your current situation regarding skills and training and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

40.9 Cost and Durability

Evidence from peer-reviewed research shows that Cost and Durability is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Cost and Durability

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cost and Durability

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cost and Durability

- Assess your current situation regarding cost and durability and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

40.10 Combining Low and High Tech

For individuals and communities, the practical implications of Combining Low and High Tech include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Combining Low and High Tech

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Combining Low and High Tech

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Combining Low and High Tech

- Assess your current situation regarding combining low and high tech and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Low-Tech and Appropriate Technology

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Low-Tech and Appropriate Technology

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Low-Tech and Appropriate Technology

> Dr. James Hansen, climate scientist: 'The science is clear: continued emissions will drive dangerous warming.'

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

Current Conditions and News: Low-Tech and Appropriate Technology

- 2023 was the hottest year on record, with global temperatures 1.48 C above pre-industrial levels (WMO, 2024).
- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.

Future Scenarios and Predictions: Low-Tech and Appropriate Technology

Under a high-emissions scenario (SSP5-8.5), low-tech and appropriate technology is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to low-tech and appropriate technology.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to low-tech and appropriate technology.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Low-Tech and Appropriate Technology

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 40

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 40

- Watts, N. et al. (2021). The 2021 report of the Lancet Countdown on health and climate change. The Lancet, 398(10311), 1619-1662.
- Romanello, M. et al. (2023). The 2023 report of the Lancet Countdown. The Lancet, 402(10419), 2346-2394.

Exercises — Chapter 40

- Reflect on how low-tech and appropriate technology affects your own community and write a one-page plan.

Chapter 41: Climate Anxiety

Theme: climate anxiety and emotional responses

This section examines this chapter in the context of climate anxiety and emotional responses. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of climate anxiety and emotional responses. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

41.1 What Is Climate Anxiety

Evidence from peer-reviewed research shows that What Is Climate Anxiety is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for What Is Climate Anxiety

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: What Is Climate Anxiety

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for What Is Climate Anxiety

- Assess your current situation regarding what is climate anxiety and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

41.2 Who Is Affected

For individuals and communities, the practical implications of Who Is Affected include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Who Is Affected

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Who Is Affected

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Who Is Affected

- Assess your current situation regarding who is affected and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

41.3 Symptoms and Impacts

Historical experience and recent disasters demonstrate that preparedness related to Symptoms and Impacts reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Symptoms and Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Symptoms and Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Symptoms and Impacts

- Assess your current situation regarding symptoms and impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

41.4 Adaptive vs Maladaptive

Policy and governance play a decisive role in Adaptive vs Maladaptive. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Adaptive vs Maladaptive

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Adaptive vs Maladaptive

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Adaptive vs Maladaptive

- Assess your current situation regarding adaptive vs maladaptive and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

41.5 Youth and Climate Anxiety

Technology can support Youth and Climate Anxiety, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Youth and Climate Anxiety

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Youth and Climate Anxiety

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Youth and Climate Anxiety

- Assess your current situation regarding youth and climate anxiety and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

41.6 Media and Anxiety

Education and training strengthen Media and Anxiety. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Media and Anxiety

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Media and Anxiety

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Media and Anxiety

- Assess your current situation regarding media and anxiety and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

41.7 Coping Strategies

Monitoring and evaluation ensure that efforts related to Coping Strategies remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Coping Strategies

- Assess local conditions and identify the most relevant hazards.

- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Coping Strategies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Coping Strategies

- Assess your current situation regarding coping strategies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

41.8 Professional Support

This section examines Professional Support in the context of climate anxiety and emotional responses. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Professional Support

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Professional Support

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Professional Support

- Assess your current situation regarding professional support and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

41.9 Community Support

Evidence from peer-reviewed research shows that Community Support is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Community Support

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community Support

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community Support

- Assess your current situation regarding community support and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

41.10 Channeling Anxiety into Action

For individuals and communities, the practical implications of Channeling Anxiety into Action include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Channeling Anxiety into Action

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Channeling Anxiety into Action

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Channeling Anxiety into Action

- Assess your current situation regarding channeling anxiety into action and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Climate Anxiety

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate Anxiety

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Climate Anxiety

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

Current Conditions and News: Climate Anxiety

- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.
- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.

Future Scenarios and Predictions: Climate Anxiety

Under a high-emissions scenario (SSP5-8.5), climate anxiety and emotional responses is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to climate anxiety and emotional responses.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to climate anxiety and emotional responses.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Climate Anxiety

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 41

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 41

- Ebi, K.L. et al. (2021). Hot weather and heat extremes: health risks. The Lancet, 398(10301), 698-708.
- Vicedo-Cabrera, A.M. et al. (2021). Burden of heat-related mortality attributable to recent human-induced climate change. Nature Climate Change, 11, 492-500.

Exercises — Chapter 41

- Reflect on how climate anxiety and emotional responses affects your own community and write a one-page plan.

Chapter 42: Trauma and Recovery

Theme: trauma and recovery after disasters

This section examines this chapter in the context of trauma and recovery after disasters. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of trauma and recovery after disasters. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

42.1 Disaster Trauma

Evidence from peer-reviewed research shows that Disaster Trauma is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Disaster Trauma

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Disaster Trauma

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Disaster Trauma

- Assess your current situation regarding disaster trauma and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

42.2 PTSD and Depression

For individuals and communities, the practical implications of PTSD and Depression include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for PTSD and Depression

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: PTSD and Depression

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for PTSD and Depression

- Assess your current situation regarding ptsd and depression and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

42.3 Grief and Loss

Historical experience and recent disasters demonstrate that preparedness related to Grief and Loss reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Grief and Loss

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Grief and Loss

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Grief and Loss

- Assess your current situation regarding grief and loss and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

42.4 Children and Trauma

Policy and governance play a decisive role in Children and Trauma. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Children and Trauma

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Children and Trauma

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Children and Trauma

- Assess your current situation regarding children and trauma and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

42.5 Community Trauma

Technology can support Community Trauma, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Community Trauma

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community Trauma

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community Trauma

- Assess your current situation regarding community trauma and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

42.6 Psychological First Aid

Education and training strengthen Psychological First Aid. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Psychological First Aid

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Psychological First Aid

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Psychological First Aid

- Assess your current situation regarding psychological first aid and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

42.7 Long-Term Recovery

Monitoring and evaluation ensure that efforts related to Long-Term Recovery remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Long-Term Recovery

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Long-Term Recovery

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Long-Term Recovery

- Assess your current situation regarding long-term recovery and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

42.8 Resilience Factors

This section examines Resilience Factors in the context of trauma and recovery after disasters. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Resilience Factors

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Resilience Factors

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Resilience Factors

- Assess your current situation regarding resilience factors and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

42.9 Support Services

Evidence from peer-reviewed research shows that Support Services is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Support Services

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Support Services

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Support Services

- Assess your current situation regarding support services and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

42.10 Preventing Re-traumatization

For individuals and communities, the practical implications of Preventing Re-traumatization include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Preventing Re-traumatization

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Preventing Re-traumatization

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Preventing Re-traumatization

- Assess your current situation regarding preventing re-traumatization and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Trauma and Recovery

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Trauma and Recovery

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Trauma and Recovery

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

Current Conditions and News: Trauma and Recovery

- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.
- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.

Future Scenarios and Predictions: Trauma and Recovery

Under a high-emissions scenario (SSP5-8.5), trauma and recovery after disasters is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to trauma and recovery after disasters.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to trauma and recovery after disasters.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Trauma and Recovery

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 42

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 42

- Hsiang, S.M. & Burke, M. (2014). Climate, conflict, and social stability. Climatic Change, 123, 39-55.
- Burke, M., Hsiang, S.M. & Miguel, E. (2015). Global non-linear effect of temperature on economic production. Nature, 527, 235-239.

Exercises — Chapter 42

- Reflect on how trauma and recovery after disasters affects your own community and write a one-page plan.

Chapter 43: Building Psychological Resilience

Theme: psychological resilience

This section examines this chapter in the context of psychological resilience. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of psychological resilience. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

43.1 Resilience Concepts

Evidence from peer-reviewed research shows that Resilience Concepts is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Resilience Concepts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Resilience Concepts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Resilience Concepts

- Assess your current situation regarding resilience concepts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

43.2 Cognitive Strategies

For individuals and communities, the practical implications of Cognitive Strategies include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Cognitive Strategies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cognitive Strategies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cognitive Strategies

- Assess your current situation regarding cognitive strategies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

43.3 Emotional Regulation

Historical experience and recent disasters demonstrate that preparedness related to Emotional Regulation reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Emotional Regulation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Emotional Regulation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Emotional Regulation

- Assess your current situation regarding emotional regulation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

43.4 Social Support

Policy and governance play a decisive role in Social Support. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Social Support

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Social Support

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Social Support

- Assess your current situation regarding social support and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

43.5 Meaning and Purpose

Technology can support Meaning and Purpose, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Meaning and Purpose

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Meaning and Purpose

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Meaning and Purpose

- Assess your current situation regarding meaning and purpose and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

43.6 Physical Health Link

Education and training strengthen Physical Health Link. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Physical Health Link

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Physical Health Link

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Physical Health Link

- Assess your current situation regarding physical health link and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

43.7 Practices and Habits

Monitoring and evaluation ensure that efforts related to Practices and Habits remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Practices and Habits

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Practices and Habits

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Practices and Habits

- Assess your current situation regarding practices and habits and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

43.8 Resilience Training

This section examines Resilience Training in the context of psychological resilience. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Resilience Training

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Resilience Training

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Resilience Training

- Assess your current situation regarding resilience training and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

43.9 Measuring Resilience

Evidence from peer-reviewed research shows that Measuring Resilience is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Measuring Resilience

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Measuring Resilience

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Measuring Resilience

- Assess your current situation regarding measuring resilience and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

43.10 Sustaining Resilience

For individuals and communities, the practical implications of Sustaining Resilience include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Sustaining Resilience

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sustaining Resilience

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sustaining Resilience

- Assess your current situation regarding sustaining resilience and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Building Psychological Resilience

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.

- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Building Psychological Resilience

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Building Psychological Resilience

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

Current Conditions and News: Building Psychological Resilience

- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.
- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.

Future Scenarios and Predictions: Building Psychological Resilience

Under a high-emissions scenario (SSP5-8.5), psychological resilience is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to psychological resilience.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to psychological resilience.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Building Psychological Resilience

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 43

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 43

- Lobell, D.B., Schlenker, W. & Costa-Roberts, J. (2011). Climate trends and global crop production since 1980. Science, 333(6042), 616-620.
- Zhao, C. et al. (2017). Temperature increase reduces global yields of major crops. PNAS, 114(35), 9326-9331.

Exercises — Chapter 43

- Reflect on how psychological resilience affects your own community and write a one-page plan.

Chapter 44: Community Mental Health

Theme: community mental health

This section examines this chapter in the context of community mental health. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of community mental health. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

44.1 Mental Health Systems

Evidence from peer-reviewed research shows that Mental Health Systems is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Mental Health Systems

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Mental Health Systems

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Mental Health Systems

- Assess your current situation regarding mental health systems and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

44.2 Access and Barriers

For individuals and communities, the practical implications of Access and Barriers include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Access and Barriers

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Access and Barriers

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Access and Barriers

- Assess your current situation regarding access and barriers and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

44.3 Stigma Reduction

Historical experience and recent disasters demonstrate that preparedness related to Stigma Reduction reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Stigma Reduction

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Stigma Reduction

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Stigma Reduction

- Assess your current situation regarding stigma reduction and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

44.4 Peer Support

Policy and governance play a decisive role in Peer Support. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Peer Support

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Peer Support

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Peer Support

- Assess your current situation regarding peer support and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

44.5 Integration with Primary Care

Technology can support Integration with Primary Care, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Integration with Primary Care

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Integration with Primary Care

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Integration with Primary Care

- Assess your current situation regarding integration with primary care and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

44.6 Crisis Services

Education and training strengthen Crisis Services. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Crisis Services

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Crisis Services

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Crisis Services

- Assess your current situation regarding crisis services and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

44.7 Workforce Development

Monitoring and evaluation ensure that efforts related to Workforce Development remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Workforce Development

- Assess local conditions and identify the most relevant hazards.

- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Workforce Development

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Workforce Development

- Assess your current situation regarding workforce development and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

44.8 Funding

This section examines Funding in the context of community mental health. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Funding

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Funding

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Funding

- Assess your current situation regarding funding and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

44.9 Monitoring and Evaluation

Evidence from peer-reviewed research shows that Monitoring and Evaluation is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Monitoring and Evaluation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Monitoring and Evaluation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Monitoring and Evaluation

- Assess your current situation regarding monitoring and evaluation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

44.10 Community-Led Initiatives

For individuals and communities, the practical implications of Community-Led Initiatives include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Community-Led Initiatives

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community-Led Initiatives

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community-Led Initiatives

- Assess your current situation regarding community-led initiatives and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Community Mental Health

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community Mental Health

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Community Mental Health

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

Current Conditions and News: Community Mental Health

- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.
- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.

Future Scenarios and Predictions: Community Mental Health

Under a high-emissions scenario (SSP5-8.5), community mental health is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to community mental health.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to community mental health.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Community Mental Health

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 44

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 44

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Exercises — Chapter 44

- Reflect on how community mental health affects your own community and write a one-page plan.

Chapter 45: International Frameworks

Theme: international climate frameworks

This section examines this chapter in the context of international climate frameworks. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of international climate frameworks. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

45.1 UNFCCC and COP

Evidence from peer-reviewed research shows that UNFCCC and COP is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for UNFCCC and COP

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: UNFCCC and COP

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for UNFCCC and COP

- Assess your current situation regarding unfccc and cop and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

45.2 Paris Agreement

For individuals and communities, the practical implications of Paris Agreement include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Paris Agreement

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Paris Agreement

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Paris Agreement

- Assess your current situation regarding paris agreement and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

45.3 Sendai Framework

Historical experience and recent disasters demonstrate that preparedness related to Sendai Framework reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Sendai Framework

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sendai Framework

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sendai Framework

- Assess your current situation regarding sendai framework and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

45.4 SDGs

Policy and governance play a decisive role in SDGs. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for SDGs

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: SDGs

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for SDGs

- Assess your current situation regarding sdgs and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

45.5 Loss and Damage

Technology can support Loss and Damage, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Loss and Damage

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Loss and Damage

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Loss and Damage

- Assess your current situation regarding loss and damage and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

45.6 Climate Finance Mechanisms

Education and training strengthen Climate Finance Mechanisms. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Climate Finance Mechanisms

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate Finance Mechanisms

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Climate Finance Mechanisms

- Assess your current situation regarding climate finance mechanisms and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

45.7 Technology Transfer

Monitoring and evaluation ensure that efforts related to Technology Transfer remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Technology Transfer

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Technology Transfer

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Technology Transfer

- Assess your current situation regarding technology transfer and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

45.8 Capacity Building

This section examines Capacity Building in the context of international climate frameworks. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Capacity Building

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Capacity Building

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Capacity Building

- Assess your current situation regarding capacity building and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

45.9 Monitoring and Reporting

Evidence from peer-reviewed research shows that Monitoring and Reporting is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Monitoring and Reporting

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Monitoring and Reporting

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Monitoring and Reporting

- Assess your current situation regarding monitoring and reporting and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

45.10 Strengthening Frameworks

For individuals and communities, the practical implications of Strengthening Frameworks include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Strengthening Frameworks

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Strengthening Frameworks

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Strengthening Frameworks

- Assess your current situation regarding strengthening frameworks and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: International Frameworks

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: International Frameworks

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: International Frameworks

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

Current Conditions and News: International Frameworks

- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.
- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).

Future Scenarios and Predictions: International Frameworks

Under a high-emissions scenario (SSP5-8.5), international climate frameworks is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to international climate frameworks.

- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to international climate frameworks.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: International Frameworks

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 45

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 45

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- Hickman, C. et al. (2021). Climate anxiety in children and young people. The Lancet Planetary Health, 5(12), e863-e873.

Exercises — Chapter 45

- Reflect on how international climate frameworks affects your own community and write a one-page plan.

Chapter 46: National Adaptation Plans

Theme: national adaptation planning

This section examines this chapter in the context of national adaptation planning. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of national adaptation planning. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

46.1 NAP Process

Evidence from peer-reviewed research shows that NAP Process is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for NAP Process

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: NAP Process

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for NAP Process

- Assess your current situation regarding nap process and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

46.2 Vulnerability Assessment

For individuals and communities, the practical implications of Vulnerability Assessment include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Vulnerability Assessment

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Vulnerability Assessment

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Vulnerability Assessment

- Assess your current situation regarding vulnerability assessment and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

46.3 Prioritization

Historical experience and recent disasters demonstrate that preparedness related to Prioritization reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Prioritization

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Prioritization

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Prioritization

- Assess your current situation regarding prioritization and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

46.4 Implementation

Policy and governance play a decisive role in Implementation. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Implementation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Implementation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Implementation

- Assess your current situation regarding implementation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

46.5 Monitoring and Evaluation

Technology can support Monitoring and Evaluation, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Monitoring and Evaluation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Monitoring and Evaluation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Monitoring and Evaluation

- Assess your current situation regarding monitoring and evaluation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

46.6 Sectoral Integration

Education and training strengthen Sectoral Integration. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Sectoral Integration

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sectoral Integration

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sectoral Integration

- Assess your current situation regarding sectoral integration and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

46.7 Local Linkages

Monitoring and evaluation ensure that efforts related to Local Linkages remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Local Linkages

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Local Linkages

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Local Linkages

- Assess your current situation regarding local linkages and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

46.8 Finance and Budgeting

This section examines Finance and Budgeting in the context of national adaptation planning. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Finance and Budgeting

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Finance and Budgeting

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Finance and Budgeting

- Assess your current situation regarding finance and budgeting and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

46.9 Stakeholder Engagement

Evidence from peer-reviewed research shows that Stakeholder Engagement is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Stakeholder Engagement

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Stakeholder Engagement

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Stakeholder Engagement

- Assess your current situation regarding stakeholder engagement and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

46.10 Challenges and Lessons

For individuals and communities, the practical implications of Challenges and Lessons include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Challenges and Lessons

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Challenges and Lessons

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Challenges and Lessons

- Assess your current situation regarding challenges and lessons and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: National Adaptation Plans

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: National Adaptation Plans

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: National Adaptation Plans

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

> Dr. Maria Neira, WHO: 'Climate change is the greatest health challenge of the 21st century.'

Current Conditions and News: National Adaptation Plans

- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).
- Coral bleaching events are becoming more frequent, threatening marine ecosystems.

Future Scenarios and Predictions: National Adaptation Plans

Under a high-emissions scenario (SSP5-8.5), national adaptation planning is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to national adaptation planning.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to national adaptation planning.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: National Adaptation Plans

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 46

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 46

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Exercises — Chapter 46

- Reflect on how national adaptation planning affects your own community and write a one-page plan.

Chapter 47: Carbon Pricing

Theme: carbon pricing and markets

This section examines this chapter in the context of carbon pricing and markets. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of carbon pricing and markets. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

47.1 Carbon Tax

Evidence from peer-reviewed research shows that Carbon Tax is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Carbon Tax

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Carbon Tax

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Carbon Tax

- Assess your current situation regarding carbon tax and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

47.2 Emissions Trading Systems

For individuals and communities, the practical implications of Emissions Trading Systems include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Emissions Trading Systems

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Emissions Trading Systems

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Emissions Trading Systems

- Assess your current situation regarding emissions trading systems and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

47.3 Carbon Markets

Historical experience and recent disasters demonstrate that preparedness related to Carbon Markets reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Carbon Markets

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Carbon Markets

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Carbon Markets

- Assess your current situation regarding carbon markets and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

47.4 Pricing Levels

Policy and governance play a decisive role in Pricing Levels. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Pricing Levels

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Pricing Levels

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Pricing Levels

- Assess your current situation regarding pricing levels and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

47.5 Revenue Use

Technology can support Revenue Use, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Revenue Use

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Revenue Use

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Revenue Use

- Assess your current situation regarding revenue use and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

47.6 Equity Considerations

Education and training strengthen Equity Considerations. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Equity Considerations

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Equity Considerations

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Equity Considerations

- Assess your current situation regarding equity considerations and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

47.7 Border Adjustments

Monitoring and evaluation ensure that efforts related to Border Adjustments remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Border Adjustments

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Border Adjustments

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Border Adjustments

- Assess your current situation regarding border adjustments and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

47.8 Voluntary Markets

This section examines Voluntary Markets in the context of carbon pricing and markets. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Voluntary Markets

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Voluntary Markets

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Voluntary Markets

- Assess your current situation regarding voluntary markets and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

47.9 Effectiveness Evidence

Evidence from peer-reviewed research shows that Effectiveness Evidence is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Effectiveness Evidence

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Effectiveness Evidence

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Effectiveness Evidence

- Assess your current situation regarding effectiveness evidence and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

47.10 Design Principles

For individuals and communities, the practical implications of Design Principles include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Design Principles

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Design Principles

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Design Principles

- Assess your current situation regarding design principles and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Carbon Pricing

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Carbon Pricing

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Carbon Pricing

> Dr. Maria Neira, WHO: 'Climate change is the greatest health challenge of the 21st century.'

> Prof. Hans-Otto Portner, IPCC: 'The window for action is closing rapidly.'

Current Conditions and News: Carbon Pricing

- Coral bleaching events are becoming more frequent, threatening marine ecosystems.
- Arctic sea ice is declining at a rate of about 13% per decade.

Future Scenarios and Predictions: Carbon Pricing

Under a high-emissions scenario (SSP5-8.5), carbon pricing and markets is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to carbon pricing and markets.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to carbon pricing and markets.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Carbon Pricing

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 47

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 47

- Cutter, S.L. et al. (2008). A place-based model for understanding community resilience. Global Environmental Change, 18(4), 598-606.
- Aldrich, D.P. & Meyer, M.A. (2015). Social capital and community resilience. American Behavioral Scientist, 59(2), 254-269.

Exercises — Chapter 47

- Reflect on how carbon pricing and markets affects your own community and write a one-page plan.

Chapter 48: Climate Finance

Theme: climate finance

This section examines this chapter in the context of climate finance. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of climate finance. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

48.1 Definition and Sources

Evidence from peer-reviewed research shows that Definition and Sources is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Definition and Sources

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Definition and Sources

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Definition and Sources

- Assess your current situation regarding definition and sources and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

48.2 Public Finance

For individuals and communities, the practical implications of Public Finance include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Public Finance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Public Finance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Public Finance

- Assess your current situation regarding public finance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

48.3 Private Finance

Historical experience and recent disasters demonstrate that preparedness related to Private Finance reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Private Finance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Private Finance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Private Finance

- Assess your current situation regarding private finance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

48.4 Green Bonds

Policy and governance play a decisive role in Green Bonds. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Green Bonds

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Green Bonds

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Green Bonds

- Assess your current situation regarding green bonds and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

48.5 Adaptation Finance

Technology can support Adaptation Finance, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Adaptation Finance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Adaptation Finance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Adaptation Finance

- Assess your current situation regarding adaptation finance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

48.6 Loss and Damage Finance

Education and training strengthen Loss and Damage Finance. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Loss and Damage Finance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Loss and Damage Finance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Loss and Damage Finance

- Assess your current situation regarding loss and damage finance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

48.7 Access and Barriers

Monitoring and evaluation ensure that efforts related to Access and Barriers remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Access and Barriers

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Access and Barriers

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Access and Barriers

- Assess your current situation regarding access and barriers and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

48.8 Transparency

This section examines Transparency in the context of climate finance. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Transparency

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Transparency

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Transparency

- Assess your current situation regarding transparency and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

48.9 Scaling Finance

Evidence from peer-reviewed research shows that Scaling Finance is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Scaling Finance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Scaling Finance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Scaling Finance

- Assess your current situation regarding scaling finance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

48.10 Innovative Instruments

For individuals and communities, the practical implications of Innovative Instruments include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Innovative Instruments

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Innovative Instruments

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Innovative Instruments

- Assess your current situation regarding innovative instruments and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Climate Finance

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate Finance

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Climate Finance

> Prof. Hans-Otto Portner, IPCC: 'The window for action is closing rapidly.'

> Dr. Debra Roberts, IPCC: 'Adaptation and mitigation must go hand in hand.'

Current Conditions and News: Climate Finance

- Arctic sea ice is declining at a rate of about 13% per decade.
- Glaciers worldwide are losing mass, contributing to sea level rise.

Future Scenarios and Predictions: Climate Finance

Under a high-emissions scenario (SSP5-8.5), climate finance is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to climate finance.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to climate finance.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Climate Finance

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 48

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 48

- Berkes, F. & Ross, H. (2013). Community resilience: toward an integrated approach. Society & Natural Resources, 26(1), 5-20.
- Folke, C. et al. (2010). Resilience thinking. Ecology and Society, 15(4), 20.

Exercises — Chapter 48

- Reflect on how climate finance affects your own community and write a one-page plan.

Chapter 49: Justice and Equity

Theme: climate justice and equity

This section examines this chapter in the context of climate justice and equity. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of climate justice and equity. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

49.1 Climate Justice Principles

Evidence from peer-reviewed research shows that Climate Justice Principles is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Climate Justice Principles

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate Justice Principles

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Climate Justice Principles

- Assess your current situation regarding climate justice principles and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

49.2 Historical Responsibility

For individuals and communities, the practical implications of Historical Responsibility include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Historical Responsibility

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Historical Responsibility

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Historical Responsibility

- Assess your current situation regarding historical responsibility and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

49.3 Vulnerable Groups

Historical experience and recent disasters demonstrate that preparedness related to Vulnerable Groups reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Vulnerable Groups

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Vulnerable Groups

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Vulnerable Groups

- Assess your current situation regarding vulnerable groups and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

49.4 Gender and Climate

Policy and governance play a decisive role in Gender and Climate. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Gender and Climate

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Gender and Climate

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Gender and Climate

- Assess your current situation regarding gender and climate and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

49.5 Indigenous Rights

Technology can support Indigenous Rights, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Indigenous Rights

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Indigenous Rights

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Indigenous Rights

- Assess your current situation regarding indigenous rights and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

49.6 Intergenerational Justice

Education and training strengthen Intergenerational Justice. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Intergenerational Justice

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Intergenerational Justice

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Intergenerational Justice

- Assess your current situation regarding intergenerational justice and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

49.7 Just Transition

Monitoring and evaluation ensure that efforts related to Just Transition remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Just Transition

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Just Transition

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Just Transition

- Assess your current situation regarding just transition and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

49.8 Equitable Adaptation

This section examines Equitable Adaptation in the context of climate justice and equity. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Equitable Adaptation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Equitable Adaptation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Equitable Adaptation

- Assess your current situation regarding equitable adaptation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

49.9 Human Rights

Evidence from peer-reviewed research shows that Human Rights is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Human Rights

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Human Rights

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Human Rights

- Assess your current situation regarding human rights and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

49.10 Advocacy and Accountability

For individuals and communities, the practical implications of Advocacy and Accountability include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Advocacy and Accountability

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Advocacy and Accountability

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Advocacy and Accountability

- Assess your current situation regarding advocacy and accountability and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Justice and Equity

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Justice and Equity

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Justice and Equity

> Dr. Debra Roberts, IPCC: 'Adaptation and mitigation must go hand in hand.'

Current Conditions and News: Justice and Equity

- Glaciers worldwide are losing mass, contributing to sea level rise.

Future Scenarios and Predictions: Justice and Equity

Under a high-emissions scenario (SSP5-8.5), climate justice and equity is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to climate justice and equity.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to climate justice and equity.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Justice and Equity

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 49

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 49

- Oppenheimer, M. et al. (2019). Sea Level Rise and Implications for Low-Lying Islands, Coasts and Communities. SROCC Chapter 4.
- Cisse, G. et al. (2022). Health, Wellbeing, and the Changing Structure of Communities. IPCC AR6 WG2 Chapter 7.

Exercises — Chapter 49

- Reflect on how climate justice and equity affects your own community and write a one-page plan.

Chapter 50: Civic Engagement

Theme: civic engagement in climate action

This section examines this chapter in the context of civic engagement in climate action. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of civic engagement in climate action. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

50.1 Why Civic Engagement Matters

Evidence from peer-reviewed research shows that Why Civic Engagement Matters is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Why Civic Engagement Matters

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Why Civic Engagement Matters

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Why Civic Engagement Matters

- Assess your current situation regarding why civic engagement matters and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

50.2 Voting and Elections

For individuals and communities, the practical implications of Voting and Elections include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Voting and Elections

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Voting and Elections

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Voting and Elections

- Assess your current situation regarding voting and elections and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

50.3 Advocacy and Lobbying

Historical experience and recent disasters demonstrate that preparedness related to Advocacy and Lobbying reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Advocacy and Lobbying

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Advocacy and Lobbying

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Advocacy and Lobbying

- Assess your current situation regarding advocacy and lobbying and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

50.4 Protests and Movements

Policy and governance play a decisive role in Protests and Movements. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Protests and Movements

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Protests and Movements

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Protests and Movements

- Assess your current situation regarding protests and movements and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

50.5 Community Organizing

Technology can support Community Organizing, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Community Organizing

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community Organizing

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community Organizing

- Assess your current situation regarding community organizing and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

50.6 Participatory Budgeting

Education and training strengthen Participatory Budgeting. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Participatory Budgeting

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Participatory Budgeting

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Participatory Budgeting

- Assess your current situation regarding participatory budgeting and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

50.7 Youth Engagement

Monitoring and evaluation ensure that efforts related to Youth Engagement remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Youth Engagement

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Youth Engagement

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Youth Engagement

- Assess your current situation regarding youth engagement and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

50.8 Digital Activism

This section examines Digital Activism in the context of civic engagement in climate action. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Digital Activism

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Digital Activism

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Digital Activism

- Assess your current situation regarding digital activism and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

50.9 Holding Leaders Accountable

Evidence from peer-reviewed research shows that Holding Leaders Accountable is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Holding Leaders Accountable

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Holding Leaders Accountable

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Holding Leaders Accountable

- Assess your current situation regarding holding leaders accountable and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

50.10 Sustaining Engagement

For individuals and communities, the practical implications of Sustaining Engagement include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Sustaining Engagement

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sustaining Engagement

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sustaining Engagement

- Assess your current situation regarding sustaining engagement and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Civic Engagement

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Civic Engagement

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Civic Engagement

> Dr. James Hansen, climate scientist: 'The science is clear: continued emissions will drive dangerous warming.'

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

Current Conditions and News: Civic Engagement

- 2023 was the hottest year on record, with global temperatures 1.48 C above pre-industrial levels (WMO, 2024).
- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.

Future Scenarios and Predictions: Civic Engagement

Under a high-emissions scenario (SSP5-8.5), civic engagement in climate action is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to civic engagement in climate action.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to civic engagement in climate action.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Civic Engagement

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 50

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 50

- WHO & UNICEF. (2023). Progress on Household Drinking Water, Sanitation and Hygiene 2000-2022. JMP.
- UNEP. (2022). Emissions Gap Report 2022. UNEP, Nairobi.

Exercises — Chapter 50

- Reflect on how civic engagement in climate action affects your own community and write a one-page plan.

Chapter 51: Case Study: Heat in South Asia

Theme: heat waves in South Asia

This section examines this chapter in the context of heat waves in South Asia. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of heat waves in South Asia. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

51.1 Regional Context

Evidence from peer-reviewed research shows that Regional Context is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Regional Context

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Regional Context

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Regional Context

- Assess your current situation regarding regional context and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

51.2 Heat Wave Trends

For individuals and communities, the practical implications of Heat Wave Trends include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Heat Wave Trends

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Heat Wave Trends

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Heat Wave Trends

- Assess your current situation regarding heat wave trends and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

51.3 Impacts on Health

Historical experience and recent disasters demonstrate that preparedness related to Impacts on Health reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Impacts on Health

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Impacts on Health

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Impacts on Health

- Assess your current situation regarding impacts on health and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

51.4 Impacts on Labor

Policy and governance play a decisive role in Impacts on Labor. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Impacts on Labor

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Impacts on Labor

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Impacts on Labor

- Assess your current situation regarding impacts on labor and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

51.5 Adaptation Responses

Technology can support Adaptation Responses, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Adaptation Responses

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Adaptation Responses

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Adaptation Responses

- Assess your current situation regarding adaptation responses and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

51.6 Policy and Governance

Education and training strengthen Policy and Governance. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Policy and Governance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Policy and Governance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Policy and Governance

- Assess your current situation regarding policy and governance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

51.7 Community Actions

Monitoring and evaluation ensure that efforts related to Community Actions remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Community Actions

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community Actions

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community Actions

- Assess your current situation regarding community actions and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

51.8 Lessons Learned

This section examines Lessons Learned in the context of heat waves in South Asia. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Lessons Learned

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Lessons Learned

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Lessons Learned

- Assess your current situation regarding lessons learned and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

51.9 Remaining Gaps

Evidence from peer-reviewed research shows that Remaining Gaps is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Remaining Gaps

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Remaining Gaps

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Remaining Gaps

- Assess your current situation regarding remaining gaps and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

51.10 Future Outlook

For individuals and communities, the practical implications of Future Outlook include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Future Outlook

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Future Outlook

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Future Outlook

- Assess your current situation regarding future outlook and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Case Study: Heat in South Asia

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Case Study: Heat in South Asia

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Case Study: Heat in South Asia

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

Current Conditions and News: Case Study: Heat in South Asia

- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.
- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.

Future Scenarios and Predictions: Case Study: Heat in South Asia

Under a high-emissions scenario (SSP5-8.5), heat waves in South Asia is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to heat waves in South Asia.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to heat waves in South Asia.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Case Study: Heat in South Asia

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 51

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 51

- IEA. (2023). World Energy Outlook 2023. International Energy Agency, Paris.
- IRENA. (2023). World Energy Transitions Outlook 2023. IRENA, Abu Dhabi.

Exercises — Chapter 51

- Reflect on how heat waves in South Asia affects your own community and write a one-page plan.

Chapter 52: Case Study: Drought in the Horn of Africa

Theme: drought in the Horn of Africa

This section examines this chapter in the context of drought in the Horn of Africa. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of drought in the Horn of Africa. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

52.1 Regional Context

Evidence from peer-reviewed research shows that Regional Context is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Regional Context

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Regional Context

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Regional Context

- Assess your current situation regarding regional context and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

52.2 Drought Trends

For individuals and communities, the practical implications of Drought Trends include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Drought Trends

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Drought Trends

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Drought Trends

- Assess your current situation regarding drought trends and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

52.3 Food Insecurity

Historical experience and recent disasters demonstrate that preparedness related to Food Insecurity reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Food Insecurity

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Food Insecurity

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Food Insecurity

- Assess your current situation regarding food insecurity and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

52.4 Water Scarcity

Policy and governance play a decisive role in Water Scarcity. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Water Scarcity

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water Scarcity

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water Scarcity

- Assess your current situation regarding water scarcity and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

52.5 Humanitarian Response

Technology can support Humanitarian Response, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Humanitarian Response

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Humanitarian Response

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Humanitarian Response

- Assess your current situation regarding humanitarian response and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

52.6 Resilience Building

Education and training strengthen Resilience Building. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Resilience Building

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Resilience Building

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Resilience Building

- Assess your current situation regarding resilience building and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

52.7 Pastoralist Adaptation

Monitoring and evaluation ensure that efforts related to Pastoralist Adaptation remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Pastoralist Adaptation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Pastoralist Adaptation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Pastoralist Adaptation

- Assess your current situation regarding pastoralist adaptation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

52.8 Policy and Finance

This section examines Policy and Finance in the context of drought in the Horn of Africa. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Policy and Finance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Policy and Finance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Policy and Finance

- Assess your current situation regarding policy and finance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

52.9 Lessons Learned

Evidence from peer-reviewed research shows that Lessons Learned is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Lessons Learned

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Lessons Learned

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Lessons Learned

- Assess your current situation regarding lessons learned and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

52.10 Future Outlook

For individuals and communities, the practical implications of Future Outlook include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Future Outlook

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Future Outlook

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Future Outlook

- Assess your current situation regarding future outlook and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

Case Study: Case Study: Drought in the Horn of Africa

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Case Study: Drought in the Horn of Africa

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Case Study: Drought in the Horn of Africa

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

Current Conditions and News: Case Study: Drought in the Horn of Africa

- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.
- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.

Future Scenarios and Predictions: Case Study: Drought in the Horn of Africa

Under a high-emissions scenario (SSP5-8.5), drought in the Horn of Africa is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to drought in the Horn of Africa.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to drought in the Horn of Africa.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Case Study: Drought in the Horn of Africa

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?

- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 52

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 52

- FAO. (2021). Climate Change: Unpacking the Burden on Food Safety. FAO, Rome.
- World Bank. (2022). What You Need to Know About Food Security and Climate Change. World Bank, Washington DC.

Exercises — Chapter 52

- Reflect on how drought in the Horn of Africa affects your own community and write a one-page plan.

Chapter 53: Case Study: Sea Level Rise in Small Island States

Theme: sea level rise in small island states

This section examines this chapter in the context of sea level rise in small island states. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of sea level rise in small island states. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

53.1 Regional Context

Evidence from peer-reviewed research shows that Regional Context is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Regional Context

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Regional Context

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Regional Context

- Assess your current situation regarding regional context and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

53.2 Sea Level Trends

For individuals and communities, the practical implications of Sea Level Trends include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Sea Level Trends

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sea Level Trends

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sea Level Trends

- Assess your current situation regarding sea level trends and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

53.3 Coastal Impacts

Historical experience and recent disasters demonstrate that preparedness related to Coastal Impacts reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Coastal Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Coastal Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Coastal Impacts

- Assess your current situation regarding coastal impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

53.4 Freshwater and Agriculture

Policy and governance play a decisive role in Freshwater and Agriculture. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Freshwater and Agriculture

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Freshwater and Agriculture

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Freshwater and Agriculture

- Assess your current situation regarding freshwater and agriculture and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

53.5 Displacement Risk

Technology can support Displacement Risk, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Displacement Risk

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Displacement Risk

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Displacement Risk

- Assess your current situation regarding displacement risk and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

53.6 Adaptation Strategies

Education and training strengthen Adaptation Strategies. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Adaptation Strategies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Adaptation Strategies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Adaptation Strategies

- Assess your current situation regarding adaptation strategies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

53.7 International Support

Monitoring and evaluation ensure that efforts related to International Support remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for International Support

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: International Support

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for International Support

- Assess your current situation regarding international support and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

53.8 Loss and Damage

This section examines Loss and Damage in the context of sea level rise in small island states. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Loss and Damage

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Loss and Damage

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Loss and Damage

- Assess your current situation regarding loss and damage and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

53.9 Lessons Learned

Evidence from peer-reviewed research shows that Lessons Learned is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Lessons Learned

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Lessons Learned

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Lessons Learned

- Assess your current situation regarding lessons learned and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

53.10 Future Outlook

For individuals and communities, the practical implications of Future Outlook include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Future Outlook

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Future Outlook

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Future Outlook

- Assess your current situation regarding future outlook and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Case Study: Sea Level Rise in Small Island States

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.

- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Case Study: Sea Level Rise in Small Island States

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Case Study: Sea Level Rise in Small Island States

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

Current Conditions and News: Case Study: Sea Level Rise in Small Island States

- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.
- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.

Future Scenarios and Predictions: Case Study: Sea Level Rise in Small Island States

Under a high-emissions scenario (SSP5-8.5), sea level rise in small island states is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to sea level rise in small island states.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to sea level rise in small island states.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Case Study: Sea Level Rise in Small Island States

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 53

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 53

- UNDP. (2022). Human Development Report 2021/22. UNDP, New York.
- IFRC. (2022). World Disasters Report 2022. IFRC, Geneva.

Exercises — Chapter 53

- Reflect on how sea level rise in small island states affects your own community and write a one-page plan.

Chapter 54: Case Study: Wildfires in Australia and North America

Theme: wildfires

This section examines this chapter in the context of wildfires. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of wildfires. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

54.1 Fire Weather Trends

Evidence from peer-reviewed research shows that Fire Weather Trends is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Fire Weather Trends

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Fire Weather Trends

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Fire Weather Trends

- Assess your current situation regarding fire weather trends and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

54.2 Ecological Impacts

For individuals and communities, the practical implications of Ecological Impacts include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Ecological Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Ecological Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Ecological Impacts

- Assess your current situation regarding ecological impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

54.3 Human Health Impacts

Historical experience and recent disasters demonstrate that preparedness related to Human Health Impacts reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Human Health Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Human Health Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Human Health Impacts

- Assess your current situation regarding human health impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

54.4 Infrastructure Damage

Policy and governance play a decisive role in Infrastructure Damage. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Infrastructure Damage

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Infrastructure Damage

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Infrastructure Damage

- Assess your current situation regarding infrastructure damage and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

54.5 Emergency Response

Technology can support Emergency Response, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Emergency Response

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Emergency Response

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Emergency Response

- Assess your current situation regarding emergency response and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

54.6 Prevention and Mitigation

Education and training strengthen Prevention and Mitigation. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Prevention and Mitigation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Prevention and Mitigation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Prevention and Mitigation

- Assess your current situation regarding prevention and mitigation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

54.7 Community Preparedness

Monitoring and evaluation ensure that efforts related to Community Preparedness remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Community Preparedness

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community Preparedness

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community Preparedness

- Assess your current situation regarding community preparedness and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

54.8 Policy and Funding

This section examines Policy and Funding in the context of wildfires. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Policy and Funding

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Policy and Funding

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Policy and Funding

- Assess your current situation regarding policy and funding and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

54.9 Lessons Learned

Evidence from peer-reviewed research shows that Lessons Learned is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Lessons Learned

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Lessons Learned

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Lessons Learned

- Assess your current situation regarding lessons learned and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

54.10 Future Outlook

For individuals and communities, the practical implications of Future Outlook include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Future Outlook

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Future Outlook

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Future Outlook

- Assess your current situation regarding future outlook and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Case Study: Wildfires in Australia and North America

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Case Study: Wildfires in Australia and North America

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Case Study: Wildfires in Australia and North America

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

Current Conditions and News: Case Study: Wildfires in Australia and North America

- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.
- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.

Future Scenarios and Predictions: Case Study: Wildfires in Australia and North America

Under a high-emissions scenario (SSP5-8.5), wildfires is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to wildfires.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to wildfires.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Case Study: Wildfires in Australia and North America

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 54

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 54

- Red Cross. (2020). Heat Waves: A Silent Killer. IFRC, Geneva.
- WMO. (2022). Early Warnings for All. WMO, Geneva.

Exercises — Chapter 54

- Reflect on how wildfires affects your own community and write a one-page plan.

Chapter 55: Case Study: Floods in Southeast Asia

Theme: floods in Southeast Asia

This section examines this chapter in the context of floods in Southeast Asia. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of floods in Southeast Asia. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

55.1 Regional Context

Evidence from peer-reviewed research shows that Regional Context is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Regional Context

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Regional Context

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Regional Context

- Assess your current situation regarding regional context and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

55.2 Flood Trends

For individuals and communities, the practical implications of Flood Trends include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Flood Trends

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Flood Trends

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Flood Trends

- Assess your current situation regarding flood trends and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

55.3 Urban Flooding

Historical experience and recent disasters demonstrate that preparedness related to Urban Flooding reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Urban Flooding

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Urban Flooding

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Urban Flooding

- Assess your current situation regarding urban flooding and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

55.4 Rural and Agricultural Impacts

Policy and governance play a decisive role in Rural and Agricultural Impacts. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Rural and Agricultural Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Rural and Agricultural Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Rural and Agricultural Impacts

- Assess your current situation regarding rural and agricultural impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

55.5 Health Impacts

Technology can support Health Impacts, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Health Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Health Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Health Impacts

- Assess your current situation regarding health impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

55.6 Early Warning

Education and training strengthen Early Warning. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Early Warning

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Early Warning

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Early Warning

- Assess your current situation regarding early warning and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

55.7 Adaptation and Infrastructure

Monitoring and evaluation ensure that efforts related to Adaptation and Infrastructure remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Adaptation and Infrastructure

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Adaptation and Infrastructure

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Adaptation and Infrastructure

- Assess your current situation regarding adaptation and infrastructure and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

55.8 Community Resilience

This section examines Community Resilience in the context of floods in Southeast Asia. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Community Resilience

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community Resilience

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community Resilience

- Assess your current situation regarding community resilience and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

55.9 Lessons Learned

Evidence from peer-reviewed research shows that Lessons Learned is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Lessons Learned

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Lessons Learned

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Lessons Learned

- Assess your current situation regarding lessons learned and identify gaps.
- Document findings and share with household or team.

- Set a timeline and assign responsibilities.

55.10 Future Outlook

For individuals and communities, the practical implications of Future Outlook include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Future Outlook

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Future Outlook

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Future Outlook

- Assess your current situation regarding future outlook and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Case Study: Floods in Southeast Asia

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Case Study: Floods in Southeast Asia

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Case Study: Floods in Southeast Asia

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

Current Conditions and News: Case Study: Floods in Southeast Asia

- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.
- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).

Future Scenarios and Predictions: Case Study: Floods in Southeast Asia

Under a high-emissions scenario (SSP5-8.5), floods in Southeast Asia is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to floods in Southeast Asia.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to floods in Southeast Asia.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Case Study: Floods in Southeast Asia

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 55

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 55

- UNESCO. (2021). Getting Every School Climate-Ready. UNESCO, Paris.
- UNESCO. (2020). Education for Sustainable Development: A Roadmap. UNESCO, Paris.

Exercises — Chapter 55

- Reflect on how floods in Southeast Asia affects your own community and write a one-page plan.

Chapter 56: Case Study: Food Crisis in Sub-Saharan Africa

Theme: food crisis in Sub-Saharan Africa

This section examines this chapter in the context of food crisis in Sub-Saharan Africa. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of food crisis in Sub-Saharan Africa. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

56.1 Regional Context

Evidence from peer-reviewed research shows that Regional Context is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Regional Context

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Regional Context

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Regional Context

- Assess your current situation regarding regional context and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

56.2 Climate and Agriculture

For individuals and communities, the practical implications of Climate and Agriculture include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Climate and Agriculture

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate and Agriculture

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Climate and Agriculture

- Assess your current situation regarding climate and agriculture and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

56.3 Food Insecurity Trends

Historical experience and recent disasters demonstrate that preparedness related to Food Insecurity Trends reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Food Insecurity Trends

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Food Insecurity Trends

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Food Insecurity Trends

- Assess your current situation regarding food insecurity trends and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

56.4 Nutrition Impacts

Policy and governance play a decisive role in Nutrition Impacts. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Nutrition Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Nutrition Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Nutrition Impacts

- Assess your current situation regarding nutrition impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

56.5 Market and Price Shocks

Technology can support Market and Price Shocks, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Market and Price Shocks

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Market and Price Shocks

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Market and Price Shocks

- Assess your current situation regarding market and price shocks and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

56.6 Humanitarian Response

Education and training strengthen Humanitarian Response. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Humanitarian Response

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Humanitarian Response

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Humanitarian Response

- Assess your current situation regarding humanitarian response and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

56.7 Resilient Agriculture

Monitoring and evaluation ensure that efforts related to Resilient Agriculture remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Resilient Agriculture

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Resilient Agriculture

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Resilient Agriculture

- Assess your current situation regarding resilient agriculture and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

56.8 Policy and Finance

This section examines Policy and Finance in the context of food crisis in Sub-Saharan Africa. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Policy and Finance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Policy and Finance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Policy and Finance

- Assess your current situation regarding policy and finance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

56.9 Lessons Learned

Evidence from peer-reviewed research shows that Lessons Learned is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Lessons Learned

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Lessons Learned

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Lessons Learned

- Assess your current situation regarding lessons learned and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

56.10 Future Outlook

For individuals and communities, the practical implications of Future Outlook include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Future Outlook

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Future Outlook

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Future Outlook

- Assess your current situation regarding future outlook and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Case Study: Food Crisis in Sub-Saharan Africa

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Case Study: Food Crisis in Sub-Saharan Africa

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Case Study: Food Crisis in Sub-Saharan Africa

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

> Dr. Maria Neira, WHO: 'Climate change is the greatest health challenge of the 21st century.'

Current Conditions and News: Case Study: Food Crisis in Sub-Saharan Africa

- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).
- Coral bleaching events are becoming more frequent, threatening marine ecosystems.

Future Scenarios and Predictions: Case Study: Food Crisis in Sub-Saharan Africa

Under a high-emissions scenario (SSP5-8.5), food crisis in Sub-Saharan Africa is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to food crisis in Sub-Saharan Africa.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to food crisis in Sub-Saharan Africa.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Case Study: Food Crisis in Sub-Saharan Africa

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 56

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 56

- OECD. (2021). Managing Climate Risks, Facing Essential Challenges. OECD, Paris.
- OECD. (2023). Climate Finance Provided and Mobilised by Developed Countries. OECD, Paris.

Exercises — Chapter 56

- Reflect on how food crisis in Sub-Saharan Africa affects your own community and write a one-page plan.

Chapter 57: Case Study: Water Stress in the Middle East

Theme: water stress in the Middle East

This section examines this chapter in the context of water stress in the Middle East. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of water stress in the Middle East. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

57.1 Regional Context

Evidence from peer-reviewed research shows that Regional Context is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Regional Context

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Regional Context

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Regional Context

- Assess your current situation regarding regional context and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

57.2 Water Scarcity Trends

For individuals and communities, the practical implications of Water Scarcity Trends include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Water Scarcity Trends

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water Scarcity Trends

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water Scarcity Trends

- Assess your current situation regarding water scarcity trends and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

57.3 Agriculture and Water

Historical experience and recent disasters demonstrate that preparedness related to Agriculture and Water reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Agriculture and Water

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Agriculture and Water

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Agriculture and Water

- Assess your current situation regarding agriculture and water and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

57.4 Urban Water Stress

Policy and governance play a decisive role in Urban Water Stress. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Urban Water Stress

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Urban Water Stress

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Urban Water Stress

- Assess your current situation regarding urban water stress and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

57.5 Transboundary Water

Technology can support Transboundary Water, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Transboundary Water

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Transboundary Water

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Transboundary Water

- Assess your current situation regarding transboundary water and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

57.6 Adaptation Strategies

Education and training strengthen Adaptation Strategies. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Adaptation Strategies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Adaptation Strategies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Adaptation Strategies

- Assess your current situation regarding adaptation strategies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

57.7 Technology and Innovation

Monitoring and evaluation ensure that efforts related to Technology and Innovation remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Technology and Innovation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Technology and Innovation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Technology and Innovation

- Assess your current situation regarding technology and innovation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

57.8 Governance

This section examines Governance in the context of water stress in the Middle East. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Governance

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Governance

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Governance

- Assess your current situation regarding governance and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

57.9 Lessons Learned

Evidence from peer-reviewed research shows that Lessons Learned is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Lessons Learned

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Lessons Learned

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Lessons Learned

- Assess your current situation regarding lessons learned and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

57.10 Future Outlook

For individuals and communities, the practical implications of Future Outlook include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Future Outlook

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Future Outlook

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Future Outlook

- Assess your current situation regarding future outlook and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Case Study: Water Stress in the Middle East

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Case Study: Water Stress in the Middle East

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Case Study: Water Stress in the Middle East

> Dr. Maria Neira, WHO: 'Climate change is the greatest health challenge of the 21st century.'

> Prof. Hans-Otto Portner, IPCC: 'The window for action is closing rapidly.'

Current Conditions and News: Case Study: Water Stress in the Middle East

- Coral bleaching events are becoming more frequent, threatening marine ecosystems.
- Arctic sea ice is declining at a rate of about 13% per decade.

Future Scenarios and Predictions: Case Study: Water Stress in the Middle East

Under a high-emissions scenario (SSP5-8.5), water stress in the Middle East is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to water stress in the Middle East.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to water stress in the Middle East.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Case Study: Water Stress in the Middle East

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 57

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 57

- IMF. (2022). World Economic Outlook: War Sets Back Global Recovery. IMF, Washington DC.
- IPCC. (2023). Climate Change 2023: Synthesis Report Summary for Policymakers. IPCC, Geneva.

Exercises — Chapter 57

- Reflect on how water stress in the Middle East affects your own community and write a one-page plan.

Chapter 58: Case Study: Storms and Hurricanes in the Caribbean

Theme: storms and hurricanes in the Caribbean

This section examines this chapter in the context of storms and hurricanes in the Caribbean. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of storms and hurricanes in the Caribbean. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

58.1 Regional Context

Evidence from peer-reviewed research shows that Regional Context is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Regional Context

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Regional Context

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Regional Context

- Assess your current situation regarding regional context and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

58.2 Storm Trends

For individuals and communities, the practical implications of Storm Trends include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Storm Trends

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Storm Trends

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Storm Trends

- Assess your current situation regarding storm trends and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

58.3 Infrastructure Damage

Historical experience and recent disasters demonstrate that preparedness related to Infrastructure Damage reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Infrastructure Damage

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Infrastructure Damage

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Infrastructure Damage

- Assess your current situation regarding infrastructure damage and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

58.4 Economic Impacts

Policy and governance play a decisive role in Economic Impacts. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Economic Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Economic Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Economic Impacts

- Assess your current situation regarding economic impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

58.5 Health Impacts

Technology can support Health Impacts, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Health Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Health Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Health Impacts

- Assess your current situation regarding health impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

58.6 Early Warning and Evacuation

Education and training strengthen Early Warning and Evacuation. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Early Warning and Evacuation

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Early Warning and Evacuation

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Early Warning and Evacuation

- Assess your current situation regarding early warning and evacuation and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

58.7 Recovery and Reconstruction

Monitoring and evaluation ensure that efforts related to Recovery and Reconstruction remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Recovery and Reconstruction

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Recovery and Reconstruction

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Recovery and Reconstruction

- Assess your current situation regarding recovery and reconstruction and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

58.8 Resilience Building

This section examines Resilience Building in the context of storms and hurricanes in the Caribbean. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Resilience Building

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Resilience Building

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Resilience Building

- Assess your current situation regarding resilience building and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

58.9 Lessons Learned

Evidence from peer-reviewed research shows that Lessons Learned is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Lessons Learned

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Lessons Learned

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Lessons Learned

- Assess your current situation regarding lessons learned and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

58.10 Future Outlook

For individuals and communities, the practical implications of Future Outlook include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Future Outlook

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Future Outlook

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Future Outlook

- Assess your current situation regarding future outlook and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Case Study: Storms and Hurricanes in the Caribbean

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Case Study: Storms and Hurricanes in the Caribbean

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Case Study: Storms and Hurricanes in the Caribbean

> Prof. Hans-Otto Portner, IPCC: 'The window for action is closing rapidly.'

> Dr. Debra Roberts, IPCC: 'Adaptation and mitigation must go hand in hand.'

Current Conditions and News: Case Study: Storms and Hurricanes in the Caribbean

- Arctic sea ice is declining at a rate of about 13% per decade.
- Glaciers worldwide are losing mass, contributing to sea level rise.

Future Scenarios and Predictions: Case Study: Storms and Hurricanes in the Caribbean

Under a high-emissions scenario (SSP5-8.5), storms and hurricanes in the Caribbean is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to storms and hurricanes in the Caribbean.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to storms and hurricanes in the Caribbean.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Case Study: Storms and Hurricanes in the Caribbean

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 58

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 58

- IPCC. (2023). Climate Change 2023: Synthesis Report. IPCC, Geneva.
- IPCC. (2021). Climate Change 2021: The Physical Science Basis (AR6 WG1). Cambridge University Press.

Exercises — Chapter 58

- Reflect on how storms and hurricanes in the Caribbean affects your own community and write a one-page plan.

Chapter 59: Case Study: Permafrost Thaw in the Arctic

Theme: permafrost thaw in the Arctic

This section examines this chapter in the context of permafrost thaw in the Arctic. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of permafrost thaw in the Arctic. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

59.1 Regional Context

Evidence from peer-reviewed research shows that Regional Context is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Regional Context

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Regional Context

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Regional Context

- Assess your current situation regarding regional context and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

59.2 Thaw Trends

For individuals and communities, the practical implications of Thaw Trends include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Thaw Trends

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Thaw Trends

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Thaw Trends

- Assess your current situation regarding thaw trends and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

59.3 Infrastructure Impacts

Historical experience and recent disasters demonstrate that preparedness related to Infrastructure Impacts reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Infrastructure Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Infrastructure Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Infrastructure Impacts

- Assess your current situation regarding infrastructure impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

59.4 Ecosystem Changes

Policy and governance play a decisive role in Ecosystem Changes. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Ecosystem Changes

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Ecosystem Changes

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Ecosystem Changes

- Assess your current situation regarding ecosystem changes and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

59.5 Indigenous Communities

Technology can support Indigenous Communities, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Indigenous Communities

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Indigenous Communities

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Indigenous Communities

- Assess your current situation regarding indigenous communities and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

59.6 Carbon Release

Education and training strengthen Carbon Release. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Carbon Release

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Carbon Release

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Carbon Release

- Assess your current situation regarding carbon release and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

59.7 Adaptation Strategies

Monitoring and evaluation ensure that efforts related to Adaptation Strategies remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Adaptation Strategies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Adaptation Strategies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Adaptation Strategies

- Assess your current situation regarding adaptation strategies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

59.8 Monitoring and Research

This section examines Monitoring and Research in the context of permafrost thaw in the Arctic. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Monitoring and Research

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Monitoring and Research

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Monitoring and Research

- Assess your current situation regarding monitoring and research and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

59.9 Lessons Learned

Evidence from peer-reviewed research shows that Lessons Learned is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Lessons Learned

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Lessons Learned

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Lessons Learned

- Assess your current situation regarding lessons learned and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

59.10 Future Outlook

For individuals and communities, the practical implications of Future Outlook include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Future Outlook

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Future Outlook

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Future Outlook

- Assess your current situation regarding future outlook and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Case Study: Permafrost Thaw in the Arctic

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Case Study: Permafrost Thaw in the Arctic

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Case Study: Permafrost Thaw in the Arctic

> Dr. Debra Roberts, IPCC: 'Adaptation and mitigation must go hand in hand.'

Current Conditions and News: Case Study: Permafrost Thaw in the Arctic

- Glaciers worldwide are losing mass, contributing to sea level rise.

Future Scenarios and Predictions: Case Study: Permafrost Thaw in the Arctic

Under a high-emissions scenario (SSP5-8.5), permafrost thaw in the Arctic is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to permafrost thaw in the Arctic.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to permafrost thaw in the Arctic.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Case Study: Permafrost Thaw in the Arctic

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 59

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 59

- IPCC. (2022). Climate Change 2022: Impacts, Adaptation and Vulnerability (AR6 WG2). Cambridge University Press.
- IPCC. (2022). Climate Change 2022: Mitigation of Climate Change (AR6 WG3). Cambridge University Press.

Exercises — Chapter 59

- Reflect on how permafrost thaw in the Arctic affects your own community and write a one-page plan.

Chapter 60: Case Study: Urban Heat in Global Cities

Theme: urban heat in global cities

This section examines this chapter in the context of urban heat in global cities. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of urban heat in global cities. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

60.1 Urban Heat Island Effect

Evidence from peer-reviewed research shows that Urban Heat Island Effect is increasingly affected by rising temperatures,

shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Urban Heat Island Effect

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Urban Heat Island Effect

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Urban Heat Island Effect

- Assess your current situation regarding urban heat island effect and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

60.2 City Case Studies

For individuals and communities, the practical implications of City Case Studies include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for City Case Studies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: City Case Studies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for City Case Studies

- Assess your current situation regarding city case studies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

60.3 Health Impacts

Historical experience and recent disasters demonstrate that preparedness related to Health Impacts reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Health Impacts

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Health Impacts

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Health Impacts

- Assess your current situation regarding health impacts and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

60.4 Infrastructure Stress

Policy and governance play a decisive role in Infrastructure Stress. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Infrastructure Stress

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Infrastructure Stress

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Infrastructure Stress

- Assess your current situation regarding infrastructure stress and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

60.5 Cooling Strategies

Technology can support Cooling Strategies, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Cooling Strategies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Cooling Strategies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Cooling Strategies

- Assess your current situation regarding cooling strategies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

60.6 Green Infrastructure

Education and training strengthen Green Infrastructure. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Green Infrastructure

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Green Infrastructure

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Green Infrastructure

- Assess your current situation regarding green infrastructure and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

60.7 Governance and Planning

Monitoring and evaluation ensure that efforts related to Governance and Planning remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Governance and Planning

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Governance and Planning

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Governance and Planning

- Assess your current situation regarding governance and planning and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

60.8 Equity

This section examines Equity in the context of urban heat in global cities. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Equity

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Equity

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Equity

- Assess your current situation regarding equity and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

60.9 Lessons Learned

Evidence from peer-reviewed research shows that Lessons Learned is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Lessons Learned

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Lessons Learned

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Lessons Learned

- Assess your current situation regarding lessons learned and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

60.10 Future Outlook

For individuals and communities, the practical implications of Future Outlook include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Future Outlook

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Future Outlook

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Future Outlook

- Assess your current situation regarding future outlook and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Case Study: Urban Heat in Global Cities

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Case Study: Urban Heat in Global Cities

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Case Study: Urban Heat in Global Cities

> Dr. James Hansen, climate scientist: 'The science is clear: continued emissions will drive dangerous warming.'

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

Current Conditions and News: Case Study: Urban Heat in Global Cities

- 2023 was the hottest year on record, with global temperatures 1.48 C above pre-industrial levels (WMO, 2024).
- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.

Future Scenarios and Predictions: Case Study: Urban Heat in Global Cities

Under a high-emissions scenario (SSP5-8.5), urban heat in global cities is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to urban heat in global cities.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to urban heat in global cities.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Case Study: Urban Heat in Global Cities

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 60

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 60

- IPCC. (2018). Global Warming of 1.5 C (SR15). IPCC, Geneva.
- IPCC. (2019). Special Report on the Ocean and Cryosphere in a Changing Climate (SROCC). IPCC, Geneva.

Exercises — Chapter 60

- Reflect on how urban heat in global cities affects your own community and write a one-page plan.

Chapter 61: 30-Day Personal Action Plan

Theme: 30-day personal action plan

This section examines this chapter in the context of 30-day personal action plan. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of 30-day personal action plan. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

61.1 Week 1: Assess and Plan

Evidence from peer-reviewed research shows that Week 1: Assess and Plan is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Week 1: Assess and Plan

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Week 1: Assess and Plan

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Week 1: Assess and Plan

- Assess your current situation regarding week 1: assess and plan and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

61.2 Week 2: Supplies and Kits

For individuals and communities, the practical implications of Week 2: Supplies and Kits include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Week 2: Supplies and Kits

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Week 2: Supplies and Kits

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Week 2: Supplies and Kits

- Assess your current situation regarding week 2: supplies and kits and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

61.3 Week 3: Skills and Health

Historical experience and recent disasters demonstrate that preparedness related to Week 3: Skills and Health reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Week 3: Skills and Health

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Week 3: Skills and Health

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Week 3: Skills and Health

- Assess your current situation regarding week 3: skills and health and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

61.4 Week 4: Community and Review

Policy and governance play a decisive role in Week 4: Community and Review. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Week 4: Community and Review

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Week 4: Community and Review

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Week 4: Community and Review

- Assess your current situation regarding week 4: community and review and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

61.5 Daily Habits

Technology can support Daily Habits, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Daily Habits

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Daily Habits

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Daily Habits

- Assess your current situation regarding daily habits and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

61.6 Tracking Progress

Education and training strengthen Tracking Progress. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Tracking Progress

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Tracking Progress

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Tracking Progress

- Assess your current situation regarding tracking progress and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

61.7 Common Obstacles

Monitoring and evaluation ensure that efforts related to Common Obstacles remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Common Obstacles

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Common Obstacles

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Common Obstacles

- Assess your current situation regarding common obstacles and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

61.8 Adapting the Plan

This section examines Adapting the Plan in the context of 30-day personal action plan. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Adapting the Plan

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Adapting the Plan

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Adapting the Plan

- Assess your current situation regarding adapting the plan and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

61.9 Involving Family

Evidence from peer-reviewed research shows that Involving Family is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Involving Family

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Involving Family

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Involving Family

- Assess your current situation regarding involving family and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

61.10 Next Steps

For individuals and communities, the practical implications of Next Steps include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Next Steps

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Next Steps

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Next Steps

- Assess your current situation regarding next steps and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: 30-Day Personal Action Plan

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: 30-Day Personal Action Plan

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: 30-Day Personal Action Plan

> Prof. Katharine Hayhoe, climate scientist: 'Climate change is not a distant threat; it is here and now.'

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

Current Conditions and News: 30-Day Personal Action Plan

- 2024 continued the trend, with multiple months exceeding 1.5 C above pre-industrial levels.
- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.

Future Scenarios and Predictions: 30-Day Personal Action Plan

Under a high-emissions scenario (SSP5-8.5), 30-day personal action plan is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to 30-day personal action plan.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to 30-day personal action plan.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: 30-Day Personal Action Plan

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 61

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 61

- IPCC. (2019). Climate Change and Land (SRCCL). IPCC, Geneva.
- NASA. (2024). Global Climate Change: Vital Signs. climate.nasa.gov.

Exercises — Chapter 61

- Reflect on how 30-day personal action plan affects your own community and write a one-page plan.

Chapter 62: 1-Year Community Action Plan

Theme: 1-year community action plan

This section examines this chapter in the context of 1-year community action plan. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of 1-year community action plan. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

62.1 Quarter 1: Organize

Evidence from peer-reviewed research shows that Quarter 1: Organize is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Quarter 1: Organize

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Quarter 1: Organize

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Quarter 1: Organize

- Assess your current situation regarding quarter 1: organize and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

62.2 Quarter 2: Assess and Train

For individuals and communities, the practical implications of Quarter 2: Assess and Train include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Quarter 2: Assess and Train

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Quarter 2: Assess and Train

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Quarter 2: Assess and Train

- Assess your current situation regarding quarter 2: assess and train and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

62.3 Quarter 3: Implement

Historical experience and recent disasters demonstrate that preparedness related to Quarter 3: Implement reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Quarter 3: Implement

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Quarter 3: Implement

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Quarter 3: Implement

- Assess your current situation regarding quarter 3: implement and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

62.4 Quarter 4: Review and Scale

Policy and governance play a decisive role in Quarter 4: Review and Scale. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Quarter 4: Review and Scale

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Quarter 4: Review and Scale

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Quarter 4: Review and Scale

- Assess your current situation regarding quarter 4: review and scale and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

62.5 Building Partnerships

Technology can support Building Partnerships, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Building Partnerships

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Building Partnerships

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Building Partnerships

- Assess your current situation regarding building partnerships and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

62.6 Funding and Resources

Education and training strengthen Funding and Resources. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Funding and Resources

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Funding and Resources

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Funding and Resources

- Assess your current situation regarding funding and resources and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

62.7 Monitoring Progress

Monitoring and evaluation ensure that efforts related to Monitoring Progress remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Monitoring Progress

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Monitoring Progress

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Monitoring Progress

- Assess your current situation regarding monitoring progress and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

62.8 Communication

This section examines Communication in the context of 1-year community action plan. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Communication

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Communication

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Communication

- Assess your current situation regarding communication and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

62.9 Sustaining Momentum

Evidence from peer-reviewed research shows that Sustaining Momentum is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Sustaining Momentum

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Sustaining Momentum

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Sustaining Momentum

- Assess your current situation regarding sustaining momentum and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

62.10 Long-Term Vision

For individuals and communities, the practical implications of Long-Term Vision include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Long-Term Vision

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Long-Term Vision

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Long-Term Vision

- Assess your current situation regarding long-term vision and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: 1-Year Community Action Plan

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: 1-Year Community Action Plan

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: 1-Year Community Action Plan

> Dr. Michael Mann, climatologist: 'The evidence for human-caused warming is overwhelming.'

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

Current Conditions and News: 1-Year Community Action Plan

- Wildfires in Canada in 2023 burned a record area, sending smoke across North America.
- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.

Future Scenarios and Predictions: 1-Year Community Action Plan

Under a high-emissions scenario (SSP5-8.5), 1-year community action plan is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to 1-year community action plan.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to 1-year community action plan.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: 1-Year Community Action Plan

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 62

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 62

- NOAA. (2024). State of the Climate. ncei.noaa.gov.
- WMO. (2023). State of the Global Climate 2023. World Meteorological Organization, Geneva.

Exercises — Chapter 62

- Reflect on how 1-year community action plan affects your own community and write a one-page plan.

Chapter 63: Checklists and Templates

Theme: checklists and templates

This section examines this chapter in the context of checklists and templates. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of checklists and templates. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

63.1 Emergency Kit Checklist

Evidence from peer-reviewed research shows that Emergency Kit Checklist is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Emergency Kit Checklist

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Emergency Kit Checklist

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Emergency Kit Checklist

- Assess your current situation regarding emergency kit checklist and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

63.2 Water Storage Checklist

For individuals and communities, the practical implications of Water Storage Checklist include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Water Storage Checklist

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water Storage Checklist

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water Storage Checklist

- Assess your current situation regarding water storage checklist and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

63.3 Food Storage Checklist

Historical experience and recent disasters demonstrate that preparedness related to Food Storage Checklist reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Food Storage Checklist

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Food Storage Checklist

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Food Storage Checklist

- Assess your current situation regarding food storage checklist and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

63.4 Evacuation Checklist

Policy and governance play a decisive role in Evacuation Checklist. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Evacuation Checklist

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Evacuation Checklist

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Evacuation Checklist

- Assess your current situation regarding evacuation checklist and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

63.5 First Aid Checklist

Technology can support First Aid Checklist, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for First Aid Checklist

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: First Aid Checklist

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for First Aid Checklist

- Assess your current situation regarding first aid checklist and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

63.6 Document Checklist

Education and training strengthen Document Checklist. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Document Checklist

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Document Checklist

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Document Checklist

- Assess your current situation regarding document checklist and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

63.7 Home Safety Checklist

Monitoring and evaluation ensure that efforts related to Home Safety Checklist remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Home Safety Checklist

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Home Safety Checklist

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Home Safety Checklist

- Assess your current situation regarding home safety checklist and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

63.8 Communication Plan Template

This section examines Communication Plan Template in the context of checklists and templates. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Communication Plan Template

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Communication Plan Template

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Communication Plan Template

- Assess your current situation regarding communication plan template and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

63.9 Risk Assessment Template

Evidence from peer-reviewed research shows that Risk Assessment Template is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Risk Assessment Template

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Risk Assessment Template

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Risk Assessment Template

- Assess your current situation regarding risk assessment template and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

63.10 After-Action Review Template

For individuals and communities, the practical implications of After-Action Review Template include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for After-Action Review Template

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: After-Action Review Template

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for After-Action Review Template

- Assess your current situation regarding after-action review template and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Checklists and Templates

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Checklists and Templates

| Indicator | Value | Source |

|---|---|---|

| Global mean temperature rise since 1850-1900 | ~1.1-1.2 C | IPCC AR6 |

| CO2 concentration (2024) | ~420 ppm | NOAA |

| Sea level rise since 1900 | ~20-25 cm | IPCC AR6 |

| Arctic sea ice decline | ~13% per decade | NASA |

| People affected by climate disasters (annual) | ~100 million+ | IFRC |

| Economic damage from climate disasters (annual) | ~$200-300 billion | World Bank |

Expert Perspectives: Checklists and Templates

> Dr. Friederike Otto, climate researcher: 'Every heat wave today is made more likely by climate change.'

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

Current Conditions and News: Checklists and Templates

- Heat waves in South Asia in 2022 and 2023 caused thousands of deaths and crop losses.
- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.

Future Scenarios and Predictions: Checklists and Templates

Under a high-emissions scenario (SSP5-8.5), checklists and templates is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to checklists and templates.

- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to checklists and templates.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Checklists and Templates

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 63

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 63

- WHO. (2023). Climate Change and Health Fact Sheet. who.int.
- WHO. (2021). COP26 Special Report on Climate Change and Health. WHO, Geneva.

Exercises — Chapter 63

- Reflect on how checklists and templates affects your own community and write a one-page plan.

Chapter 64: Glossary

Theme: glossary of climate and survival terms

This section examines this chapter in the context of glossary of climate and survival terms. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of glossary of climate and survival terms. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

64.1 A-C

Evidence from peer-reviewed research shows that A-C is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for A-C

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: A-C

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for A-C

- Assess your current situation regarding a-c and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

64.2 D-F

For individuals and communities, the practical implications of D-F include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for D-F

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: D-F

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for D-F

- Assess your current situation regarding d-f and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

64.3 G-I

Historical experience and recent disasters demonstrate that preparedness related to G-I reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for G-I

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: G-I

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for G-I

- Assess your current situation regarding g-i and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

64.4 J-L

Policy and governance play a decisive role in J-L. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for J-L

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: J-L

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for J-L

- Assess your current situation regarding j-l and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

64.5 M-O

Technology can support M-O, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for M-O

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: M-O

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for M-O

- Assess your current situation regarding m-o and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

64.6 P-R

Education and training strengthen P-R. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for P-R

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).

- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: P-R

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for P-R

- Assess your current situation regarding p-r and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

64.7 S-U

Monitoring and evaluation ensure that efforts related to S-U remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for S-U

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: S-U

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for S-U

- Assess your current situation regarding s-u and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

64.8 V-Z

This section examines V-Z in the context of glossary of climate and survival terms. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for V-Z

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: V-Z

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for V-Z

- Assess your current situation regarding v-z and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Glossary

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Glossary

| Health Impact | Estimate | Source |

|---|---|---|

| Heat-related deaths annually | ~500,000+ | WHO/Lancet |

| Vector-borne disease expansion | Increasing range | WHO |

| Air pollution deaths annually | ~7 million | WHO |

| Climate anxiety prevalence (youth) | ~50%+ | Lancet Planetary Health |

| Mental health burden | Rising | WHO |

Expert Perspectives: Glossary

> Prof. Johan Rockstrom, Earth system scientist: 'We are pushing the planet beyond its safe operating space.'

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

Current Conditions and News: Glossary

- Flooding in Pakistan in 2022 affected 33 million people and caused over $30 billion in damage.
- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.

Future Scenarios and Predictions: Glossary

Under a high-emissions scenario (SSP5-8.5), glossary of climate and survival terms is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to glossary of climate and survival terms.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to glossary of climate and survival terms.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Glossary

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?

- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 64

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 64

- FAO. (2023). The State of Food Security and Nutrition in the World. FAO, Rome.
- FAO. (2022). The State of the World's Land and Water Resources (SOLAW 2021). FAO, Rome.

Exercises — Chapter 64

- Reflect on how glossary of climate and survival terms affects your own community and write a one-page plan.

Chapter 65: Extended References

Theme: extended references and further reading

This section examines this chapter in the context of extended references and further reading. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Chapter Overview

This chapter provides a detailed examination of extended references and further reading. It combines scientific evidence, practical guidance, and case-based learning. Readers are encouraged to consult the reference papers listed at the end of the chapter.

65.1 Climate Science

Evidence from peer-reviewed research shows that Climate Science is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Climate Science

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Climate Science

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Climate Science

- Assess your current situation regarding climate science and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

65.2 Health and Disasters

For individuals and communities, the practical implications of Health and Disasters include changes in daily routines, resource

use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for Health and Disasters

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Health and Disasters

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Health and Disasters

- Assess your current situation regarding health and disasters and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

65.3 Water and Food

Historical experience and recent disasters demonstrate that preparedness related to Water and Food reduces harm. Communities that invested in knowledge, infrastructure, and social ties recovered faster.

Key Considerations for Water and Food

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Water and Food

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Water and Food

- Assess your current situation regarding water and food and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

65.4 Economics and Policy

Policy and governance play a decisive role in Economics and Policy. Effective interventions combine regulation, finance, and participation, while inequitable policies deepen existing gaps.

Key Considerations for Economics and Policy

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Economics and Policy

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Economics and Policy

- Assess your current situation regarding economics and policy and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

65.5 Community and Social

Technology can support Community and Social, but it is not a substitute for social resilience. The most robust solutions integrate low-tech and high-tech approaches and remain accessible to all.

Key Considerations for Community and Social

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Community and Social

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Community and Social

- Assess your current situation regarding community and social and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

65.6 Technology

Education and training strengthen Technology. When people understand the science and practice the skills, they make better decisions before, during, and after crises.

Key Considerations for Technology

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Technology

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Technology

- Assess your current situation regarding technology and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

65.7 Mental Health

Monitoring and evaluation ensure that efforts related to Mental Health remain effective over time. Indicators should be simple, locally relevant, and regularly updated.

Key Considerations for Mental Health

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Mental Health

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Mental Health

- Assess your current situation regarding mental health and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

65.8 Case Studies

This section examines Case Studies in the context of extended references and further reading. Understanding this dimension is essential for survival under global warming, because it shapes both risk and response.

Key Considerations for Case Studies

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Case Studies

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Case Studies

- Assess your current situation regarding case studies and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

65.9 Education

Evidence from peer-reviewed research shows that Education is increasingly affected by rising temperatures, shifting precipitation, and more frequent extremes. These changes are not uniform; they vary by region, season, and population.

Key Considerations for Education

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Education

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for Education

- Assess your current situation regarding education and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

65.10 General Reading

For individuals and communities, the practical implications of General Reading include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Key Considerations for General Reading

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: General Reading

| Indicator | Current Status | Future Outlook |

|---|---|---|

| Risk level | Moderate to high | Increasing |

| Preparedness | Uneven | Needs improvement |

| Resource availability | Declining in stressed regions | Variable |

| Community capacity | Moderate | Can be strengthened |

Practical Steps for General Reading

- Assess your current situation regarding general reading and identify gaps.
- Document findings and share with household or team.
- Set a timeline and assign responsibilities.

Case Study: Extended References

For individuals and communities, the practical implications of the case study include changes in daily routines, resource use, and long-term planning. Ignoring these implications increases vulnerability.

Lessons from the Case Study

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

Data and Evidence: Extended References

| Food & Water | Estimate | Source |

|---|---|---|

| People facing food insecurity | ~800 million+ | FAO |

| People lacking safe water | ~2 billion | WHO/UNICEF |

| Crop yield decline (wheat, per 1 C) | ~6% | IPCC/Science |

| People displaced by climate (annual) | ~20 million+ | UNHCR |

| Economic loss from droughts | Billions USD | World Bank |

Expert Perspectives: Extended References

> Dr. Saleemul Huq, climate adaptation expert: 'Adaptation is not optional; it is survival.'

> Prof. Nicholas Stern, economist: 'The cost of inaction far exceeds the cost of action.'

Current Conditions and News: Extended References

- Drought in the Horn of Africa from 2020-2023 was the worst in 40 years, affecting 23 million people.
- Sea level rise is accelerating, with rates doubling in recent decades (IPCC AR6).

Future Scenarios and Predictions: Extended References

Under a high-emissions scenario (SSP5-8.5), extended references and further reading is projected to intensify significantly by 2050 and beyond. Under a low-emissions scenario (SSP1-2.6), impacts are reduced but not eliminated due to committed warming. These scenarios are based on IPCC AR6 and should be used for planning, not prediction of exact outcomes.

Scenario A: High Emissions (Business as Usual)

- Greater frequency and intensity of hazards related to extended references and further reading.
- Larger affected populations and economic losses.
- Higher adaptation costs and residual risk.

Scenario B: Low Emissions (Strong Mitigation)

- Reduced but not eliminated impacts related to extended references and further reading.
- More manageable adaptation needs.
- Greater social and economic stability.

Academic Debate and Discussion Points: Extended References

- What is the strength of evidence for the projected impacts, and what are the key uncertainties?
- How do different socioeconomic pathways alter the outcomes?
- What are the ethical implications for vulnerable populations?
- Which adaptation strategies have the strongest evidence base?
- How should trade-offs between mitigation and adaptation be resolved?

Key Takeaways — Chapter 65

- Assess local conditions and identify the most relevant hazards.
- Prioritize actions that protect the most vulnerable first.
- Build redundancy into critical systems (water, food, energy, communication).
- Practice plans regularly and update them after each event.
- Share knowledge and coordinate with neighbors and local institutions.

References — Chapter 65

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Exercises — Chapter 65

- Reflect on how extended references and further reading affects your own community and write a one-page plan.

Appendix A: Lampiran (Checklists & Panduan Penulisan/Debat)

Master Emergency Checklist

- Item 1: Verify supplies, documents, and communication plans.
- Item 2: Verify supplies, documents, and communication plans.
- Item 3: Verify supplies, documents, and communication plans.
- Item 4: Verify supplies, documents, and communication plans.
- Item 5: Verify supplies, documents, and communication plans.
- Item 6: Verify supplies, documents, and communication plans.
- Item 7: Verify supplies, documents, and communication plans.
- Item 8: Verify supplies, documents, and communication plans.
- Item 9: Verify supplies, documents, and communication plans.
- Item 10: Verify supplies, documents, and communication plans.

Panduan Penulisan dan Debat Akademik

Buku ini dirancang untuk digunakan dalam diskusi akademik. Saat memperdebatkan atau memperluas klaimnya:

- Selalu sitasi sumber primer (paper peer-reviewed, laporan IPCC).
- Bedakan antara observasi, atribusi, dan proyeksi.
- Akui ketidakpastian dan tingkat kepercayaan.
- Pertimbangkan berbagai jalur sosioekonomi (SSP).
- Sertakan dimensi keadilan dan kesetaraan.
- Perbarui klaim seiring munculnya bukti baru.

Appendix B: Glosarium

- Adaptation: Adjusting to actual or expected climate impacts.
- Mitigation: Reducing greenhouse gas emissions.
- Resilience: Capacity to absorb disturbance and reorganize.
- Vulnerability: Susceptibility to harm.
- Exposure: Presence of people or assets in hazard-prone areas.
- Sensitivity: Degree to which a system is affected by climate.
- Adaptive capacity: Ability to adjust to climate change.
- Tipping point: Threshold beyond which change becomes self-sustaining.
- Greenhouse effect: Trapping of heat by atmospheric gases.
- Carbon footprint: Total greenhouse gas emissions caused by an entity.
- SSP: Shared Socioeconomic Pathway (scenario framework).
- RCP: Representative Concentration Pathway.
- Committed warming: Warming locked in by past emissions.

Appendix C: Indeks

- Adaptation
- Biodiversity
- Carbon footprint
- Carbon pricing
- Climate finance
- Committed warming
- Conflict
- Drought
- Early warning
- Ecosystem
- Equity
- Exposure
- Flood
- Food security
- Greenhouse effect
- Health
- Heat wave
- Indigenous knowledge
- Infrastructure
- Justice
- Mental health
- Migration
- Mitigation
- Mutual aid
- RCP
- Renewable energy
- Resilience
- SSP
- Sea level rise
- Sensitivity
- Social capital
- Tipping point

- Vulnerability
- Water security

Appendix D: Daftar Pustaka (Extended References)

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Appendix E: Biografi Penulis

Setyo Wisodo adalah profesional yang menggabungkan keahlian dalam Pengembangan AI, Manajemen Produk, dan Pemasaran Digital. Aktif dalam pengembangan solusi berbasis AI dan produk digital. Lokasi: Indonesia.

End of Book